

CYBER SECURITY

UNIT 1 – COMPLETE DETAILED NOTES

Cyber Crime • Challenges of Cyber Crime • Classifications of Cyber Crimes

Email Spoofing | Spamming | Internet Time Theft | Salami Attack

Book-style notes with Practical Examples, Definitions, Flowcharts,
Diagrams, Important Points, Exam Notes & Memory Tricks

UNIT 1 – SYLLABUS (As per Examination)

S.No	Topic	Covered In
1	Cyber Security (Introduction, objectives, components, benefits)	Part 1
2	Cyber Attack (types, process, reasons, prevention)	Part 2
3	Introduction of Cyber Crime	Part 3
4	Challenges of Cyber Crime	Part 4
5	Classifications of Cyber Crimes	Part 5
5.1	Email Spoofing	Section 5.1
5.2	Spamming	Section 5.2
5.3	Internet Time Theft	Section 5.3
5.4	Salami Attack / Salami Technique	Section 5.4

How to read this book: Har topic me sabse pehle ek **long practical example** diya hai, phir **short definition**, phir **flowchart / diagram**, phir **important points** aur aakhir me **exam notes**. Isi order me padhna — definition yaad aasan ho jayegi.

CONTENTS

UNIT 1 – SYLLABUS (As per Examination)	2
PART 1 – CYBER SECURITY (Complete)	5
1.1 Cyber Security Kya Hoti Hai?	5
1.2 Cyber Security Kiski Security Karti Hai?	5
1.3 Cyber Security Ki Zarurat Kyu Hai?	5
1.4 Cyber Security Ke Objectives (Goals)	6
1.5 Cyber Security Ke Main Components	6
1.6 Cyber Security Kahan Use Hoti Hai?	6
1.7 Cyber Security Ke Benefits	6
1.8 Agar Cyber Security Na Ho To?	6
PART 2 – CYBER ATTACK (Complete)	8
2.1 Cyber Attack Kya Hota Hai?	8
2.2 Cyber Attack Me 3 Main Characters Hote Hain	8
2.3 Cyber Attack Kyu Kiya Jata Hai? (Motives)	9
2.4 Cyber Attack Ka Process (Attack Lifecycle)	9
2.5 Cyber Attack Kin Cheezon Par Ho Sakta Hai?	9
2.6 Cyber Attack Ke Main Categories	9
APPENDIX C – CYBER SECURITY MASTER TREE	11
APPENDIX B – MALWARE TREE	12
APPENDIX A – SECURITY SERVICES (CIA TRIAD)	13
2.7 Cyber Attack Ke Possible Results	14
2.8 Cyber Attack Ko Rokne Ke Tarike (Prevention)	14
PART 3 – INTRODUCTION OF CYBER CRIME	15
3.1 Long Practical Example (Real Life)	15
3.2 Definition	15
3.3 Flowchart	15
3.4 Elements of Cyber Crime (5 Cheezein Zaroori)	15
3.5 Cyber Crime vs Traditional Crime	16

3.6 Classifications of Cyber Crimes (Overview)	16
3.7 Characteristics of Cyber Crime	16
3.8 Who Commits Cyber Crime? (Types of Cyber Criminals)	16
3.9 Real World Examples (Chote Chote)	17
PART 4 – CHALLENGES OF CYBER CRIME	18
4.1 Long Practical Example (Real Life)	18
4.2 Definition	18
4.3 Flowchart – Challenges Ki Chain	18
4.4 Detailed Challenges (Sabhi Point Detail Me)	18
4.5 Diagram – Challenge vs Reason vs Solution	20
PART 5 – CLASSIFICATIONS OF CYBER CRIMES	21
5.0 Overview – Cyber Crimes Ki Classification	21
5.1 EMAIL SPOOFING	21
5.2 SPAMMING	22
5.3 INTERNET TIME THEFT	24
5.4 SALAMI ATTACK / SALAMI TECHNIQUE	25
APPENDIX D – QUICK REVISION SHEET (UNIT 1)	29

PART 1 – CYBER SECURITY (Complete)

Unit 1 ki pehli building block. Cyber Security samajhne ke liye pehle ye janna zaroori hai ki wo kya hoti hai, kisko protect karti hai, aur kyu zaroori hai. Aage ke saare chapters (Cyber Attack, Cyber Crime) isi par khade hain.

1.1 Cyber Security Kya Hoti Hai?

PRACTICAL EXAMPLE – REAL LIFE

Socho tumhare Papa ne ek naya ghar banaya. Us ghar me bahut important cheezein hain: Rs. 5 lakh cash, gold jewellery, laptop, documents (Aadhaar, PAN, Property Papers), TV, mobile, family photos.

Ab tumhare Papa **bina lock lagaye** ghar chhod dete hain. Kya ho sakta hai? Chor andar aa sakta hai, paise chori ho sakte hain, jewellery le ja sakta hai, documents jala sakta hai, ghar ka samaan tod sakta hai.

Isliye Papa security lagate hain: **strong door, lock, CCTV, alarm, security guard, fingerprint lock**. Ab chor ke liye ghar me ghusna bahut mushkil ho gaya.

PRACTICAL EXAMPLE – REAL LIFE

Aaj har insaan ka ek **Digital Ghar** hota hai. Usme kya hota hai? Gmail, WhatsApp, Instagram, Facebook, College Portal, Bank Account, UPI, ATM Details, Aadhaar, PAN, Photos, Passwords, Company Files.

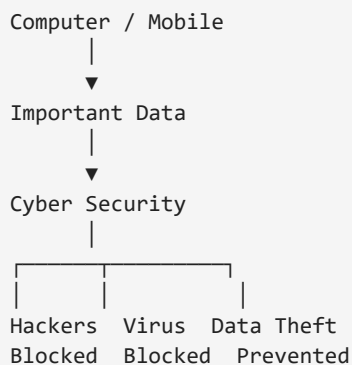
Ab socho agar koi hacker is digital ghar me ghus jaye to wo kya kar sakta hai? Password chori, bank account se paise transfer, photos leak, Gmail hack, Instagram account delete, college marks change, company ka data chori.

Bilkul waise hi jaise chor tumhare ghar me ghus gaya tha. Is digital ghar ko bachane ke liye jo security lagayi jaati hai, usse **Cyber Security** kehte hain.

DEFINITION

Cyber Security woh process hai jo computers, mobiles, servers, networks aur data ko hackers, viruses aur unauthorized access se protect karti hai.

FLOWCHART



1.2 Cyber Security Kiski Security Karti Hai?

Cyber Security **sirf computer ki security nahi karti**. Ye protect karti hai:

- Computer, Mobile, Network, Cloud, Database, Server
- Email, Online Banking, Online Shopping
- Hospital Records, College Data, Company Information

1.3 Cyber Security Ki Zarurat Kyu Hai?

Agar Cyber Security na ho to:

- Password chori ho sakta hai.

- ATM account hack ho sakta hai, WhatsApp hack ho sakta hai.
- Company ka data leak ho sakta hai.
- Hospital records delete ho sakte hain.
- Government websites hack ho sakti hain.

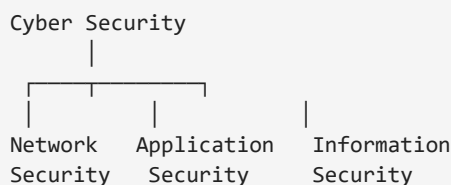
Daily Life Examples: ATM PIN, Phone Lock, Face Unlock, OTP, Gmail Password, Antivirus, Firewall — ye sab choti-choti security hi to hain jo hamari digital life protect karti hain.

1.4 Cyber Security Ke Objectives (Goals)

- **Data ko safe rakhna** — jaise bank balance.
- **Privacy bachana** — jaise WhatsApp chats.
- **Unauthorized access rokna** — sirf owner hi account khol sake.
- **Virus rokna** — computer ko damage hone se bachana.
- **Online fraud rokna** — fake websites aur scams se protection.

1.5 Cyber Security Ke Main Components

FLOWCHART



Aur bhi components hote hain: **Cloud Security, Mobile Security, IoT Security, Endpoint Security.**

1.6 Cyber Security Kahan Use Hoti Hai?

Jagah	Kyu
Bank	Money protect karne ke liye
Hospital	Patient records protect karne ke liye
College	Student data aur marks protect karne ke liye
Amazon / Flipkart	Customer payment secure karne ke liye
Airport	Flight systems ko secure rakhne ke liye
Army	Secret military information protect karne ke liye

1.7 Cyber Security Ke Benefits

- Data Safe, Privacy Safe, Money Safe, Identity Safe
- Business Safe, Trust Increase
- Online Transactions Safe

1.8 Agar Cyber Security Na Ho To?

Imagine ek din: tumhara Gmail hack, WhatsApp delete, Instagram kisi aur ke control me, bank account se Rs. 50,000 transfer, college portal me marks change, laptop ki files encrypt. **Ye sab ek hi din me ho sakta hai** — isi liye Cyber Security bahut important hai.

EXAM NOTES (2 MARKS / VIVA)

Exam Definition (2 Marks): Cyber Security is the practice of protecting computers, networks, devices and data from unauthorized access, cyber attacks and digital threats.

Viva Q1. Cyber Security kya protect karti hai?

Ans: Computer, Network, Server, Mobile aur Data.

Viva Q2. Cyber Security ka main purpose kya hai?

Ans: Data ko unauthorized access aur attacks se protect karna.

MEMORY TRICK

Cyber Security = Digital Ghar ki Security. Jaise ghar ko lock, CCTV aur guard protect karte hain, waise hi Cyber Security hamare digital data ko hackers aur attacks se protect karti hai.

PART 2 – CYBER ATTACK (Complete)

Foundation chapter. Cyber Attack ka concept samajhna zaroori hai kyunki Cyber Crime (Unit 1) aur Active/Passive Attack dono isi ki branches hain. Dhyani se padho — ye sabki root hai.

2.1 Cyber Attack Kya Hota Hai?

PRACTICAL EXAMPLE – REAL LIFE

Socho tumhare school me ek **Computer Lab** hai jisme 40 computers hain. Usme save hai: students ki marksheet, fees records, teachers ki salary, exam papers, principal ke documents, CCTV recordings.

Ab maan lo ek din ek hacker internet ke through school ke server se connect ho gaya. Wo kya kar sakta hai?

Situation 1: Sabhi students ke marks change kar deta hai — $90 \rightarrow 40$, $40 \rightarrow 95$.

Situation 2: Principal ka password change kar deta hai — ab Principal khud login nahi kar pa rahe.

Situation 3: Exam paper internet par leak kar deta hai.

Situation 4: Saari files delete kar deta hai.

Situation 5: School ki website band kar deta hai.

Situation 6: Students ki personal information (mobile, Aadhaar, address, parents details) chori kar leta hai.

Socho, school ko kitna bada loss hoga. Ye jo hacker ne **jaan-boojhkar** system ko nuksan pahunchaya, ise **Cyber Attack** kehte hain.

PRACTICAL EXAMPLE – REAL LIFE

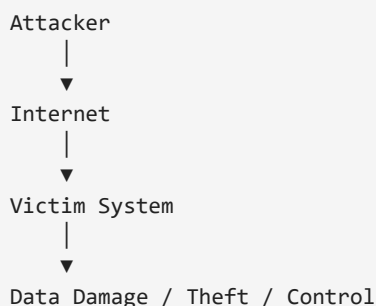
Tumhara Gmail password hai **Vinay@123**. Ek hacker ne kisi tarah password jaan liya. Fir usne Gmail login kiya, password change kar diya, recovery email badal di aur OTP number bhi badal diya.

Ab Gmail tumhara tha, lekin control hacker ke paas chala gaya. **Ye bhi Cyber Attack hai.**

DEFINITION

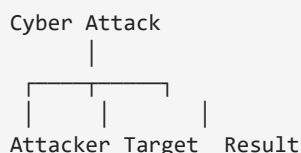
Cyber Attack ek intentional (jaan-boojhkar) attempt hota hai jisme attacker computer, network, server ya data ko access, steal, damage, modify ya destroy karne ki koshish karta hai.

FLOWCHART



2.2 Cyber Attack Me 3 Main Characters Hote Hain

DIAGRAM



- **Attacker** — jo attack karta hai: Hacker, Criminal, Terrorist, Insider Employee.
- **Target (Victim)** — jis par attack hota hai: Mobile, Laptop, Bank, Hospital, School, Company, Government Website.
- **Result** — attack ke baad kya hua: Data chori, Password leak, Website band, Virus install, Money transfer.

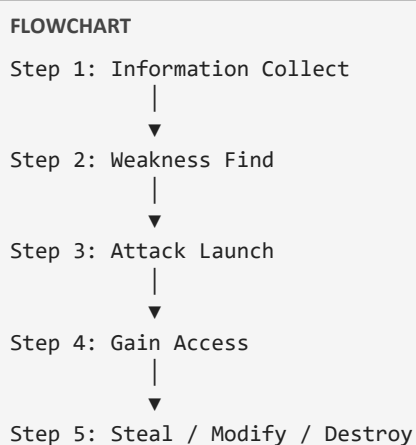
2.3 Cyber Attack Kyu Kiya Jata Hai? (Motives)

Bahut log sochte hain hacker sirf **maze ke liye** hack karta hai. Reality me reasons alag hote hain:

Motives	Example
1. Money	Bank account hack, UPI fraud, Credit Card fraud (sabse common)
2. Data Chori	Company ke secret documents chori karna
3. Identity Chori	Tumhare naam se fake account banana
4. Revenge	Ex-employee company ka database delete kar deta hai
5. Fame	Kuch hackers famous hone ke liye website hack karte hain
6. Political Reason	Ek country dusri country ke systems par attack karti hai
7. Terrorism	Power grid, airport ya railway system ko target karna

2.4 Cyber Attack Ka Process (Attack Lifecycle)

Har attack ek hi second me nahi hota. Generally ye steps hote hain:

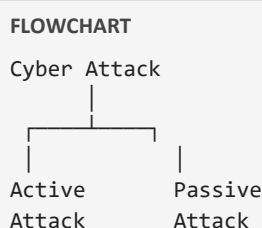


- **Step 1 – Information Collection:** Hacker pehle victim ke baare me info collect karta hai — naam, email, mobile, website, IP address.
- **Step 2 – Weakness Find:** Dekhta hai system me vulnerability kahan hai — weak password, old software, no antivirus.
- **Step 3 – Attack Launch:** Ab hacker attack karta hai.
- **Step 4 – Access:** System ke andar ghus jata hai.
- **Step 5 – Damage:** Ab jo chahe karta hai — delete, copy, encrypt, leak.

2.5 Cyber Attack Kin Cheezon Par Ho Sakta Hai?

Computer, Mobile, Wi-Fi, Server, Website, Email, Cloud, ATM, Smart TV, CCTV, IoT Devices — **koi bhi digital device target ban sakta hai.**

2.6 Cyber Attack Ke Main Categories

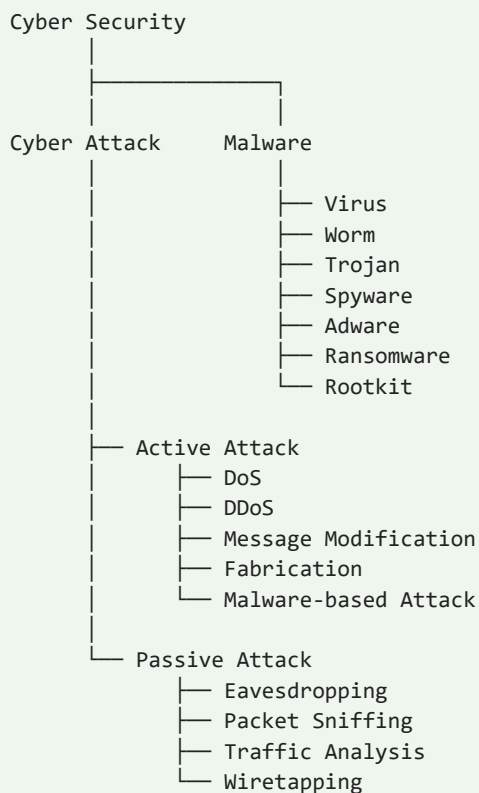


Active Attack (jaise DoS, DDoS, Message Modification, Masquerade) aur **Passive Attack** (jaise Eavesdropping, Packet Sniffing, Traffic Analysis) — inhe aage ki books me detail me padha jata hai. Cyber Crime ke liye bas itna yaad rakho ki **attack hi crime ka jariya hota hai**.

APPENDIX C – CYBER SECURITY MASTER TREE

Cyber Security ka **poora map** ek hi diagram me — isme Cyber Attack ki types (Active/Passive) aur Malware ki types dono dikhti hain. Ye exam me diagram ke liye ekdum ready hai.

DIAGRAM – CYBER SECURITY MASTER TREE



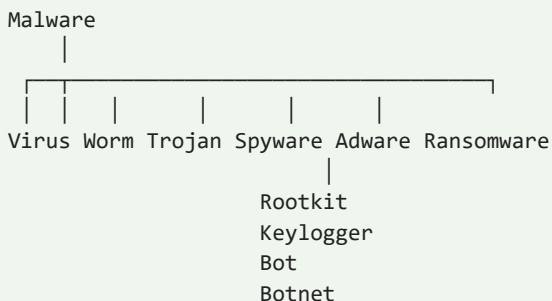
Active Attack = jo system me **badlaav** karta hai (modify/destroy) — jaise DoS, DDoS, Message Modification, Fabrication, Malware-based Attack. **Passive Attack** = jo sirf **dekh-ta hai** (data churakar padhta hai, badlaav nahi karta) — jaise Eavesdropping, Packet Sniffing, Traffic Analysis, Wiretapping.

Attack	Type	Kya hota hai
DoS / DDoS	Active	System par overload karke service band karna
Message Modification	Active	Message ke andar data badal dena
Fabrication	Active	Jhootha / fake message ghusa dena
Malware-based Attack	Active	Virus/ransomware install kar ke nuksan
Eavesdropping	Passive	Chupke se baat-chit sunna
Packet Sniffing	Passive	Network traffic ke packets padhna
Traffic Analysis	Passive	Data ko padhe bina uske patterns se info nikalna
Wiretapping	Passive	Kisi aur ki line/communication tap karna

APPENDIX B – MALWARE TREE

Malware = **Malicious software** — wo programs jo computer ko nuksan pahunchate hain. Ye exact tree exam ke diagram sawal ke liye ready hai.

DIAGRAM – MALWARE CLASSIFICATION TREE



Ye tree exam me 3 tarah se poochha jata hai: (1) Malware kya hai — definition, (2) Malware ki types — tree diagram, (3) kisi ek type ko explain karo — table se.

Malware	Kya karta hai	Example
Virus	Files me chhip kar phailta hai, damage karta hai	ILOVEYOU
Worm	Bina kisi file ke khud copy hoke network par phailta hai	Morris Worm
Trojan	Achha software ban ke andar ghus jata hai	Banking trojans
Spyware	Bina bataye aapki activity dekhta hai	Keyloggers, trackers
Adware	Zabardasti ads dikhata hai	Pop-up ads
Ransomware	Data lock (encrypt) karke paisa maangta hai	WannaCry
Rootkit	System ki jad me chhip jata hai, detection se bachta hai	Hiding in kernel
Keylogger	Har key press record karta hai (password churata hai)	Bank password steal
Bot / Botnet	Aapka system hacker ke remote control me	Spam-sending botnets

APPENDIX A – SECURITY SERVICES (CIA TRIAD)

Cyber Security ka poora foundation **CIA Triad** par khada hai. Exam me ye sabse poocha jaane wala diagram hai. Pehle **exact diagram**, phir explanation.

DIAGRAM – CIA TRIAD

CIA

- Confidentiality → Kaun dekh sakta hai?
- Integrity → Data badla ya nahi?
- Availability → Zarurat par mil raha hai?

DIAGRAM – BAQI SERVICES

Baaki Security Services:

- Authentication → Tum kaun ho?
- Authorization → Tum kya kar sakte ho?
- Accountability → Kisne kya kiya?
- Non-Repudiation → Kiya hai to mana nahi kar sakte.
- Accessibility → Authorized user aasani se use kar sake.
- Privacy → Personal information ko protect karna.

DEFINITION

Confidentiality: Data ko sirf authorized (allowed) log dekh saken.

Integrity: Data ko koi badal / corrupt na kar sake.

Availability: Authorized user ko zarurat par data hamesha milta rahe.

Service	Sawaal	Matlab	Attack Jo Todta Hai
Confidentiality	Kaun dekh sakta hai?	Data sirf authorized log hi dekh saken	Eavesdropping, Sniffing
Integrity	Data badla ya nahi?	Data original aur galat nahi hona chahiye	Message Modification
Availability	Zarurat par mil raha hai?	Authorized user ko kabhi bhi data mile	DoS / DDoS

Service	Sawaal	Matlab
Authentication	Tum kaun ho?	Pehchan verify karna (password, OTP, fingerprint)
Authorization	Tum kya kar sakte ho?	Access level decide karna (kya allowed hai)
Accountability	Kisne kya kiya?	Har kaam ka record hona chahiye
Non-Repudiation	Kiya hai to mana nahi kar sakte	Kiya hua kaam deny nahi kar sakte (digital signature)
Accessibility	Aasani se use kar sake	System sirf aapke liye kaam kare
Privacy	Personal info protect	Personal data kisi aur ke haath na jaye

2.7 Cyber Attack Ke Possible Results

Result	Example
Data Theft	Password chori
Data Modification	Marks badal dena
Data Deletion	Files delete
Financial Loss	Bank fraud
Service Down	Website crash
Privacy Loss	Personal photos leak

2.8 Cyber Attack Ko Rokne Ke Tarike (Prevention)

- Strong Password, 2-Factor Authentication (2FA)
- Antivirus, Firewall, Software Update, Backup
- Unknown links par click na karna, Public Wi-Fi par sensitive kaam na karna

EXAM NOTES (2 MARKS / VIVA)

Exam Definition (2 Marks): Cyber Attack is an intentional attempt to gain unauthorized access to a computer system, network, or data in order to steal, modify, destroy, or disrupt information or services.

Viva Q1. Cyber Attack kya hai?

Ans: Kisi computer ya network par unauthorized access ya damage pahunchane ki koshish ko Cyber Attack kehte hain.

Viva Q2. Cyber Attack ka sabse common objective kya hai?

Ans: Data theft, financial gain, aur system ko damage karna.

MEMORY TRICK

Cyber Attack = Digital Chor. Jaise ghar me chor ghuskar paisa ya samaan chori karta hai, waise hi cyber attack me attacker digital system me ghuskar data, password ya paise chori karta hai ya system ko nuksan pahunchata hai.

PART 3 – INTRODUCTION OF CYBER CRIME

Unit 1 ka asli syllabus yahan se shuru hota hai. Is chapter me hum samjhenge ki Cyber Crime kya hai, iske elements kya hain, ye traditional crime se kaise alag hai, aur kis kis category me divide hota hai.

3.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Meera Delhi me ek bank me kaam karti hai. Ek din uske phone par WhatsApp message aaya:

“Congratulations! Aapne Rs. 5,00,000 ki lottery jeeti hai. Claim karne ke liye is link par click karein.”

Meera ne link click kiya. Ek page khula jisme likha tha — **“Processing fee Rs. 2,000 bharo”**. Usne UPI se bhar diye. Phir aise hi do aur fees bhari. Total **Rs. 12,000** chale gaye. Lottery ka paisa kabhi nahi aaya. **Ye ek Cyber Crime hai (Online Fraud).**

Usi hafte aur bhi kuch hua:

- Meera ke friend **Rohit** ke Instagram account se uske followers ko “free iPhone” messages gaye — account hack ho gaya.
- College portal me kisi ne **marks change** kar diye.
- Ek kaam-wali didi ke **Aadhaar se fake SIM** nikal li gayi.

Ye saare incidents ek cheez me same hain — **computer / mobile / internet ka istemal karke criminal kaam hua**. Isi ko hum Cyber Crime kehte hain.

3.2 Definition

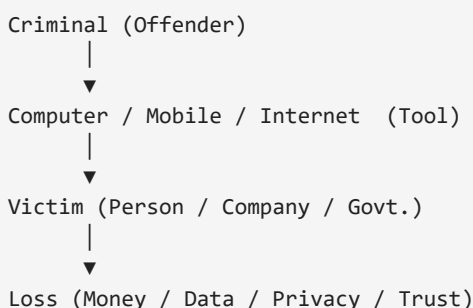
DEFINITION

Cyber Crime woh criminal activity hai jisme computer, mobile ya internet ka istemal karke koi illegal kaam kiya jata hai — jaise paisa chori karna, data chori karna, kisi ko dhamkana, ya kisi system ko nuksan pahunchana.

Simple English: Any criminal activity that involves a computer or the internet is called Cyber Crime.

3.3 Flowchart

FLOWCHART



3.4 Elements of Cyber Crime (5 Cheezein Zaroori)

Kisi bhi cyber crime me ye 5 elements hote hain — agar ek bhi missing ho to cyber crime nahi banega:

Element	Kya hai	Example
1. Offender	Jo galat kaam karta hai	Hacker, fraud gang, insider employee
2. Tool	Jis se galat kaam hota hai	Computer, internet, mobile, software
3. Victim	Jis par galat kaam hota hai	Person, company, bank, government

4. Act	Illegal activity khud	Hacking, fraud, spoofing, spam
5. Loss / Impact	Hone wala nuksan	Money, data, privacy, reputation

3.5 Cyber Crime vs Traditional Crime

Cyber Crime ko samajhne ka sabse aasan tarika hai ise **normal (traditional) crime** se compare karna:

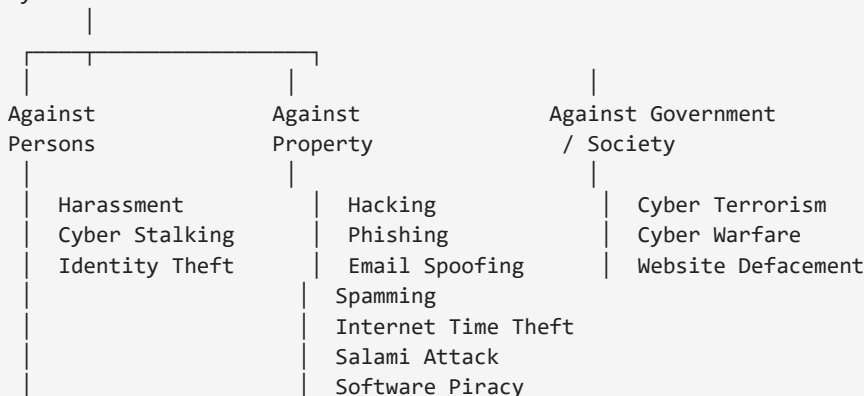
Basis	Traditional Crime	Cyber Crime
Jagah	Sirf ek jagah (local)	Internet hai to poore world me
Tool	Physical (chaku, bandook)	Computer, internet, software
Criminal ki pehchan	Pata chalti hai (face, CCTV)	Chhupi hoti hai (fake ID, VPN)
Evidence	Physical (fingerprint, weapon)	Digital (logs, IP, files) — delete ho sakta hai
Speed	Dheere hota hai	Seconds me khatam
Distance	Criminal ko paas hona zaroori	Duniya ke kisi bhi kone se
Jurisdiction	Ek hi desh/court ka law	Kaunsa desh ka law lagega — confusion
Record	Bahut kam records	Sab kuch digital record rehta hai

3.6 Classifications of Cyber Crimes (Overview)

Cyber Crimes ko target ke hisaab se **3 bade categories** me divide kiya jata hai:

FLOWCHART

Cyber Crime



IMPORTANT POINTS

Unit 1 syllabus me **4 classifications** detail me padhenge: **Email Spoofing, Spamming, Internet Time Theft, Salami Attack** — ye sab 'Against Property' category ke cyber crimes hain. Inhe Part 5 me poore detail me padha jayega.

3.7 Characteristics of Cyber Crime

- **Anonymity** — criminal apni pehchan chhupa kar kaam karta hai.
- **Global reach** — duniya ke kisi bhi country ko target kar sakta hai.
- **Low cost** — bas ek computer + internet chahiye.
- **No boundaries** — borders nahi maante.
- **Instant** — seconds me kaam ho jata hai.
- **Hard to detect** — evidence delete karna aasan hai.
- **High impact** — ek attack se lakhon logon ko nuksan ho sakta hai.

3.8 Who Commits Cyber Crime? (Types of Cyber Criminals)

Criminal Type	Kaun hai	Kya karta hai
Hackers	White / Black / Grey hat	System me ghusne wale technical log
Organized Criminals	Gangs	Bank fraud, card fraud, extortion
Insiders	Company ke andar ka employee	Andar ki information leak karta hai
Terrorists	Desh ke khilaf groups	Cyber terrorism, critical systems par attack
Disgruntled Employees	Ex-employee	Badla lene ke liye data delete/leak
Script Kiddies	Naye log	Ready-made tools use karke khelne wale

3.9 Real World Examples (Chote Chote)

- **WannaCry Ransomware (2017)** — duniya bhar ke hospitals aur laptops band kar diye.
- **Bank SIM-swap frauds** in India — SIM block karke OTP chura kar paise nikalna.
- **Aadhaar data leaks** — personal data market me bikna.
- **Deepfake videos** — kisi ki fake video banakar blackmail.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes – One Line Definitions:

1. Cyber Crime = computer/internet ke through ki gayi criminal activity.
2. Cyber Crime ke 5 elements = Offender, Tool, Victim, Act, Loss.
3. Cyber Crime ki 3 categories = Against Person, Against Property, Against Government.
4. Cyber crime me criminal chhupa rehta hai, par digital evidence rehta hai.

Viva Q1. Cyber Crime me 'tool' kya hota hai?

Ans: Computer, mobile, ya internet jiske through crime hota hai.

Viva Q2. Cyber crime ko traditional crime se alag kya banata hai?

Ans: Anonymity, global reach, speed aur digital evidence — criminal ko victim ke paas hone ki zaroorat nahi.

MEMORY TRICK

Cyber Crime = Digital Daku. Jaise highway par daku gaadi rok kar lootta hai, waise hi internet par cyber criminal system rok kar data aur paise lootta hai — bas farak itna hai ki daku ko face dikhata hai, cyber criminal nahi.

PART 4 – CHALLENGES OF CYBER CRIME

Crime hota hai, lekin usse pakadna itna aasan nahi hota. Is chapter me hum wo saari problems padhenge jo cyber crimes ko rokne, pakadne aur saza dilane me aati hain.

4.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ravi ke bank account se Rs. 2,00,000 gayab ho gaye. Police ne case khola. Lekin investigation me har step par rukawat aayi:

1. Account se paisa 3 alag accounts me gaya, phir 50 aur accounts me transfer hua.
2. Sab transactions **VPN** se hui — IP address fake tha.
3. Attacker South Africa me tha, server America me, victim India me.
4. Paisa cryptocurrency (Bitcoin) me badal diya gaya — track karna near impossible.
5. Jab tak police ko pata chala, 6 mahine nikal chuke the — digital records delete ho chuke the.

Police ke saamne problem: **criminal pakadna mushkil, saza dilana aur bhi mushkil.** Yehi problems — jo cyber crime se nipatne me aati hain — **Challenges of Cyber Crime** kehlati hain.

4.2 Definition

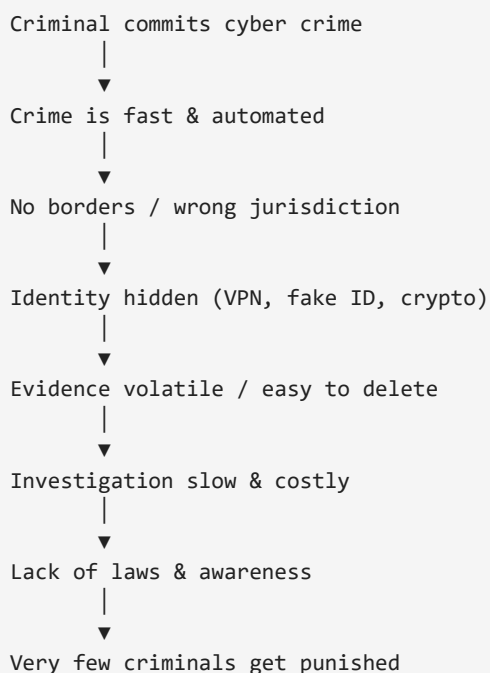
DEFINITION

Challenges of Cyber Crime wo difficulties / problems hain jo cyber crimes ko **prevent (rokne), detect (pakadne), investigate (jaanch) aur punish (saza dilane)** me saamne aati hain.

Simple: Crime karna aasan, lekin cyber crime ko pakadna — kyunki criminal chhupa hua hai, country ke bahar hai, aur evidence mit jata hai.

4.3 Flowchart – Challenges Ki Chain

FLOWCHART



4.4 Detailed Challenges (Sabhi Point Detail Me)

1. Anonymity (Pehchan Chhupana)

Attacker **VPN, Tor browser, fake accounts aur stolen SIM** istemal karta hai jisse uski asli pehchan kabhi saamne nahi aati. **Example:** fraud karne ke baad account fake naam se khula milta hai, asli banda kisi aur country me hota hai.

2. Jurisdiction Problem (Kaunsi Police Case Kare)

Crime **India** me hua, server **USA** me hai, attacker **Russia** me baitha hai. Ab kaunsi country ki police case kare? Har desh ke laws alag hote hain, aur cross-border cooperation me mahino lag jaate hain.

3. Global / Borderless Nature

Internet par koi border nahi hota. Ek click me attacker duniya ke kisi bhi system par attack kar sakta hai.

Example: ek country ke hacker dusre country ke bank par hamla — bina wahan gaye.

4. Evidence Collection (Saboot Jutana Mushkil)

Digital evidence (logs, cookies, IP records, files) **bahut nazuk** hota hai — computer band karo ya system restart karo to data jaa sakta hai. Delete karna bhi ek click ka kaam hai.

5. Fast & Automated Attacks

Bots aur scripts ek second me lakhon attacks kar dete hain. Police/security ke paas itna time hi nahi hota ki wo har attack ko rok saken.

6. Encryption (Data Lock)

Attacker apna data **encrypt** kar leta hai (jise bina key ke koi nahi padh sakta). Police ke liye encrypted data ko padhna aur prove karna bahut mushkil ho jata hai.

7. Rapidly Changing Technology

Roz nayi technology aur naye tareeke ke crimes bante hain. **Laws aur police piche reh jaate hain** kyunki naya crime samajhne me time lagta hai.

8. Lack of Awareness (Sabse Bada Weak Point)

Zyaada tar log OTP share kar dete hain, unknown links par click karte hain, weak passwords rakhte hain. **Insaan hi sabse bada weak link hai.** Cyber security itni strong nahi hoti ki user ko bacha sake.

9. Lack of Technical Skills in Investigation

Aam police ko cyber ka training nahi hoti. **Cyber cell / forensic lab mahangi aur kam** hoti hai, isliye investigation slow aur incomplete rehti hai.

10. Cross-border Legal Differences

Kuch desh me cyber crime ke **strict laws hi nahi hain** — aise desh cyber criminals ke liye safe haven ban jaate hain jahan se wo bina dar ke kaam karte hain.

11. High Cost of Security

Strong security (firewall, SIEM, cyber insurance, trained staff) bahut mahangi hai. Chhoti companies aur schools afford nahi kar paati — isliye wo aasan target ban jati hain.

12. Difficult Prosecution (Saza Dilana Mushkil)

Sahi saboot, sahi jurisdiction aur international cooperation ke bina **court me conviction rate bahut kam** hota hai. Criminal pakda bhi jaye to punish karna mushkil hai.

4.5 Diagram – Challenge vs Reason vs Solution

Exam me likhne ke liye ye compact table sabse useful hai:

Challenge	Problem Kyu Hai	Kya Solution Ho Sakta Hai
Anonymity	Criminal ki pehchan chhupi rehti hai	Digital footprint tracking, cyber forensics
Jurisdiction	Crime ka desh clear nahi	International treaties, mutual legal assistance
Borderless nature	Koi border nahi	Global cyber security cooperation
Evidence	Digital data volatile / deletable	Forensic imaging, proper chain of custody
Fast attacks	Seconds me ho jate hain	Automated detection, AI monitoring
Encryption	Data lock ho jata hai	Legal decryption procedures, key escrow
Technology change	Laws piche reh jate hain	Regular law updates
Lack of awareness	User khud dhoka kha jata hai	Cyber awareness campaigns, education
Low conviction	Proof + jurisdiction mushkil	Stronger laws, cyber courts

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Challenge = cyber crime ko rokne / pakadne me aane wali problem.
2. Sabse bada challenge = **Jurisdiction + Anonymity** (crime ka desh aur criminal ki pehchan).
3. Evidence digital hota hai — mit sakta hai, isliye jaldi collect karna zaroori.

Viva Q1. Cyber crime ko pakadna itna mushkil kyu hai?

Ans: Criminal chhupa hota hai (VPN/fake ID), crime country se bahar hota hai (jurisdiction), aur digital evidence mit jata hai.

Viva Q2. Insaan sabse bada weak point kyu?

Ans: Kyonki user OTP/share karta hai, unknown links par click karta hai aur weak passwords rakhta hai.

MEMORY TRICK

Challenge = Chhup-kar kaam karne wala criminal. Jaise chor raat ke andhere me chhup jata hai, waise cyber criminal VPN ke andhere me chhup jata hai — pakadna mushkil isliye kyunki koi face hi nahi dikhta.

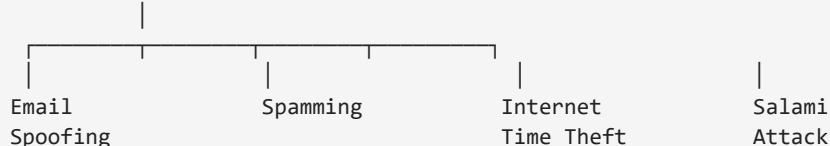
PART 5 – CLASSIFICATIONS OF CYBER CRIMES

Unit 1 ka sabse important chapter. Isme hum syllabus ke 4 classifications ko ek-ek karke detail me padhenge: **Email Spoofing, Spamming, Internet Time Theft, Salami Attack.** Har ek me pattern same rahega — Example → Definition → Diagram/Flowchart → Prevention → Exam Notes.

5.0 Overview – Cyber Crimes Ki Classification

FLOWCHART

Classifications of Cyber Crimes



Iske alawa bhi kai classifications hain (Phishing, Hacking, Cyber Stalking, Software Piracy, etc.) — lekin syllabus me yahi 4 main hain, aur inhi ko exam me detail me likhna hai.

5.1 EMAIL SPOOFING

PRACTICAL EXAMPLE – REAL LIFE

Riya ko ek email mila jo **bilkul uske bank se dikh raha tha** — sender: **support@sbibank.com**.

Email me likha tha: *“Aapka account block hone wala hai. Verify karne ke liye is link par click karein.”*

Riya ne link par click kiya — ek page khula jo bank ke page jaisa hi tha. Usne apni user ID, password aur OTP daal diye. Agle hi minute uske account se **Rs. 40,000** nikal gaye.

Sach: wo email bank ne nahi bheja tha. Sender ka address **fake** tha — attacker ne bank jaisa dikhne ke liye bana diya tha. **Isi ko Email Spoofing kehte hain.**

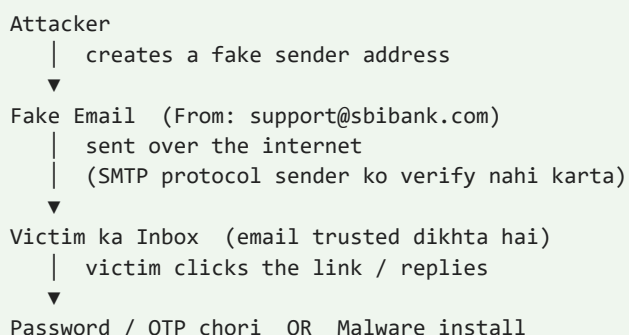
DEFINITION

Email Spoofing ek cyber crime hai jisme attacker email ke **sender address ko fake** karke aisa dikhata hai jaise email kisi trusted person / company se aaya ho — taaki victim dhoka kha kar apni sensitive information (password, OTP, bank details) de de ya malware install kar le.

Simple: Jisne bheja hai uska naam jhootha likhna, taaki tumhe lagge ki trusted logon ne bheja hai.

Kaise Kaam Karta Hai (Diagram)

DIAGRAM – HOW EMAIL SPOOFING WORKS



IMPORTANT POINTS

Ye kyun possible hai? Email bhejne wala protocol **SMTP (Simple Mail Transfer Protocol)** sender ka address **verify nahi karta**. Jaise koi letter par bina pakke hue jhootha naam likh de — post office rokta nahi. Waise hi email system bhi sender ko check nahi karta.

Types of Email Spoofing

Type	Kya hota hai	Example
1. Display Name Spoofing	Sirf naam trusted dikhta hai, email ID alag hoti hai	Riya Verma <riya123@gmail.com>
2. Legitimate Domain Spoofing	Same company ka domain hack karke use karta hai	support@sbibank.com (hacked)
3. Look-alike Domain Spoofing	Similar dikhne wala domain	sbibank.com vs sbibank-secure.com
4. Reply-to Spoofing	Reply jab karo to attacker ke paas jata hai	Reply-to: fraud@fake.com

Email Spoofing vs Phishing

Spoofing = sender address fake karna (technique). **Phishing = dhoka dena (purpose).** Spoofing ka istemal phishing karne ke liye hota hai. Matlab — spoofing chabi hai, phishing taala hai jo wo khulta hai. Dono aksar saath saath aate hain.

Prevention (Bachao Ke Tarike)

- Sender ka email address **dhyan se dekho** (domain check karo).
- Link par **hover** karke URL check karo — click karne se pehle.
- Urgent / dhamki wale messages par suspicious raho.
- **OTP kabhi kisi ko na do** — bank bhi OTP nahi maangta.
- Company/bank par call karke verify karo.
- Organizations apne email par **SPF, DKIM, DMARC** records enable karein.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Email Spoofing = sender ka address fake karna.
2. Target = trust jeet kar password/OTP churana.
3. Ye isliye possible hai kyunki **SMTP sender ko verify nahi karta**.

Viva Q1. Email spoofing me kya fake hota hai?

Ans: Sender ka address / identity.

Viva Q2. Spoofing aur phishing me kya relation hai?

Ans: Spoofing ek technique hai jo phishing (dhoka) karne ke liye use hoti hai.

5.2 SPAMMING

PRACTICAL EXAMPLE – REAL LIFE

Aman ka mobile har din **30 se zyada spam messages** se bhara rehta hai:

- “Aapko sirf Rs. 99 me smartphone milega!”
- “Congratulations! Prize jitne ke liye reply karein”
- Random loan / gambling / casino ads

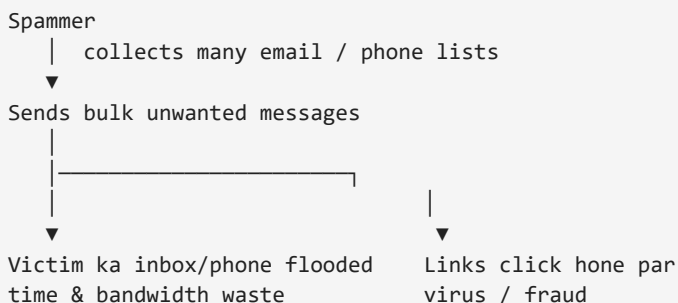
Aur uske Gmail me bhi 100 unread promotional emails — asli emails dhoondna mushkil ho jata hai.

Agar Aman inme se kisi link par click kar de — malware ya fraud. **Yehi Spamming hai.**

DEFINITION

Spamming ek cyber crime hai jisme kisi ko **bina uski permission ke bulk (bahut zyada) unwanted messages, emails ya advertisements** bheje jaate hain.

Simple: Kisi aur ka inbox/mobile bina pooche-jhadhe ads aur bekaar messages se bhar dena.

FLOWCHART**Types of Spam**

Type	Kya hai
Email Spam	Bulk promotional / junk emails
SMS Spam	Bulk messages on mobile
Comment Spam	Blogs / forums par fake comments (link ke saath)
Search Spam	Search results me fake pages dikhana
Social Media Spam	Fake accounts se bulk messages / tags

Spamming Ka Nuksan (Harm)

- **Time waste** — real messages dhoondne me time lagta hai.
- **Bandwidth / data khatam** — mobile data aur internet bill badta hai.
- **Real email miss** ho jate hain (important mail spam me dab jati hai).
- **Phishing aur malware ka zariya** — spam ke andar harmful links hote hain.
- **Fraud ka gate** — spam me fake offers lottery scams bheje jate hain.

Spam vs Scam

Spam = koi bhi unwanted bulk message. **Scam** = us message ke andar ka dhoka (paisa chori ki koshish). Spam scam ka **carrier** ban sakta hai — spam message ke andar scam ka link hota hai.

Prevention (Bachao Ke Tarike)

- **Spam filter ON** rakho (email apps me default hota hai).
- Apna email / phone number **public jagah share na karo** (social media, random sites).
- Trusted sites par hi **unsubscribe** karo.
- **Unknown links par click mat karo** — reply bhi mat karo.
- Spam ko **report** karo.
- Sign-ups ke liye **alag email ID** banao (junk email).

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Spamming = bina permission ke bulk unwanted messages bhejna.
2. Spam ka sabse bada nuksan = time/bandwidth waste + phishing ka zariya.
3. Spam vs Scam = spam carrier hai, scam dhoka hai.

Viva Q1. Spamming ko cyber crime kyu mana jata hai?

Ans: Kyunki ye bina permission ke doosron ke inbox/data par bojh daalta hai aur fraud ka zariya banta hai.

Viva Q2. Spam se bachne ka best tarika?

Ans: Spam filter, email public na karna, unknown links par click na karna.

5.3 INTERNET TIME THEFT

PRACTICAL EXAMPLE – REAL LIFE

Ek company har month internet bill deti hai — Rs. 50,000. Employees ka kaam 6 ghante ka hai. Lekin ek employee **Suresh** roz 3 ghante **office internet par online movies dekhta hai, games download karta hai, file sharing karta hai.** Iska asar:

- Office ka internet slow ho jata hai — baaki employees ka kaam rukta hai.
- Company ka bandwidth aur paisa bekaar jaata hai.
- Company ke data par risk aata hai (pirated content download).

Yahan company ka **paid internet time aur bandwidth bina permission ke personal kaam me use hua — Internet Time Theft.**

PRACTICAL EXAMPLE – REAL LIFE

Kabir ke neighbour ne Kabir ka **Wi-Fi password guess** kar liya. Wo roz Kabir ke internet se downloads karta hai. Ab:

- Kabir ka internet slow ho jata hai.
- Data / FUP khatam ho jata hai, bill badhta hai.

Sabse khatarnak baat — agar neighbour ne internet par koi **galat kaam** kiya (illegal download, fake post), to IP address ke hisaab se **police Kabir ke ghar aayegi** kyunki login Kabir ke connection se hua tha. Kabir ko apni bezati ke saath apni bezosai sabit karni padegi.

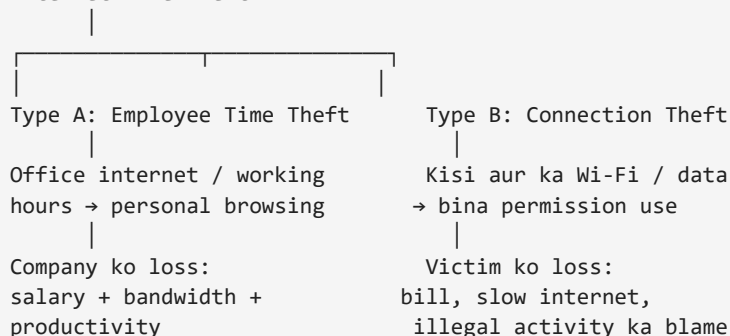
DEFINITION

Internet Time Theft ek cyber crime hai jisme koi person kisi doosre ki **paid internet connection, bandwidth, ya kaam ka samay bina permission ke apne personal / illegal kaam me use** karta hai.

Simple: Doosre ka internet / doosre ka kaam ka time — bina pooche apne kaam me lagaana.

FLOWCHART

Internet Time Theft



Examples

- Office me duty hours par **online gaming / movie** dekhna.
- Neighbour ka **Wi-Fi password hack / guess** karke use karna.
- Kisi ka **SIM data balance** bina permission use karna.
- Shared connection (network spoofing) bina permission use karna.

Prevention (Bachao Ke Tarike)

- **Strong Wi-Fi password + WPA2/WPA3** encryption.
- Company me **internet usage policy aur monitoring**.
- **MAC address filtering** — sirf known devices hi connect hon.
- **Guest network** — visitors ke liye alag network.
- Default password turant **change** karo, time-to-time password change karo.
- Office kaam aur personal kaam ke liye **alag connections**.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Internet Time Theft = doosre ka paid internet/bandwidth/time bina permission use karna.
2. Do types = Employee Time Theft aur Connection Theft.
3. Danger = illegal activity ka IP blame victim par aata hai.

Viva Q1. Internet Time Theft me kaun kaun loss me hota hai?

Ans: Company (salary/bandwidth/productivity) ya victim (bill/slow internet/blame).

Viva Q2. Isse bachne ke liye sabse zaroori kya hai?

Ans: Strong password + WPA2/WPA3 aur usage monitoring.

5.4 SALAMI ATTACK / SALAMI TECHNIQUE

PRACTICAL EXAMPLE – REAL LIFE

Ek bank me **10 lakh logon ke accounts** hain. Har account par interest credit hota hai.

Interest nikalte waqt exact value aati hai, jaise: Rs. 100.0047, Rs. 55.2358 ...

Bank aam taur par **paise tak round** karta hai: Rs. 100.0047 → Rs. 100.00 credit.

Attacker (employee ya hacker) system me **chhota sa code** daal deta hai jo round karne ke baad bacha fraction **(0.0047, 0.2358) alag account me transfer** kar deta hai.

- 1 account se kuch nahi — sirf Rs. 0.0047.
- 10 lakh accounts se — $0.0047 \times 10,00,000 = \text{Rs. } 4,700$ chupchaap.
- Agar fraction 0.50 se 0.90 ka ho to — **lakhon rupaye**.

Kisi bhi account holder ko notice nahi hota — kyunki usko hamesha puri value hi mil rahi thi (round off). Total bekaar me kai lakh rupaye chori — **ye Salami Attack hai**.

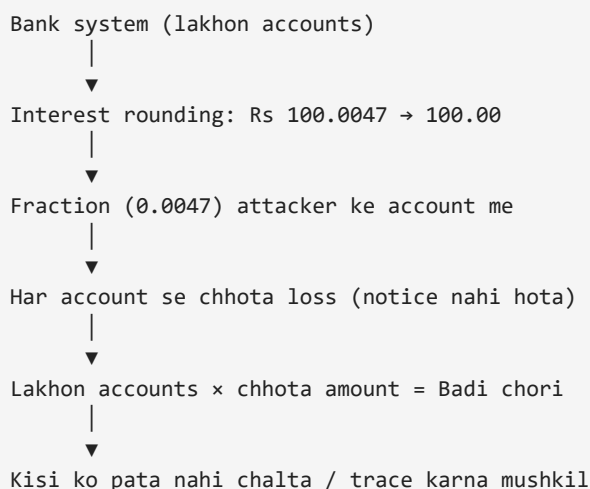
DEFINITION

Salami Attack (Salami Technique) ek cyber crime hai jisme bahut se accounts / files se **bahut chhote-chhote amounts** is tarah cut kiye jaate hain ki **koi notice na kare**, lekin **total badi amount** ban jati hai.

Simple: Har jagah se thoda-thoda — kisi ko pata na chale — lekin total milakar bada gola.

Naam 'Salami' Kyu?

Salami (sausage) ko **bahut patle-patle slices** me kaata jata hai — har slice kisi ko nazar nahi aata, lekin dheere-dheere **poori sausage khatam** ho jaati hai. Bilkul isi tarah is attack me **chhote-chhote hisse** kaate jaate hain — isliye naam **Salami Attack**.

FLOWCHART**Formula (Diagram)****DIAGRAM**

Chhota Amount × Bahut Saare Accounts = Badi Chori

Rs. 0.0047 × 10,00,000 accounts = Rs. 4,700

Rs. 0.50 × 10,00,000 accounts = Rs. 50,00,000

(Har account ko loss itna chhota ki wo notice hi na kare)

Salami Technique Ke Aur Examples

- **Billing system** — har bill par Rs. 1 extra (rounding), 1 lakh bills = Rs. 1 lakh chori.
- **Currency exchange** — fraction ka paisa (paisa value) wapas na dena, collect karte jaana.
- **Payroll / pension** — har employee ke salary se Rs. 2-3 katna — kisi ko pata nahi chalta.

Prevention (Bachao Ke Tarike)

- **Audit trails** — har transaction ka pura record.
- Rounding / fraction ke changes par **monitoring**.
- **Internal controls + dual approval** — ek banda akela change na kar sake.
- **Regular reconciliation** — saare accounts match karna (deposits vs records).
- Fraction data ko bhi **log** karna.
- **Whistle-blower policy** — employees bina dar ke report karein.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Salami Attack = bahut chhote-chhote amounts, bahut saare accounts se, notice na hote hue — total bada.
2. Naam 'Salami' isliye — jaise sausage ke patle slices kaatte hain, waise hi chhote hisse kaate jaate hain.
3. Formula: Chhota amount $\times$ Lakhon accounts = Badi chori.
4. Ye itna dangerous hai kyunki **detection bahut mushkil** hota hai — koi bhi ek account owner notice nahi karta.

Viva Q1. Salami attack me har account ko kitna loss hota hai?

Ans: Bahut chhota (paisa-fraction) — isliye notice nahi hota.

Viva Q2. Salami attack ko pakadna mushkil kyu?

Ans: Kyunki har jagah amount na ke barabar hai, sirf total bada hota hai — aur rounded values ki wajah se kisi ko difference dikhta nahi.

MEMORY TRICK

Salami = Sausage ki patli-slice. Jaise patle-patle slices se poori sausage khatam — waise hi chhote-chhote cut se poori daulat khatam. Socho: **1 rupaya $\times$ 1 crore accounts = 1 crore rupaye** — kisi ko pata bhi nahi chala.

APPENDIX D – QUICK REVISION SHEET (UNIT 1)

Sabse Pehle Yaad Rakho — One-Line Definitions

Topic	Ek Line Me
Cyber Security	Computers, networks, data ko attacks se protect karne ka process
Cyber Attack	System/data ko access, steal, damage karne ka intentional attempt
Cyber Crime	Computer/internet ke through ki gayi criminal activity
Email Spoofing	Sender ka address fake karke trusted dikhana
Spamming	Bina permission ke bulk unwanted messages bhejna
Internet Time Theft	Doosre ka paid internet/bandwidth/time bina permission use karna
Salami Attack	Chhote-chhote amounts, bahut saare accounts se, notice na hote hue

Memory Tricks

- Cyber Security = **Digital Ghar ki Security**.
- Cyber Attack = **Digital Chor**.
- Cyber Crime = **Digital Daku**.
- Salami = **Sausage ki patli-slice** — 1 rupaya × 1 crore accounts = 1 crore रुपये.
- CIA = **Confidentiality, Integrity, Availability**.
- Active Attack = **Badlaav karta hai**, Passive Attack = **Sirf dekhta hai**.

Rapid Fire Viva (10 Questions)

Q	Answer
Cyber security kya protect karti hai?	Computer, network, server, mobile, data
Cyber attack ka sabse common motive?	Paisa (financial gain)
Cyber crime ke 5 elements?	Offender, Tool, Victim, Act, Loss
Cyber crime ki 3 categories?	Against Person, Against Property, Against Government
Email spoofing me kya fake hota hai?	Sender ka address
Email spoofing kyu possible hai?	SMTP sender ko verify nahi karta
Spam ka carrier kya hota hai?	Scam / phishing / malware
Internet Time Theft ke 2 types?	Employee Time Theft, Connection Theft
Salami attack me kitna loss per account?	Na ke barabar (chhota) — isliye notice nahi hota
Sabse mushkil cyber crime pakadna?	Salami attack (koi notice nahi karta)

Note: Ye Unit 1 ka complete notes hai. Aage ke chapters (Active/Passive Attack detail, Malware types, Phishing, Virus etc.) same pattern par alag book me banenge.

CYBER SECURITY

UNIT 2 – COMPLETE DETAILED NOTES

Web Jacking • Online Frauds • Software Piracy • Computer Network Intrusions •
Password Sniffing • Identity Theft

Cyber Terrorism • Virtual Crime • Perception of Cyber Criminals • Web Server Hacking • Session Hijacking

Book-style notes with Practical Examples, Definitions, Flowcharts,
Diagrams, Important Points, Exam Notes & Memory Tricks

UNIT 2 – SYLLABUS (As per Examination)

S.No	Unit 2 Topics	Marks
1	Web Jacking	---
2	Online Frauds	---
3	Software Piracy	---
4	Computer Network Intrusions	---
5	Password Sniffing	---
6	Identity Theft	---
7	Cyber Terrorism	---
8	Virtual Crime	---
9	Perception of Cyber Criminals (Hackers, Insurgents & Extremist Groups)	---
10	Web Server Hacking	---
11	Session Hijacking	---

How to read this book: Har topic me sabse pehle ek **long practical example** diya hai, phir **short definition**, phir **flowchart / diagram**, phir **important points** aur aakhir me **exam notes**. Isi order me padhna — definition yaad aasan ho jayegi.

CONTENTS

UNIT 2 – SYLLABUS (As per Examination)	2
PART 1 – WEB JACKING	6
1.1 Long Practical Example (Real Life)	6
1.2 Definition	6
1.3 Flowchart – Web Jacking Kaise Hota Hai	6
1.4 Web Jacking Ke Tarike (Kaise Hota Hai)	6
1.5 Web Jacking vs Website Defacement	6
1.6 Prevention (Bachao Ke Tarike)	7
PART 2 – ONLINE FRAUDS	8
2.1 Long Practical Example (Real Life)	8
2.2 Definition	8
2.3 Flowchart – Online Fraud Kaise Hota Hai	8
2.4 Types of Online Frauds	8
2.5 Red Flags (Fraud Kaise Pehchano)	9
2.6 Prevention (Bachao Ke Tarike)	9
PART 3 – SOFTWARE PIRACY	10
3.1 Long Practical Example (Real Life)	10
3.2 Definition	10
3.3 Flowchart	10
3.4 Types of Software Piracy	10
3.5 Piracy Ka Nuksan (Impact)	10
3.6 Prevention (Bachao Ke Tarike)	11
PART 4 – COMPUTER NETWORK INTRUSIONS	12
4.1 Long Practical Example (Real Life)	12
4.2 Definition	12
4.3 Flowchart – Intrusion Process	12
4.4 Types of Network Intrusions	12
4.5 Intrusion Kaise Hoti Hai (Methods)	12
4.6 Detection aur Prevention	13

PART 5 – PASSWORD SNIFFING	14
5.1 Long Practical Example (Real Life)	14
5.2 Definition	14
5.3 Flowchart – Sniffing Process	14
5.4 Ye Kyu Possible Hai?	14
5.5 Sniffing Tools (Exam Me Naam Aate Hain)	14
5.6 Prevention (Bachao Ke Tarike)	14
PART 6 – IDENTITY THEFT	16
6.1 Long Practical Example (Real Life)	16
6.2 Definition	16
6.3 Flowchart – Identity Theft Process	16
6.4 Identity Info Kaise Chori Hoti Hai (Methods)	16
6.5 Impact (Nuksan)	16
6.6 Prevention (Bachao Ke Tarike)	17
PART 7 – CYBER TERRORISM	18
7.1 Long Practical Example (Real Life)	18
7.2 Definition	18
7.3 Flowchart	18
7.4 Targets of Cyber Terrorism	18
7.5 Cyber Crime vs Cyber Terrorism	19
7.6 Prevention (Bachao Ke Tarike)	19
PART 8 – VIRTUAL CRIME	20
8.1 Long Practical Example (Real Life)	20
8.2 Definition	20
8.3 Flowchart	20
8.4 Types of Virtual Crimes	20
8.5 Important Points	20
8.6 Prevention (Bachao Ke Tarike)	21
PART 9 – PERCEPTION OF CYBER CRIMINALS	22
9.1 Long Practical Example (Real Life)	22
9.2 Definition	22

9.3 Flowchart – Cyber Criminals Ki Perception Categories	22
9.4 Hackers (Detail Me)	22
9.5 Insurgents (Detail Me)	22
9.6 Extremist Groups (Detail Me)	23
9.7 Baaki Cyber Criminals (Ek Nazar)	23
PART 10 – WEB SERVER HACKING	24
10.1 Long Practical Example (Real Life)	24
10.2 Definition	24
10.3 Flowchart	24
10.4 Common Methods (Vulnerabilities)	24
10.5 Impact (Nuksan)	25
10.6 Prevention (Bachao Ke Tarike)	25
PART 11 – SESSION HIJACKING	26
11.1 Long Practical Example (Real Life)	26
11.2 Definition	26
11.3 Flowchart	26
11.4 Types of Session Hijacking	26
11.5 Prevention (Bachao Ke Tarike)	26
APPENDIX – QUICK REVISION SHEET (UNIT 2)	29

PART 1 – WEB JACKING

Unit 2 ka pehla topic. Web Jacking me website ki **poori control** chali jati hai — website khud nahi badalti, par uske visitors ko kahin aur bhej diya jata hai. Detail me samjhte hain.

1.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek chhoti company **Bandhan Store** ki website thi — bandhanstore.com — jahan se log clothes kharidte the aur card se payment karte the.

Ek din ek hacker ne company ke **domain/DNS account ka password** hack kar liya. Ab usne:

1. Website ka saara control apne paas le liya.
2. Visitors jab bhi bandhanstore.com kholte the, wo seedha **fake page** par chale jaate the jo bilkul asli jaisa dikhta tha.
3. Fake page par customer ne card details daali — details hacker ke paas chali gayi.
4. Kuch visitors ko virus wali file download bhi ho gayi.

Company ko pata bhi nahi chala ki uski website ab hacker ke control me hai. **Website ko hi 'jacked' (control me le liya) kar ke visitors ko fake jagah bhejna — yehi Web Jacking hai.**

1.2 Definition

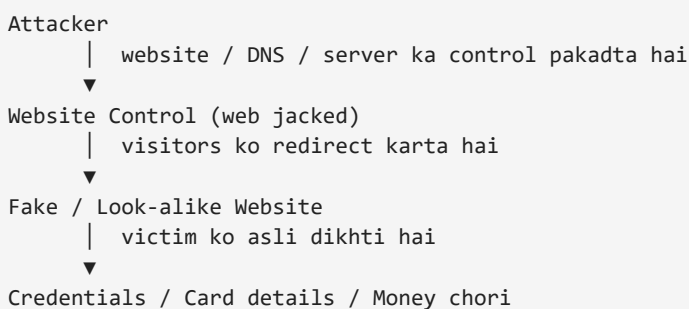
DEFINITION

Web Jacking ek attack hai jisme attacker kisi website ka **control haath me le leta hai** aur uske saare visitors ko bina unki jaankari ke kisi **fake ya illegal website par redirect** kar deta hai — taaki data, paise ya credentials chura sake.

Simple: Website ki gaddi ka steering wheel chura kar uske sawari ko kahin aur le jaana.

1.3 Flowchart – Web Jacking Kaise Hota Hai

FLOWCHART



1.4 Web Jacking Ke Tarike (Kaise Hota Hai)

- **DNS Hijacking** — domain/DNS settings badal kar website ka traffic kahin aur bhejna.
- **Web server compromise** — server hack karke redirect rules (htaccess/config) change karna.
- **Malicious redirects** — site me code/malware daal kar visitors ko fake page par le jaana.
- **Look-alike domain + traffic steal** — milta-julta domain (bandhanstore.net) banakar confusion paida karna.

1.5 Web Jacking vs Website Defacement

Basis	Web Jacking	Defacement
Kya hota hai	Website ka control + visitors redirect	Sirf page ka content badal dena

Visitors	Fake site par le jaaye jaate hain	Asli site par rehte hain
Data theft	Credentials/card details chori hoti hai	Aam taur par sirf message/chinta
Damage	Financial loss, identity theft	Reputation + trust loss

1.6 Prevention (Bachao Ke Tarike)

- Domain registrar account par **strong password + 2FA**.
- **DNSSEC** enable karo (DNS ko fake responses se bachata hai).
- Website par **HTTPS** lagao aur hosting passwords strong rakho.
- Server/software ko **update** karte raho, backups rakho.
- Website aur DNS records ko **regularly monitor** karo.
- Visitors ko bhi sikhana: URL dhyan se dekho, koi bhi link par OTP/card daalne se pehle check karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Web Jacking = website ka control le kar visitors ko fake site par redirect karna.
2. Main tarika = DNS hijacking ya web server compromise.
3. Defacement me sirf content badalta hai, Web Jacking me poora control + redirect.

Viva Q1. Web jacking me visitor ko kya dikhta hai?

Ans: Fake/look-alike page jo asli jaisa dikhta hai.

Viva Q2. Web jacking se kaise bachein?

Ans: DNSSEC, strong password + 2FA, HTTPS, monitoring.

MEMORY TRICK

Web Jacking = Website ki hijack. Jaise plane hijack kar ke usse kahin aur le jaate hain, waise hi website hijack kar ke uske visitors ko fake page par le jate hain.

PART 2 – ONLINE FRAUDS

Online Fraud sabse aam cyber crime hai — **paisa lootne ke liye dhoka**. Is part me hum types aur bachao ka tarika detail me dekhenge.

2.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Priya ko WhatsApp par message aaya: “Amazon se aapki ek order wapas ho gayi hai. Rs. 2,499 ka refund milega. Form bharein.”

Usne form bhara — usme **card number, expiry, CVV, OTP** maanga gaya. Dalte hi uske card se **Rs. 15,000** kat gaye.

Kuch din baad usne ek website se **Rs. 999 ka smartphone** dekha (asli price Rs. 9,000). Payment online kiya — phone kabhi nahi aaya, website gayab. Ye dono **Online Frauds** hain.

Ek aur example: Ravi ko job ka email mila — “Registration fee Rs. 5,000 bharo, job pakki.” Fee bhari, interview ki jagah form bharnе ko bola, phir baat hi khatam. **Ye bhi Online Fraud hai.**

2.2 Definition

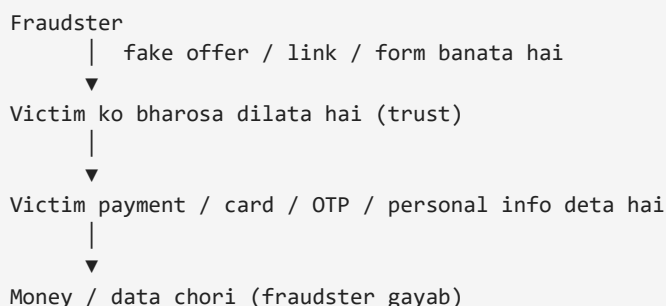
DEFINITION

Online Fraud woh dhoka (fraud) hai jo **internet ke through** kiya jata hai — jisme victim ko jhootha bharosa de kar usse **paise, goods, ya personal information** illegally haasil ki jati hai.

Simple: Internet par bana hua jhootha bharosa — paisa lootne ki koshish.

2.3 Flowchart – Online Fraud Kaise Hota Hai

FLOWCHART



2.4 Types of Online Frauds

Type	Kya hota hai
Phishing	Fake email/site se credentials churana
Lottery / Prize scam	Jhootha inaam dikha kar fee vasoolna
UPI / UPI-refund fraud	Fake UPI link ya refund ke bahane OTP lena
Card fraud	Card details + OTP chura kar kharidaari
Job scam	Job ke naam par registration fee lena
Auction / Shopping scam	Fake site par payment lena, samaan nahi dena
Romance scam	Pyara wala chat karke paise maangna
Donation scam	Fake charity ke naam par daan maangna

2.5 Red Flags (Fraud Kaise Pehchano)

- **Too good to be true** — itna sasta / itna bada inaam aam nahi milta.
- **Jaldi bharo** wali pressure (time limit).
- **OTP / CVV / PIN** maangna — asli company kabhi nahi maangti.
- Payment sirf **personal account** me bhejna.
- **Fake URL / site** — spelling check karo (amaz0n.com).
- **Unverified phone/email** — Google par search karke verify karo.

2.6 Prevention (Bachao Ke Tarike)

- OTP/PIN/CVV **kabhi kisi ko na do** — bank/company bhi nahi.
- Sirf **trusted sites** se online payment karo.
- Website ka **URL + padlock** (HTTPS) check karo.
- **Unexpected links** par click na karo.
- Shopping ke liye **verified sellers** hi chuno.
- Fraud hone par turant **bank + cyber cell (1930)** ko report karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Online Fraud = internet ke through kiya gaya dhoka, motive = paisa/info.
2. Sabse aam = phishing, UPI fraud, lottery scam, job scam.
3. Golden rule: **OTP/CVV/PIN kabhi share mat karo.**

Viva Q1. Online fraud me victim ko kya dilaya jata hai?

Ans: Jhootha bharosa (trust).

Viva Q2. Fraud hone par kya karein?

Ans: Turant bank ko inform + 1930 par report + evidence save karna.

MEMORY TRICK

Online Fraud = Jhootha wala dosti. Jaise koi dost ban ke baad loot le — waise hi scammer bharosa jeet kar paisa lootta hai. Rule: **kisi ko bhi OTP mat do.**

PART 3 – SOFTWARE PIRACY

Software Piracy me software ka **illegal copy** banakar usse bechna / use karna hota hai. Ye developer company ka **legal right (copyright)** todta hai.

3.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Aman ek computer shop chalata hai. Wo har naye laptop par **Windows + Photoshop** install karta hai — bina license kharede. Customer ko laptop mil jata hai, shop ko sale mil jaati hai, lekin Microsoft aur Adobe ko **unka paisa nahi milta**.

Ek baar Aman ne isi software ko **WhatsApp group me bhej diya** taki 20 customers khud install kar lein. Ek customer ka **crack wala file download** karte hi uske PC par virus aaya aur uske bank details chori ho gayi.

Software company ne Aman ke khilaf **legal case** kiya — jaise kisi aur ki cheez churana illegal hai, waise software ka copy banana bhi illegal hai. **Yehi Software Piracy hai**.

3.2 Definition

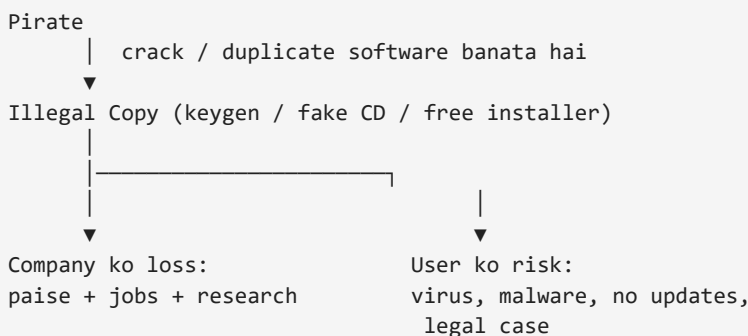
DEFINITION

Software Piracy software ka **unauthorized copying, distribution, sale ya use** hai — bina uske owner (developer company) ki proper license / permission ke.

Simple: Jo software paise de kar kharidna chahiye, usse free me copy/bech kar company ka paisa kamaana / bachana.

3.3 Flowchart

FLOWCHART



3.4 Types of Software Piracy

Type	Kya hota hai
Softlifting	1 license ko bahut saare computers par install karna
Client-server overuse	Network me allowed se zyada users server par use karna
Hard-disk loading	Shop naye PC par bina license software install kar deti hai
Counterfeiting	Nakli CD/DVD (fake look-alike) bechna
Online piracy	Crack / keygen / free download sites se distribution

3.5 Piracy Ka Nuksan (Impact)

- **Developer company** ko paise nahi milte — research aur jobs khatam.

- **User ke liye khatra** — cracked files me virus/malware chhupa hota hai.
- **No updates / support** — software risky rehta hai.
- **Legal action** — fine aur jail ho sakta hai.
- Desh ki **economy** ko bhi loss.

3.6 Prevention (Bachao Ke Tarike)

- Sirf **licensed software** kharido (official website / authorised shop).
- **Product key / activation** ke systems ka use karo.
- **DRM (Digital Rights Management)** — software ko copy hone se rokna.
- Company me **software audit** — kaun sa licensed, kaun sa nahi.
- Public awareness — piracy bhi chori hai.
- Open-source / free alternatives (Linux, LibreOffice) use karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Software Piracy = software ka bina license copy / bech / use karna.
2. Types = softlifting, hard-disk loading, counterfeiting, online piracy, client-server overuse.
3. Risk = company ka paisa loss + user ke PC par malware.

Viva Q1. Softlifting kya hai?

Ans: Ek license ko kai computers par install karna.

Viva Q2. Piracy se user ko kya risk hai?

Ans: Cracked file me virus/malware, no updates, legal case.

MEMORY TRICK

Software Piracy = Book ki photo-copy. Jaise book ka photocopy banakar bechna illegal hai, waise software ka copy banakar bechna/use karna illegal hai.

PART 4 – COMPUTER NETWORK INTRUSIONS

Network Intrusion matlab **bina permission ke kisi network ke andar ghusna**. Jaise ghar me andar ghus kar samaan dekhna, waise network me ghus kar data par nazar aur haath.

4.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek company ke office me 100 computers ek network se jude hain. Ek din ek hacker ne **network scan** kiya aur paaya ki ek **purana server** khula (open port) hai jiska password **admin123** tha.

Hacker ne us password se login kiya — ab wo **network ke andar** tha. Usne:

1. Salary records chura liye.
2. Ek system par **backdoor** install kar diya.
3. Aur do computers se bhi paise-transfer files nikali.

Company ko pata hi nahi chala. Jab tak software update hua, data nikal chuka tha. **Bina permission ke network ke andar ghus kar data churana / malware daalna — Computer Network Intrusion.**

4.2 Definition

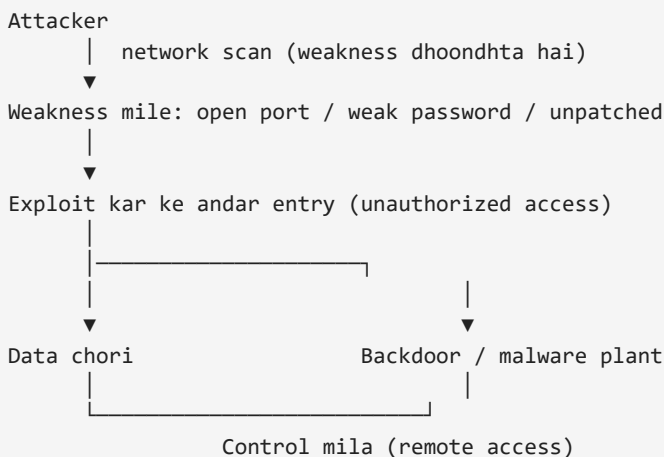
DEFINITION

Computer Network Intrusion kisi computer network ya system me **unauthorized access** hai — jisme attacker security tod kar andar ghus kar data churata hai, malware daalta hai, ya network ko apne control me le leta hai.

Simple: Network ki deewar (security) tod kar andar ghusna.

4.3 Flowchart – Intrusion Process

FLOWCHART



4.4 Types of Network Intrusions

Type	Kya hota hai
External Intrusion	Bahar se hacker network me ghusna
Internal Intrusion	Andar ka employee/insider misuse karna
Host Intrusion	Kisi ek computer/system par unauthorized access
Network Intrusion	Poore network me unauthorized entry

4.5 Intrusion Kaise Hoti Hai (Methods)

- **Weak passwords** — admin123, password, 123456.
- **Unpatched / old software** — known vulnerabilities khuli hain.
- **Open ports** — services internet par khuli hain.
- **Phishing / social engineering** — employee se login details nikalna.
- **Brute force** — password guess karna.
- **Malware** — keylogger/trojan se andar ka rasta.

4.6 Detection aur Prevention

- **IDS / IPS** (Intrusion Detection / Prevention System) lagaana.
- **Firewall** — bahar ka traffic filter karna.
- Software ko **patch / update** karte rehna.
- **Strong passwords + MFA.**
- **Network segmentation** — sensitive part alag rakhna.
- **Logs monitoring** — suspicious activity track karna.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Network Intrusion = bina permission ke network/system me unauthorized access.
2. Sabse aam tarika = weak password / open port / unpatched software.
3. Detect karte hain = IDS/IPS, firewall, log monitoring.

Viva Q1. Intrusion me attacker entry ke baad kya karta hai?

Ans: Data chori, malware/backdoor plant, remote control.

Viva Q2. Intrusion se bachne ka best tarika?

Ans: Firewall + IDS/IPS + patch updates + strong password.

MEMORY TRICK

Network Intrusion = Chor ka ghar me ghusna. Jaise chor diwaar tod kar ghus kar samaan le jata hai, waise hacker security tod kar data le jata hai.

PART 5 – PASSWORD SNIFFING

Password Sniffing me attacker **chupke se network traffic sunta hai** aur login password pakad leta hai — victim ko kuch pata nahi chalta.

5.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Rahul ek **cafe ki free WiFi** par baitha tha. Usi cafe me ek hacker bhi tha jisne apna laptop laga rakha tha jisme **packet sniffing tool (Wireshark)** chal raha tha.

Rahul ne apne account par login kiya — lekin wo site **HTTP** par chal rahi thi (secure nahi).

Hacker ke tool ne Rahul ke **username aur password** ko network ke packets me padh liya — kyunki data bina encryption ke travel kar raha tha.

Agle hi din hacker ne Rahul ke account se login kiya aur **Rs. 8,000** ka recharge/fraud kar diya.

Chupke se network traffic se password pakadna — Password Sniffing.

5.2 Definition

DEFINITION

Password Sniffing network traffic ko monitor karke **passwords aur login credentials ko pakadna** hai — bina victim ko pata chale. Ye aam taur par **unencrypted (plain text) data** par hota hai.

Simple: Victim ka password, hacker ke kaano me gir jata hai — jaise koi baat-chit sun li.

5.3 Flowchart – Sniffing Process

FLOWCHART

```

Attacker same network par (WiFi / LAN)
  |
  | sniffing tool (Wireshark, tcpdump) chalata hai
  |
  ▼
Victim login karta hai (HTTP / unencrypted)
  |
  | password plain text me travel karta hai
  |
  ▼
Attacker packets ko padh leta hai
  |
  |
  ▼
Username + Password captured
  |
  |
  ▼
Victim ke account me login / misuse
  
```

5.4 Ye Kyu Possible Hai?

IMPORTANT POINTS

Jab bhi data **encrypt nahi hota** (jaise HTTP, Telnet, FTP, old email), to wo network par **plain text** me travel karta hai. Sniffer us traffic ko aasaan se padh leta hai. **HTTPS/SSH encryption hone par data padha nahi ja sakta** — isliye encrypted connection zaroori hai.

5.5 Sniffing Tools (Exam Me Naam Aate Hain)

Wireshark, tcpdump, Cain & Abel, Ettercap — ye network packets pakad kar dikhate hain.

5.6 Prevention (Bachao Ke Tarike)

- **HTTPS / SSL** wali sites hi use karo (URL me padlock).
- **VPN** use karo — traffic encrypt ho jata hai.

- **Public WiFi** par sensitive kaam (banking) na karo.
- WiFi par **WPA2/WPA3** encryption lagao.
- **2FA** lagao — password milne par bhi extra step.
- Passwords har jagah **same mat rakho**.

EXAM NOTES (2 MARKS / VIVA)**Exam Notes (2 Marks):**

1. Password Sniffing = network traffic se passwords pakadna (bina permission).
2. Ye tab possible hai jab data **unencrypted (plain text)** ho.
3. Bachao = HTTPS, VPN, WPA2/WPA3, public WiFi avoid.

Viva Q1. Sniffing me attacker ko kya chahiye?

Ans: Victim ke same network par hona + unencrypted traffic.

Viva Q2. Sniffing se bachne ka best tarika?

Ans: Encrypted connection (HTTPS/SSH/VPN) use karna.

MEMORY TRICK

Sniffing = Kaan lagana. Jaise koi aapke paas aakar baat sun leta hai, waise sniffer network par aakar password sun leta hai. Isliye **baat (data) encrypt karo**.

PART 6 – IDENTITY THEFT

Identity Theft me **aapki pehchan (identity)** chori ho jaati hai — aapka naam, Aadhaar, PAN, bank details — aur usse aapke naam par fraud kiya jata hai.

6.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Deepak ko ek din bank ka message aaya: *“Aapke loan ki Rs. 25,000 ki EMI baki hai.”*

Deepak ne koi loan nahi liya tha! Jaankari lene par pata chala:

1. Kisi ne Deepak ke **Aadhaar aur PAN** ki copy ka istemal karke ek loan online liya tha.
2. Loan ki raqam ek **fake bank account** me gayi.
3. Deepak ka **credit score** kharab ho gaya.

Deepak ne shayad apne documents ki photo kisi fake site par daali thi. **Kisi aur ki personal identity chura kar uske naam par fraud karna — Identity Theft.**

6.2 Definition

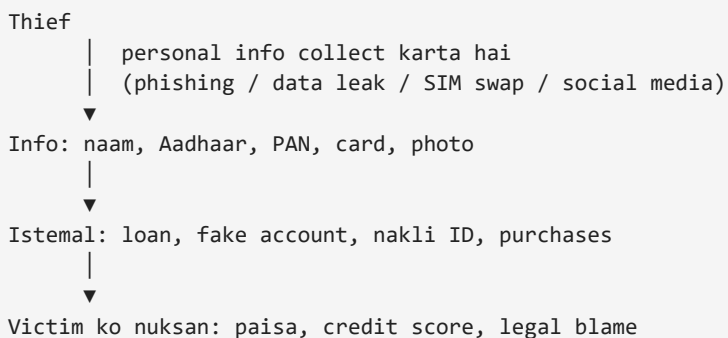
DEFINITION

Identity Theft kisi aur ki **personal identity information** (naam, Aadhaar, PAN, bank/card details, photos, phone number) ko bina permission ke churana aur usse **fraud ya doosre crimes** karne ke liye istemal karna hai.

Simple: Aapka mask pehen kar aapke naam par lootna.

6.3 Flowchart – Identity Theft Process

FLOWCHART



6.4 Identity Info Kaise Chori Hoti Hai (Methods)

- **Phishing / fake sites** — documents maang kar nikalna.
- **Data breaches** — company ka data leak.
- **SIM swap** — SIM block kar ke OTP churana.
- **Skimming** — ATM/card reader se card data.
- **Social media oversharing** — Aadhaar/photos public karna.
- **Dumpster diving** — pheke hue documents se info.

6.5 Impact (Nuksan)

Impact	Kya hota hai
Financial loss	Aapke naam par loan / purchases

Credit score kharab	Future loan mushkil
Legal problems	Crime aapke naam par ho gaya
Reputation	Lokjivan me badnaami

6.6 Prevention (Bachao Ke Tarike)

- Documents (Aadhaar/PAN) ko **public share na karo**.
- **OTP kabhi share na karo**.
- Bank statements / cards ko **regularly check** karo.
- Lost documents ki **turant report** karo.
- Social media par **privacy settings** tight karo.
- Documents ko pheko to **shred** karo.
- **Credit report** time-to-time check karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Identity Theft = kisi ki personal identity info chura kar uske naam par fraud.
2. Methods = phishing, data breach, SIM swap, skimming, oversharing.
3. Golden rule: documents + OTP kabhi public/share mat karo.

Viva Q1. Identity theft me kya chori hota hai?

Ans: Personal identity (Aadhaar, PAN, bank info, photos).

Viva Q2. Identity theft hone par victim ko kya nuksan?

Ans: Paisa, credit score, legal blame, reputation.

MEMORY TRICK

Identity Theft = Nakli mask. Jaise koi aapka mask pehen kar aapke naam par lootta hai, waise thief aapki identity pehen kar fraud karta hai.

PART 7 – CYBER TERRORISM

Cyber Terrorism me motive **paisa nahi** hota — motive hota hai **dar, tabahi aur desh ko nuksan**. Isliye ye normal cyber crime se zyada khatarnak hai.

7.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek group kisi desh ki **power company ke control system** par cyber attack karta hai.

Attackers ne:

1. Power grid ke software me virus dal diya.
2. Lakhon gharon ki bijli band kar di.
3. Hospitals, airports aur metro bhi affected hue.

Poori public dar me aa gayi. Government ko emergency declare karni padi.

Ye attack **paisa lene ke liye nahi** tha — motive tha **desh ko dhamkana aur nuksan pahunchana**.

Isi tarah ka aur example: Government ki website par terror ka message likh kar deface karna, ya army ke defense secrets churana. **In sabko Cyber Terrorism kehte hain.**

7.2 Definition

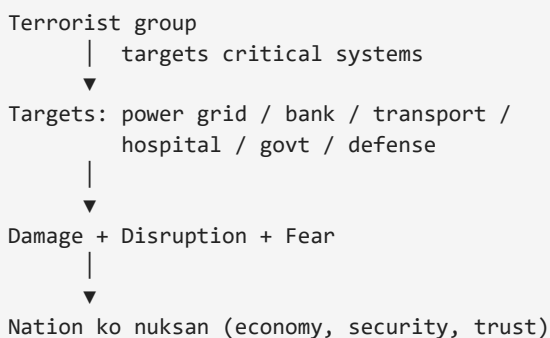
DEFINITION

Cyber Terrorism politically ya ideologically motivated attack hai jo **computer systems, networks ya digital infrastructure** par kiya jata hai — jiska maqsad **dar (fear), bade-scale damage, ya kisi nation / critical infrastructure ko nuksan** pahunchana hai.

Simple: Paisa nahi, **desh aur logon ka dar** lootna.

7.3 Flowchart

FLOWCHART



7.4 Targets of Cyber Terrorism

Target	Kyu
Power grid	Bijli band karne se poori public par asar
Banking network	Economy ka theka system
Transport (airport/rail)	Traffic aur safety
Hospitals	Records aur emergency services
Government websites	Deface + information leak

Defense systems	Secret military data
-----------------	----------------------

7.5 Cyber Crime vs Cyber Terrorism

Basis	Cyber Crime	Cyber Terrorism
Motive	Paisa / personal gain	Fear + political/ideological
Target	Person / company	Nation / critical infrastructure
Effect	Individual loss	Large-scale social/national harm
Who	Cyber criminals, gangs	Terror groups, extremist states

7.6 Prevention (Bachao Ke Tarike)

- **Critical infrastructure protection** — power/bank/defense ki apni security.
- **National cyber security strategy** aur agencies (CERT).
- System par **continuous monitoring** aur threat intelligence.
- **International cooperation** — countries mil kar attack rokna.
- **Redundancy / backups** — ek system fail ho to doosra chal sake.
- Cyber attack hone par **quick response team**.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Cyber Terrorism = dar aur political motive se critical systems par attack.
2. Target = power grid, banks, transport, defense, govt.
3. Cyber Crime = paisa, Cyber Terrorism = nation ko dar/nuksan.

Viva Q1. Cyber terrorism ka motive kya hai?

Ans: Fear + political/ideological motive, nation ko nuksan.

Viva Q2. Cyber crime aur cyber terrorism me farak?

Ans: Crime ka motive paisa, terrorism ka motive dar aur desh ko nuksan.

MEMORY TRICK

Cyber Terrorism = Digital aatank. Jaise aatanki darail ke liye bomb phodta hai, waise cyber terrorist nation ke systems par hamla kar ke dar phailata hai.

PART 8 – VIRTUAL CRIME

Virtual Crime **virtual world** me hone wale crimes hain — online games, metaverse, virtual economies. Isme koi physical cheez nahi hoti, par **real loss** hota hai.

8.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Karan ek online game khelta hai jisme log virtual coins se **virtual items** kharidte hain (sword, skins, pets) jo real paise se kharede jaate hain.

Ek hacker ne Karan ke game account ko hack kiya aur:

1. Uske **virtual coins aur items** chura liye.
2. Unhe real currency me bech diya.
3. Karan ke account se fake chats bhej kar doosre players ko bhi dhoka diya.

Karan ne real paise lagaye the — lekin koi physical cheez chori nahi hui. Fir bhi use **real loss** hua.

Virtual world me hone wala crime (chori, dhoka, harassment) — Virtual Crime.

8.2 Definition

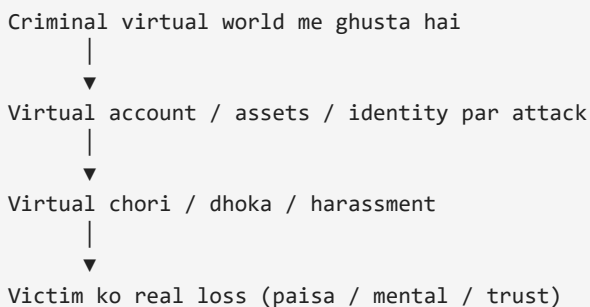
DEFINITION

Virtual Crime wo criminal acts hain jo **virtual environment** (online games, metaverse, virtual worlds, cyberspace) me kiye jaate hain — jisme physical object nahi hota, lekin victim ko **real financial ya emotional loss** hota hai.

Simple: Asli duniya ka dhoka, virtual duniya ka maidan.

8.3 Flowchart

FLOWCHART



8.4 Types of Virtual Crimes

Type	Kya hota hai
Virtual theft	Virtual coins/items/currency churana
Virtual harassment	Avatar/game me dhamkana, gali dena
Virtual fraud	Virtual cheezon ke naam par paisa dhoka
Money laundering	Real paisa virtual currency me white karna
Virtual identity theft	Virtual pehchan/account churana
Virtual property crime	Virtual land/items par kabza

8.5 Important Points

IMPORTANT POINTS

1. Virtual crime me **physical object nahi hota**, par loss real hota hai.
2. **Jurisdiction problem** — virtual world me kiska law lagega, clear nahi.
3. Virtual property ka **ownership** bhi legal issue hai.
4. Isliye game companies account security aur reporting systems rakhti hain.

8.6 Prevention (Bachao Ke Tarike)

- Game / virtual account par **strong password + 2FA**.
- Virtual items real paise se kharidte waqt **trusted platforms** hi use karo.
- Kisi ko bhi **account login share na karo**.
- Suspicious **in-game links** par click na karo.
- Virtual dhande me dhoka ho to **platform ko report** karo.

EXAM NOTES (2 MARKS / VIVA)**Exam Notes (2 Marks):**

1. Virtual Crime = virtual world (games/metaverse) me hone wala crime, real loss ke saath.
2. Types = virtual theft, virtual fraud, harassment, money laundering.
3. Challenge = jurisdiction aur virtual property ka ownership clear nahi.

Viva Q1. Virtual crime me physical object kyu nahi hota?

Ans: Kyunki wo virtual world me hota hai — lekin loss real hota hai.

Viva Q2. Virtual crime ka legal challenge kya?

Ans: Kaunsa desh/law lagega (jurisdiction) aur virtual property ka ownership.

MEMORY TRICK

Virtual Crime = Dream me chori. Jaise sapne me chori ho aur jagne par asar ho — waise virtual world me chori hoti hai aur real loss hota hai.

PART 9 – PERCEPTION OF CYBER CRIMINALS

Perception ka matlab hai — log kisi ko kaise dekhte hain. Ek hi hacker ko ek jagah **hero** mana jata hai, doosri jagah **criminal**. Is part me hum cyber criminals ke types aur unke baare me logon ki soch samjhenge.

9.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek 17 saal ke ladke **Arjun** ne apne school ki website hack kar ke marks change kar diye.

Logon ki alag-alag soch:

- Uske **doston** ko wo **genius / hero** lagta hai — “kaisa cool hack kiya!”
- **School** ke liye wo **criminal** hai — exam system ke saath dhoka.
- **Media** ne usse “super smart hacker” likh diya.
- **Police / cyber cell** usse **juvenile cyber criminal** maanti hai.

Ek hi ladka — **perception** ke hisaab se hero, criminal, ya specialist. Isi tarah **hackers, insurgents aur extremist groups** ko bhi log alag tarah se dekhte hain.

9.2 Definition

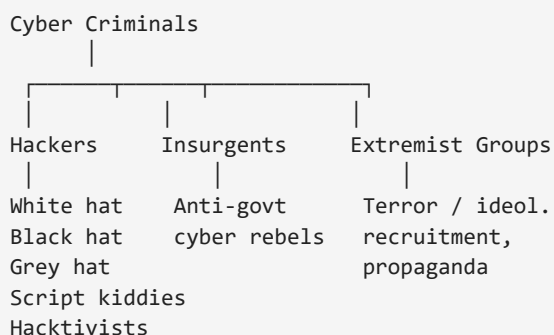
DEFINITION

Perception of Cyber Criminals — society, media, law enforcement aur public ka **nazariya** hai un logon ke baare me jo cyber crimes karte hain — unke **motives, backgrounds, skills aur threat level** ke hisaab se.

Simple: Cyber criminal ko log kaise dekhte hain — hero, criminal, ya threat — motive aur tarah ke hisaab se.

9.3 Flowchart – Cyber Criminals Ki Perception Categories

FLOWCHART



9.4 Hackers (Detail Me)

Type	Kya karta hai	Perception
White Hat	Legal + ethical hacker, security ke liye	Hero / security expert
Black Hat	Criminal motive — data/money chori	Criminal
Grey Hat	Be-ijazat ghusta hai par bina nuksan	Mixed perception
Script Kiddie	Ready tools se khelne wala naya banda	Naive / nuisance
Hactivist	Social/political protest ke liye hack	Kuch ko hero, kuch ko criminal

9.5 Insurgents (Detail Me)

Insurgents wo groups hain jo apni **government ke khilaf** ladte hain — aur cyber ka istemal karte hain:

- Government ki websites aur systems par attack.
- Apna **propaganda** phailana aur logon ko bhadkana.
- Information ko **leak** karke sarkar ko badnaam karna.

Perception: government inhe **criminal/terrorist** maanti hai; inke supporters inhe **freedom fighters** keh sakte hain.

9.6 Extremist Groups (Detail Me)

Extremist groups (terror organizations) internet ka istemal karte hain:

- **Recruitment** — naye logon ko apne me shaamil karna.
- **Propaganda & radicalization** — dushman ke khilaf nafrat phailana.
- **Funding** — donations / crypto se paise jama karna.
- **Attacks** — critical systems par hamla (cyber terrorism).

Perception: pure **terrorist threat** — national security ke liye khatra.

9.7 Baaki Cyber Criminals (Ek Nazar)

- **Organized criminals** — gangs jo financial fraud karti hain.
- **Insiders** — company ke andar ka employee, andar se data leak.
- **Disgruntled employees** — badla lene wale ex-employees.
- **State-sponsored actors** — desh ke support se spying/attacks.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Perception = log cyber criminals ko kaise dekhte hain (hero ya criminal).
2. Hackers: White hat (legal/hero), Black hat (criminal), Grey hat (beech), Script kiddie, Hactivist.
3. Insurgents = anti-government cyber rebels; Extremist groups = terror/ideological (recruitment, propaganda, attacks).

Viva Q1. White hat aur black hat hacker me farak?

Ans: White hat legal/ethical security ke liye; black hat criminal motive se attack karta hai.

Viva Q2. Extremist groups internet kaise use karte hain?

Ans: Recruitment, propaganda, funding, aur cyber attacks.

MEMORY TRICK

Perception = Raai ka nazariya. Jaise ek hi insaan kisi ko hero, kisi ko criminal lagta hai — waise hi hacker bhi motive ke hisaab se alag-alag perception paata hai. Trick: **White=legal, Black=criminal, Grey=beech me.**

PART 10 – WEB SERVER HACKING

Web Server Hacking me attacker **web server par unauthorized access** le leta hai — jis par websites chalti hain. Fir wo data, website ya server ko apne control me le leta hai.

10.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek company ki website **schoolonline.com** thi jisme students login karte the. Login page par **SQL Injection** ki vulnerability thi.

Hacker ne login box me normal password ki jagah **special SQL code** daal diya:

- Database ne attacker ka code asli command maan liya.
- **Saari students ki table** (usernames, passwords, Aadhaar, parents info) dump ho gayi.
- Hacker ne website ka **home page deface** kar diya.
- Ek **backdoor** bhi install kar diya taaki baar-baar andar aa sake.

Web server (website) par unauthorized access le kar data churana, deface karna ya backdoor daalna — Web Server Hacking.

10.2 Definition

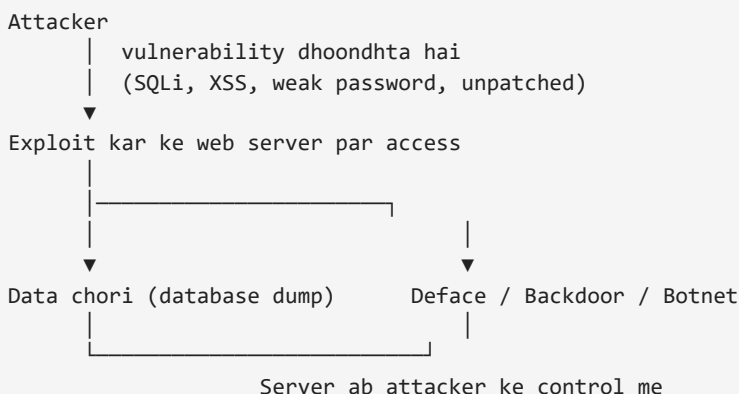
DEFINITION

Web Server Hacking kisi **web server** (jis par websites chalti hain) par **unauthorized access** lena hai — data steal karne, websites deface karne, malware/backdoor install karne, ya server ko aage ke attacks ke liye use karne ke liye.

Simple: Website ki building (server) ka darwaza tod kar andar ghusna.

10.3 Flowchart

FLOWCHART



10.4 Common Methods (Vulnerabilities)

Method	Kya hota hai
SQL Injection	Login/form me SQL code daal kar database se data nikalna
XSS (Cross-site Scripting)	Website par malicious script daal kar users se data lena
Directory Traversal	Server ki secret files tak rasta dhoondhna
Default / weak credentials	admin/admin, 123456 jaisi passwords
Unpatched software	Purane server/plugin me known bug

Misconfiguration

Galat settings se files/backups public ho jana

10.5 Impact (Nuksan)

- **Data breach** — user data market me.
- **Website defacement** — reputation kharab.
- **Backdoor / botnet** — server ab attacker ka.
- **Financial loss** — fines, recovery, business band.
- **Legal action** — data privacy law (DPDP) ke under.

10.6 Prevention (Bachao Ke Tarike)

- **Input validation** — SQL injection/XSS ko roko (parameterized queries).
- Server/software ko **patch & update** karte rehna.
- **Strong admin passwords + 2FA.**
- **WAF** (Web Application Firewall) lagaana.
- **Least privilege** — server files ko kam log hi access karein.
- Website/backups ko **monitor aur test** karte rehna.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Web Server Hacking = web server par unauthorized access le kar data/website/server ko control karna.
2. Sabse aam tarika = SQL injection, XSS, weak password, unpatched software.
3. Prevention = input validation, patching, WAF, strong password.

Viva Q1. SQL injection me kya hota hai?

Ans: Login/form me SQL code daal kar database se data nikalna.

Viva Q2. Defacement kya hai?

Ans: Website ka content bina permission ke badal dena.

MEMORY TRICK

Web Server Hacking = Building ka gate todna. Jaise darwaza tod kar andar ka maal nikalte hain, waise server ka gate (vulnerability) tod kar data nikalte hain.

PART 11 – SESSION HIJACKING

Session Hijacking me attacker **aapki active login session** chura leta hai — password janne ki zaroorat nahi, bina password ke aapke account me ghus jata hai.

11.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Sneha ek cafe ki WiFi par baith kar apne **bank account** me login karti hai.
Login hone par server usse ek **session cookie** deta hai (chhoti file jisse server pehchanta hai ki Sneha logged-in hai).
Usi WiFi par ek attacker **packet sniffer** se wo cookie pakad leta hai — kyunki connection secure nahi tha.
Ab attacker apne browser me wo cookie daal deta hai — aur **bina password ke usi logged-in session par ghus jata hai**.
Wo paise transfer kar deta hai — jab tak Sneha ko pata chalta hai, nuksan ho chuka hota hai.

Kisi user ke active session (cookie/session ID) ko chura kar uske naam par ghusna — Session Hijacking.

11.2 Definition

DEFINITION

Session Hijacking kisi logged-in user ke **active session identifier** (session ID / cookie) ko churana hai — aur usse use kar ke **bina password ke user ke account par take-over** karna hai.

Simple: Aapka ticket (session) chura kar aapki jagah train me baith jaana.

11.3 Flowchart

FLOWCHART

```

graph TD
    A[User login karta hai] --> B[server session ID / cookie deta hai]
    B --> C[Attacker session ID pakadta hai  
(sniffing / XSS / fixation / prediction)]
    C --> D[Attacker session ID ko use karta hai]
    D --> E[Server samajhta hai: yehi user hai]
    E --> F[Attacker user ban jaata hai (take-over)]
  
```

11.4 Types of Session Hijacking

Type	Kya hota hai
Session Sidejacking	Network par packet sniffing se cookie pakadna
Session Fixation	Attacker khud session ID bana kar victim se usse use karwana
Session Prediction	Session ID ka pattern nikal kar agla ID guess karna
Cookie Theft (XSS)	Website par script daal kar cookie nikalwana
Session Replay	Pakda hua request/cookie wapas bhejna

11.5 Prevention (Bachao Ke Tarike)

- **HTTPS** use karo — cookie encrypted tarike se jaaye.

- Cookie par **Secure + HttpOnly** flags — script se nahi pakdi ja sake.
- Login ke baad **session ID regenerate** karo.
- **Session timeout** — kuch der baad session khatam.
- Public WiFi par sensitive kaam na karo.
- **2FA** — session chori hone par bhi extra protection.

EXAM NOTES (2 MARKS / VIVA)**Exam Notes (2 Marks):**

1. Session Hijacking = logged-in user ka session ID/cookie chura kar bina password account take-over.
2. Types = sidejacking, fixation, prediction, cookie theft (XSS).
3. Prevention = HTTPS, Secure+HttpOnly cookie, session regenerate, timeout, 2FA.

Viva Q1. Session hijacking me password ki zaroorat kyu nahi?

Ans: Kyunki attacker server ko wahi session ID de deta hai jo pehle se login hai.

Viva Q2. Session fixation me kya hota hai?

Ans: Attacker apna session ID victim ko use karwa deta hai.

MEMORY TRICK

Session Hijacking = Ticket chori. Jaise koi aapka movie ka ticket chura kar aapki jagah andar ghus jata hai — waise hi session ID chura kar account me ghus jaate hain.

APPENDIX – QUICK REVISION SHEET (UNIT 2)

Sabse Pehle Yaad Rakho — One-Line Definitions

Topic	Ek Line Me
Web Jacking	Website ki poori visitors traffic ko redirect kar ke hamla karna
Online Fraud	Internet par dhokha — card cloning, phishing, advance fee, fake sites
Software Piracy	Software ki illegal copy — activation crack karna, illegal copy banana
Network Intrusion	Network par bina permission ka access — ke andar ghusna
Password Sniffing	Network par password chup-chaap pakadna
Identity Theft	Doosre ki pehchan chura kar uske naam par kaam karna
Cyber Terrorism	Dar + political motive se critical systems par attack
Virtual Crime	Virtual world (games/metaverse) me crime, real loss ke saath
Web Server Hacking	Web server par unauthorized access le kar data/deface/backdoor
Session Hijacking	Logged-in user ka session ID chura kar account take-over

Memory Tricks

- Web Jacking = **Uturna (redirect)** — traffic udta hai.
- Online Fraud = **Digital Dhoka**.
- Software Piracy = **Soft COPY bina license**.
- Network Intrusion = **Andar ghusna**.
- Password Sniffing = **Sungh kar sungha lena**.
- Identity Theft = **Doosre ka mask pehenna**.
- Cyber Terrorism = **Digital aatank**.
- Virtual Crime = **Dream me chori**.
- Web Server Hacking = **Building ka gate todna**.
- Session Hijacking = **Ticket chori**.

Rapid Fire Viva (15 Questions)

Q	Answer
Web jacking me kya hack hota hai?	Website ki traffic / domain
Web jacking ke 3 tarike?	Vulnerability, spam email, hijacked website
Online fraud ke 2 examples?	Card cloning, phishing
Card cloning me kya banaya jata hai?	Duplicate card (skimmed info se)
Phishing me attacker kya banata hai?	Fake page / fake link
Software piracy ke 3 types?	Soft-lifting, counterfeiting, internet piracy
Network intrusion ke 2 types?	External + insider (internal) attack
Intrusion prevention system?	IPS

Password sniffing me attacker kya use karta hai?	Sniffer / packet capture tool
Password sniffing ka desi solution?	HTTPS, 2FA, password manager
Identity theft ke 2 types?	Financial + Criminal identity theft
Identity theft ka sabse bada nuksan?	Aapke naam par illegal kaam + CIBIL kharab
Cyber terrorism ka motive?	Fear + political/ideological motive
Web server hacking ka sabse common method?	SQL injection
Session hijacking me password ki zaroorat?	Nahi — session ID hi kaafi hai

Note: Ye Unit 2 ka complete notes hai. Unit 1 aur 2 ke saare important points isi series me cover ho chuke hain.

CYBER SECURITY

UNIT 3 – COMPLETE DETAILED NOTES

Cyber Crime & IT Act 2000 • Hacking • Teenage Web Vandals • Cyber Fraud & Cheating • Defamation • Harassment & E-mail Abuse
Other IT Act Offences • Monetary Penalties • Jurisdiction & Cyber Crimes • Nature of Criminality • Strategies to Tackle Cyber Crime & Trends

Book-style notes with Practical Examples, Definitions, Flowcharts, Diagrams, Important Points, Exam Notes & Memory Tricks

UNIT 3 – SYLLABUS (As per Examination)

S.No	Unit 3 Topics	Marks
1	Concept of Cyber Crime and the IT Act, 2000	---
2	Hacking	---
3	Teenage Web Vandals	---
4	Cyber Fraud and Cheating	---
5	Defamation	---
6	Harassment and E-mail Abuse	---
7	Other IT Act Offences	---
8	Monetary Penalties	---
9	Jurisdiction and Cyber Crimes	---
10	Nature of Criminality	---
11	Strategies to Tackle Cyber Crime and Trends	---

How to read this book: Har topic me sabse pehle ek **long practical example** diya hai, phir **short definition**, phir **flowchart / diagram**, phir **important points** aur aakhir me **exam notes**. Isi order me padhna — definition yaad aasan ho jayegi.

CONTENTS

UNIT 3 – SYLLABUS (As per Examination)	2
PART 1 – CONCEPT OF CYBER CRIME AND THE IT ACT, 2000	6
1.1 Long Practical Example (Real Life)	6
1.2 Definition	6
1.3 Flowchart	6
1.4 IT Act, 2000 — Key Points	6
1.5 Cyber Crime vs Cyber Law	7
1.6 Prevention (Bachao Ke Tarika)	7
PART 2 – HACKING	8
2.1 Long Practical Example (Real Life)	8
2.2 Definition	8
2.3 Flowchart	8
2.4 Hacking ke Types	8
2.5 Hacking vs Cracking	9
2.6 Prevention (Bachao Ke Tarika)	9
PART 3 – TEENAGE WEB VANDALS	10
3.1 Long Practical Example (Real Life)	10
3.2 Definition	10
3.3 Flowchart	10
3.4 Kyun Karte Hain (Motives)	10
3.5 Important Points	10
3.6 Prevention (Bachao Ke Tarika)	11
PART 4 – CYBER FRAUD AND CHEATING	12
4.1 Long Practical Example (Real Life)	12
4.2 Definition	12
4.3 Flowchart	12
4.4 Common Frauds (Types)	12
4.5 Fraud vs Cheating (Ek Nazar)	13
4.6 Prevention (Bachao Ke Tarika)	13

PART 5 – DEFAMATION	14
5.1 Long Practical Example (Real Life)	14
5.2 Definition	14
5.3 Flowchart	14
5.4 Types of Defamation	14
5.5 Important Points	14
5.6 Prevention (Bachao Ke Tarike)	15
PART 6 – HARASSMENT AND E-MAIL ABUSE	16
6.1 Long Practical Example (Real Life)	16
6.2 Definition	16
6.3 Flowchart	16
6.4 Types of Harassment	16
6.5 Important Points	17
6.6 Prevention (Bachao Ke Tarike)	17
PART 7 – OTHER IT ACT OFFENCES	18
7.1 Long Practical Example (Real Life)	18
7.2 Definition	18
7.3 Flowchart	18
7.4 Key Other Offences (Table)	18
7.5 Important Points	19
PART 8 – MONETARY PENALTIES	20
8.1 Long Practical Example (Real Life)	20
8.2 Definition	20
8.3 Flowchart	20
8.4 Key Penalty Sections (Table)	20
8.5 Important Points	20
8.6 Prevention (Bachao Ke Tarike)	21
PART 9 – JURISDICTION AND CYBER CRIMES	22
9.1 Long Practical Example (Real Life)	22
9.2 Definition	22
9.3 Flowchart	22

9.4 Jurisdiction Rules (Table)	22
9.5 Important Points	23
9.6 Prevention (Bachao Ke Tarike)	23
PART 10 – NATURE OF CRIMINALITY	24
10.1 Long Practical Example (Real Life)	24
10.2 Definition	24
10.3 Flowchart	24
10.4 Characteristics (Table)	24
10.5 Important Points	25
10.6 Prevention (Bachao Ke Tarike)	25
PART 11 – STRATEGIES TO TACKLE CYBER CRIME AND TRENDS	26
11.1 Long Practical Example (Real Life)	26
11.2 Definition	26
11.3 Flowchart	26
11.4 Strategies (Detail)	26
11.5 Trends (Aaj Kal Ke Attacks)	27
11.6 Prevention (Bachao Ke Tarike)	27
APPENDIX – QUICK REVISION SHEET (UNIT 3)	29

PART 1 – CONCEPT OF CYBER CRIME AND THE IT ACT, 2000

Cyber crime wo crime hai jo **computer / internet ke through** hota hai. Aur India me isko rokne ke liye **IT Act, 2000** banaya gaya — jo cyber crimes ki saza aur penalty batata hai.

1.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Rahul ne online shopping site par apna bank card use kiya. Ek din uske account se **50,000 रुपये** gayab ho gaye — usne card use nahi kiya tha.

Jaane par pata chala: ek website ne uski card details (skimming se) chura li thi aur doosre state me paisa nikala gaya.

Rahul ne **cyber cell** me complaint ki. Police ne **IT Act, 2000** ke under case dara:

- Section 66 — hacking (computer system me ghus kar damage).
- Section 66C — identity theft (card details).

Court ne hacker ko **jail + fine** ki saza di. Rahul ko kuch paisa wapas mila.

Isliye IT Act chahiye — cyber crime karne wale ko law ke through pakda jaye.

1.2 Definition

DEFINITION

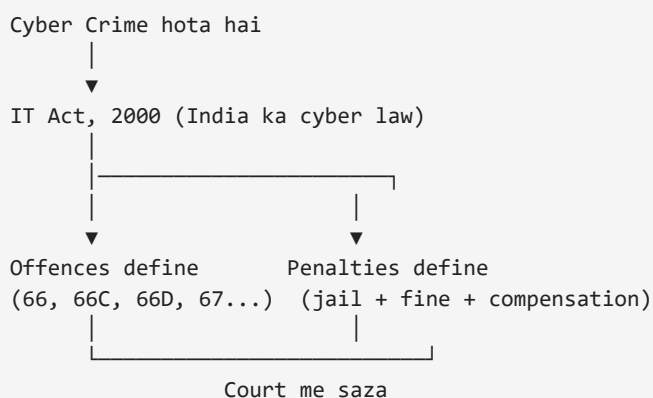
Cyber Crime — computer, internet ya digital device ke through ki gayi criminal activity (chori, dhoka, gali, data tampering, hacking, aadi).

IT Act, 2000 — India ka cyber law jo electronic transactions ko legal maanyata deta hai aur cyber crimes ke liye **offences + penalties** define karta hai.

Simple: Cyber Crime = Digital gunaaah; IT Act = Digital gunahon ki book (kya saza milegi).

1.3 Flowchart

FLOWCHART



1.4 IT Act, 2000 — Key Points

- **Kab banaya:** 2000 me. **Kyun:** internet se hone wale crimes aur electronic kaam ke liye.
- Ye **digital signature, e-commerce, e-governance** ko legal banata hai.
- **Section 66** — hacking ki saza: 3 saal jail + 5 lakh fine.
- **Section 66C** — identity theft: 3 saal jail + 1 lakh fine.
- **Section 66D** — personation se cheating: 3 saal jail + 1 lakh fine.

- **Section 67** — obscene content publish karna.
- **Section 43** — computer system ko damage karne par penalty up to 1 crore.
- **66A** (offensive messages) ko Supreme Court ne 2015 me khatam (Shreya Singhal case) kar diya.

1.5 Cyber Crime vs Cyber Law

Basis	Cyber Crime	IT Act (Cyber Law)
Kya hai	Crime jo computer se hota hai	Law jo us crime ki saza deta hai
Example	Hacking, identity theft	Section 66, 66C, 67
Kaam	Hota hai (action)	Rokta hai (rule)
Result	Nuksan	Saza / penalty

1.6 Prevention (Bachao Ke Tarike)

- Cyber crime hone par turant **cyber cell / cybercrime.gov.in** par report karo.
- Apni **complaint aur proof (screenshots)** sambhal ke rakho.
- Bank fraud ho to **bank ko turant batao** aur card block karo.
- Cyber crime ka **18/100 prati se hone wala** risk kam karo — strong passwords + 2FA.
- Kisi ke saath **cards/OTP share na karo**.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Cyber Crime = computer/internet se criminal activity.
2. IT Act 2000 = India ka cyber law — offences + penalties.
3. Section 66 = hacking (3 yr + 5 lakh), 66C = identity theft (3 yr + 1 lakh), 67 = obscene content.
4. 66A ko 2015 me SC ne khatam kar diya (Shreya Singhal case).

Viva Q1. IT Act kab aur kyu banaya?

Ans: 2000 me — electronic kaam legal aur cyber crimes punishable banane ke liye.

Viva Q2. 66A kyu hataya gaya?

Ans: Ye free speech par pabandi thi, SC ne isse unconstitutional declare kiya.

MEMORY TRICK

Cyber Crime = Digital gunaah, IT Act = Digital gunahon ki book. Trick: **6-6-6 ki mithai** — 66=hacking, 66C=identity, 66D=cheating, sab ~3 saal jail.

PART 2 – HACKING

Hacking ka matlab hai kisi ke system / account me **bina permission ghusna** — chahe paisa churana ho ya sirf data dekhna. IT Act me iski saza Section 66 ke under hai.

2.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek student **Vicky** ko pata chala ki school ke result portal ki password **admin123** hai.

Usne login kiya aur:

- Apne aur doston ke **marks change** kar diye.
- Doosre students ke **results dekhne** laga.
- Ek din school ke **system ka data delete** kar diya.

School ne complaint ki. Cyber cell ne Vicky ko pakda — **Section 66 (Hacking)** ke under case bana.

Saza: 3 saal jail + 5 lakh fine. Vicky ko apni galti ka pata chala.

Bina permission system me ghus kar data badalna/dekhna/delete karna — Hacking.

2.2 Definition

DEFINITION

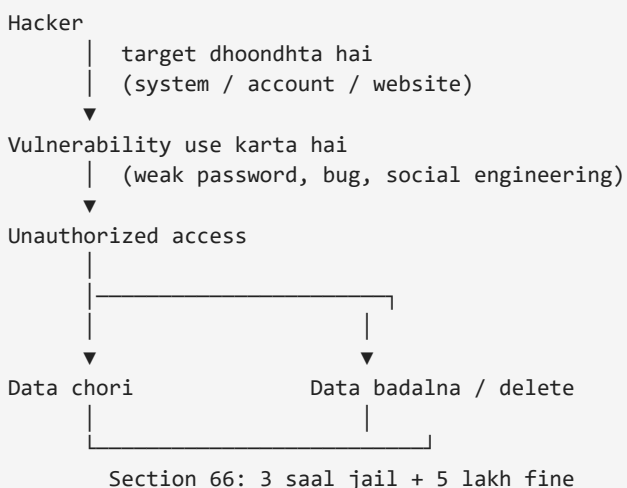
Hacking kisi ke **computer system, network ya account** me bina permission access lena hai — data churana, badalna, delete karna ya system ko nuksan pahunchana.

IT Act ke **Section 66** me hacking: system me access le kar data change/delete/damage — saza 3 saal jail + 5 lakh fine.

Simple: Apne dil se ghusna mana hai — hacking wo ghusna jo bina ijaazat hota hai.

2.3 Flowchart

FLOWCHART



2.4 Hacking ke Types

Type	Kya hota hai
Website hacking	Website deface / data nikalna
Account hacking	Email/social/bank account ka take-over

System hacking	Computer/server par ghusna
WiFi hacking	WiFi password todna, network me ghusna
Mobile hacking	Phone/WhatsApp ka access lena

2.5 Hacking vs Cracking

Basis	Hacking	Cracking
Kaun karta hai	Hacker	Cracker
Motive	Knowledge / security (kabhi kabhi bad)	Hamesha criminal motive
Law	White-hat legal ho sakta hai	Hamesha illegal

2.6 Prevention (Bachao Ke Tarike)

- **Strong password** + 2FA har account par.
- Software / apps ko **update** karte raho (bugs patched).
- Unknown **links / attachments** na kholo.
- Apni **WiFi password** strong rakho.
- Kisi ko **login / OTP share na karo**.
- System par **antivirus + firewall**.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Hacking = bina permission system me access + data change/delete/damage.
2. IT Act Section 66 = hacking ki saza (3 saal + 5 lakh).
3. White-hat = legal (security), Black-hat = criminal, Cracker = hamesha criminal.

Viva Q1. Hacking aur cracking me farak?

Ans: Hacking knowledge/security ke liye ho sakti hai; cracking hamesha criminal motive se.

Viva Q2. Section 66 ki saza?

Ans: 3 saal jail + 5 lakh fine.

MEMORY TRICK

Hacking = Bina ijaazat ghusna. Trick: **66 = 6-6** — 3 saal jail + 5 lakh fine. Yaad rakho: system ka darwaza toda, Section 66 laga.

PART 3 – TEENAGE WEB VANDALS

Teenage Web Vandals — 13–19 saal ke bachche jo internet par **website deface**, **hacks** ya **nuksan** karte hain — aksar mazak, show-off ya badle ke liye.

3.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek 16 saal ka student **Aman** ne school ki website ki login password (**admin**) dhoondh li.

Usne home page par apna naam aur mazak ka message likh diya — “**Aman was here!**”

Sabhi students ne dekha aur use cool samjha. School ko sharam aayi — website 2 din band rahi.

Cyber cell ne Aman ko pakda:

- Section 66 (hacking) + Section 66D (cheating).
- Bhaale wo minor tha — **Juvenile Court** me case gaya.

Teenage bachche websites par aise hacks/mazak karte hain — Teenage Web Vandals.

3.2 Definition

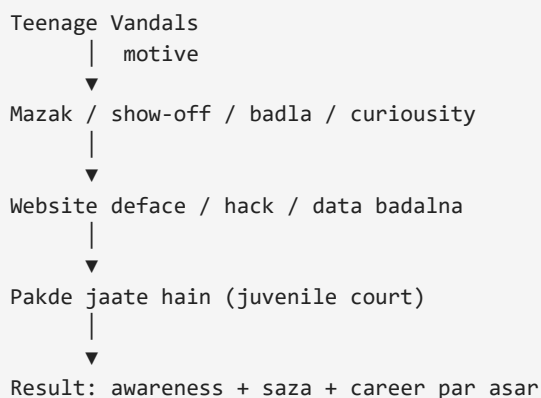
DEFINITION

Teenage Web Vandals wo **young (teenage) individuals** hain jo internet par **websites deface karte hain, data badalte hain, ya vandals (nuksan karna)** jaise hacks karte hain — aksar attention, mazak, show-off ya badle ke liye, na ki professional cyber crime ke liye.

Simple: Bade criminals paisa maangte hain; teenage vandals **attention maangte hain**.

3.3 Flowchart

FLOWCHART



3.4 Kyun Karte Hain (Motives)

Motive	Matlab
Show-off	Doston ko dikhana ki 'main hacker hoon'
Curiosity	Yeh kaise kaam karta hai, test karna
Badla / revenge	School ya teacher se gussa
Attention	Famous hona chahte hain
Peer pressure	Doston ke kehne par

3.5 Important Points

IMPORTANT POINTS

1. Ye **minor hote hain** (18 se kam) — isliye juvenile justice ke under deal hota hai.
2. Aksar ye **sochte nahi** ki wo crime kar rahe hain.
3. **Harm less lagta hai** par legal consequences bade hote hain.
4. Inhe **school + parents ke through** guide karna best hai — cyber safety education.
5. Kuch aage ja ke **ethical hackers / security experts** bhi bante hain.

3.6 Prevention (Bachao Ke Tarike)

- Schools me **cyber safety / cyber ethics** ki padhai.
- Parents bachcho ke **internet activity** par nazar rakhein.
- Bachcho ko batana ki **hack ka matlab crime** hai — mazak nahi.
- Hacking ki jagah **ethical hacking / coding** ka rasta dikhana.
- Website par **strong admin password + logging** rakho.
- Deface hote hi **report karo** — deri se case weak hota hai.

EXAM NOTES (2 MARKS / VIVA)**Exam Notes (2 Marks):**

1. Teenage Web Vandals = teens jo websites deface/hack karte hain mazak ya show-off ke liye.
2. Motive = attention, curiosity, revenge, peer pressure.
3. Ye minors hain — juvenile court me case, lekin crime legal hai.
4. Best bachao = cyber ethics education + parents ka dhyan.

Viva Q1. Vandals professional criminals se kaise alag?

Ans: Unka motive paisa nahi — attention/mazak/show-off hota hai.

Viva Q2. Inse kaise niplein?

Ans: Awareness + cyber ethics + ethical hacking ka rasta.

MEMORY TRICK

Teenage Web Vandals = Internet ke naughty bachche. Trick: Vandals = **wall par naam likhne wale** — bas wall (website) internet ki hai aur saza Section 66 ki.

PART 4 – CYBER FRAUD AND CHEATING

Cyber Fraud / Cheating — internet par dhokha de kar paisa ya cheez haasil karna. Ye India ka sabse zyada hone wala cyber crime hai.

4.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Rohit ko WhatsApp par ek **job offer** aaya — “part-time, aaram se 5000/day kamaiye.”
 Usne form bhara. Phir **acche kaam ke liye advance paisa** maanga gaya — “registration fee.”
 Rohit ne 20,000 bhej diye. Phir aur paisa maanga — “tax clear karo.”
 Jab aur paisa maangne lage to Rohit ko shak hua. Ab number band — paisa gaya.

Ye **Online Job Fraud** hai. IT Act Section 66D (cheating by personation) + IPC me bhi cheating ka case.

Internet par jhootha bharosa de kar paisa lena — Cyber Fraud and Cheating.

4.2 Definition

DEFINITION

Cyber Fraud wo **dhokha** hai jo internet/computer ke through kiya jata hai — jisme victim se **jhootha bharosa karwa ke paisa ya property** haasil ki jati hai.

IT Act **Section 66D** — personation (fake pehchan) se cheating: 3 saal jail + 1 lakh fine.

Simple: Jhooth ka khel — screen ke piche ka fraudster.

4.3 Flowchart

FLOWCHART

```

  Fraudster fake story banata hai
  | (job / prize / bank help / lottery)
  ▼
  Victim ko message / call / website
  |
  ▼
  Victim ko bharosa ho jata hai
  |
  ▼
  Victim paisa / OTP / details deta hai
  |
  ▼
  Fraudster ghaayab (Section 66D + IPC)
  
```

4.4 Common Frauds (Types)

Fraud	Kya hota hai
Job fraud	Fake job offer ke naam par fee lena
Prize/lottery fraud	‘Aapko crore mila’ — pehle fee maangna
Bank / KYC fraud	Bank wala ban kar OTP/details nikalwana
Online shopping fraud	Product nahi bhejna, paisa le lena
Matrimonial fraud	Fake profile se relationship + paisa
Investment / crypto fraud	Fake scheme me paisa daalwana

4.5 Fraud vs Cheating (Ek Nazar)

Basis	Fraud	Cheating
Matlab	Dhokha de kar gain lena	Jhooth se bharosa banakar nuksan karna
IT Act	66D (personation)	IPC Section 415/420 + 66D
Example	Fake bank call	Fake job / fake shopping site

4.6 Prevention (Bachao Ke Tarike)

- **Kisi ko bhi OTP / PIN / password na do** — bank kabhi nahi maangta.
- Job/prize ke liye **advance paisa kabhi na bhejo**.
- Link/site verify karo — **official URL** dekho.
- Bina jaanch **investment/crypto scheme** me paisa na daalo.
- Dhoke me aa jao to **turant report** + bank card block.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Cyber Fraud = internet par dhoka de kar paisa/property haasil karna.
2. Types = job, prize, KYC, shopping, investment fraud.
3. IT Act: 66D (personation cheating) 3 saal + 1 lakh; IPC 415/420 bhi lagta hai.
4. Sabse important rule = **OTP share karo, fraud hote hi report karo**.

Viva Q1. Section 66D kya punish karta hai?

Ans: Fake pehchan (personation) se online cheating — 3 saal jail + 1 lakh fine.

Viva Q2. Sabse common online fraud?

Ans: Job fraud aur bank/KYC fraud — OTP nikalwana.

MEMORY TRICK

Cyber Fraud = Jhooth ki diwali. Trick: Koi bhi **pehle paisa maange** — **wo fraud hai**. OTP maange to seedha suspect karo.

PART 5 – DEFAMATION

Defamation — kisi ki **izzat par hamla** — aisa jhooth/be-izzati ki baat jo online (social media, email) me phailayi jaye. Online defamation **Cyber Defamation** kehlati hai.

5.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Anita aur Priya ki dosti toot gayi. Gussa me Priya ne Anita ki **fake photo + jhoothi baatein** Instagram par post kar di — “Anita ne ye kiya!”

- Post viral ho gayi — doosre log bhi share karne lage.
- Anita ke rishtedaar, office — sabne dekha.
- Anita ka **reputation kharab** ho gaya, kaam par bhi issue hua.

Anita ne **cyber defamation** ki case ki:

- IT Act Section 66A tha (ab nahi) + IPC Section 499/500 (defamation).

Court ne Priya ko saza + Anita ko compensation dilwaya.

Online jhooth bol kar kisi ki izzat girana — Cyber Defamation.

5.2 Definition

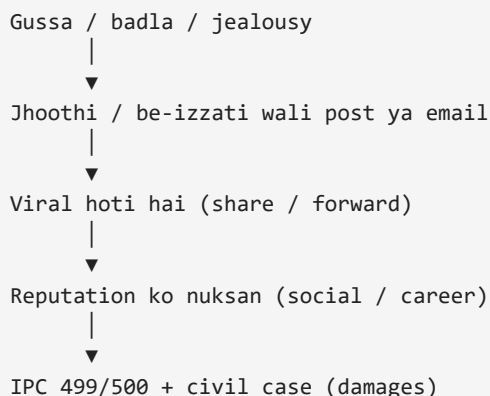
DEFINITION

Defamation kisi insaan ke **reputation (izzat) ko nuksan** pahunchana hai — aisi baat/photos/videos jo false hon aur public me phail jayein. Jab ye **internet (social media, email, blogs)** par hota hai — usse **Cyber Defamation** kehte hain.

Simple: Screen par izzat ka attack — jab wo baat viral ho jaye.

5.3 Flowchart

FLOWCHART



5.4 Types of Defamation

Type	Kya hota hai
Libel	Written / visible form (post, photo, video)
Slander	Spoken form (rumor, voice note)
Cyber defamation	Internet par libel/slander (social media, email)

5.5 Important Points

IMPORTANT POINTS

1. Online defamation ki case me **proof** aasaan hota hai — post/email screenshots saved rehti hain.
2. Ek baar viral, **hamesha ke liye** — isliye social media par pehle soch, phir post.
3. **IP / account** se offender pakda jata hai.
4. Bharat me defamation **criminal (IPC 499/500) + civil (damages)** dono me hai.
5. True but private info bhi defamation nahi — **sirf false + harmful** baat defamation hai.

5.6 Prevention (Bachao Ke Tarike)

- Post karne se pehle socho — **sab log padhenge**.
- Gussa me koi **post/rumor na phailao**.
- Apni **privacy settings** strong rakho.
- Apni hi defamation ho to **screenshots save** karke report karo.
- Bina saboot **kisi par ilzaam na lagao**.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Defamation = kisi ki izzat ko false + harmful baat se nuksan.
2. Online = cyber defamation; Libel = written, Slander = spoken.
3. Laws: IPC 499/500 (defamation) + civil damages.
4. Proof internet par aasaan — screenshots/account se offender pakda jata hai.

Viva Q1. Defamation ke 2 types?

Ans: Libel (written/visual) aur Slander (spoken).

Viva Q2. Online defamation ka proof kya hota hai?

Ans: Post/email ke screenshots, IP, account details.

MEMORY TRICK

Defamation = Izzat ki LOL. Trick: Defamation = **Defame** = naam kharab. Post viral = izzat viral (galat tarike se).

PART 6 – HARASSMENT AND E-MAIL ABUSE

Harassment and E-mail Abuse — online par pareshan karna, gali dena, dhamkana ya email/account ka galat use karna. Ye cyber bullies, ex-partners aur stalkers karte hain.

6.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Pooja apne ex-boyfriend Arjun se alag ho gayi. Arjun gusse me:

- Roz Pooja ko **gali wale emails** bhejta.
- Uske **fake account** bana kar galat messages bhejta.
- Uske **photo/phone number** group me share kar deta.
- Har jagah usse **follow + dhoki** — social media par.

Pooja bahut dar gayi. Usne cyber cell me complaint ki.

Case bana:

- IT Act Section 67 (obscene content) + IPC 354D (stalking) + IPC 499 (defamation).

Online par kisi ko pareshan karna, dhamkana ya account ka galat use — Harassment & E-mail Abuse.

6.2 Definition

DEFINITION

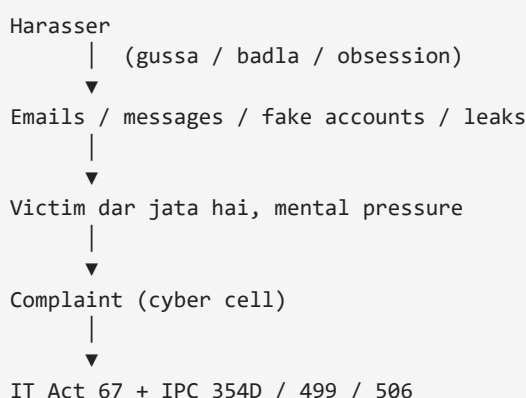
Cyber Harassment kisi ko **internet ke through pareshan karna** hai — threats, gali, stalking, fake accounts, private info leak, ya embarrassing content post karna.

E-mail Abuse — email ka galat use: threatening emails, fake emails, spamming, ya kisi aur ke naam par email bhejna.

Simple: Bullying jo ab school se internet par shift ho gayi hai.

6.3 Flowchart

FLOWCHART



6.4 Types of Harassment

Type	Kya hota hai
Cyberstalking	Baar-baar follow, messages, tracking
Cyberbullying	Juniors/classmates ko online gali/pareshan
Trolling	Jaan-boojh kar provoke karna

Doxxing	Private info (address/phone) public karna
Sextortion	Private photos ke through dhamkana
Fake accounts	Kisi aur ki pehchan se galat kaam

6.5 Important Points

IMPORTANT POINTS

1. Harassment me **ek baar ka incident nahi** — repeat behaviour matter karta hai.
2. **Screenshots rakhna** zaroori hai — warna proof nahi.
3. Harasser ko **pehle block** karo, reply na do.
4. Bhartiya law: **IT Act 67** (obscene) + **IPC 354D** stalking + **IPC 506** threat + **IPC 509** word/gesture insult.
5. Shame nahi — **complaint karna** hi solution hai.

6.6 Prevention (Bachao Ke Tarike)

- Harasser ko turant **block + report** karo.
- **Private info** (phone/address/photos) online mat daalo.
- Accounts par **privacy settings** lock rakho.
- Social media par **2FA** — fake account se bachav.
- Koi dhamkaye to **cybercrime.gov.in** par complaint karo.
- Family aur doston ko **batao** — alone fight mat karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Harassment = online pareshan karna (stalking, bullying, doxxing, sextortion).
2. E-mail abuse = threatening/fake email, spam, galat use.
3. Laws: IT Act 67 (obscene), IPC 354D (stalking), 506 (threat), 509 (insult).
4. Proof = screenshots + repeat behaviour.

Viva Q1. Cyberstalking kya hai?

Ans: Online baar-baar follow, messages, tracking karke pareshan karna.

Viva Q2. Harassment me sabse pehle kya karein?

Ans: Block karo, reply na do, screenshots save karke complaint karo.

MEMORY TRICK

Harassment = Online darr. Trick: **Harassment = Harrass = Haraas** — bilkul. Proof: Screenshots hi asli haathiyar.

PART 7 – OTHER IT ACT OFFENCES

Hacking, fraud, defamation ke alawa IT Act me **aur bhi offences** hain — obscene content, identity theft, data tampering, privacy violation — jo regular cyber cases me ate hain.

7.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ramesh ne ek raat me:

- Apne ex-boss ki email **hack** kar li (Section 66).
- Uske naam se **fake profile** bana kar galat baatein bheji (Section 66D).
- Boss ki **private photos** online daal di (Section 66E - privacy).
- Gali wale content bhi post kar diye (Section 67 - obscene).

Cyber cell ne case bana — **4 alag-alag IT Act offences** ek hi case me.

Ye sab '**Other IT Act Offences**' kehte hain — jo har roj cyber cases me aate hain.

7.2 Definition

DEFINITION

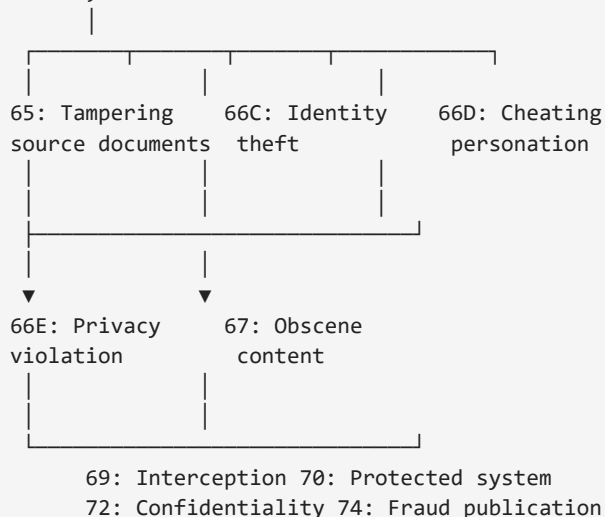
Other IT Act Offences — IT Act, 2000 ke wo **crime sections** jo hacking/fraud ke alawa hain: tampering, identity theft, personation, privacy violation, obscene content, cyber terrorism, breach of confidentiality, aadi.

Simple: IT Act ki doosri police station (sections) — ek section se zyada crimes.

7.3 Flowchart

FLOWCHART

IT Act, 2000 - Offences



7.4 Key Other Offences (Table)

Section	Offence	Saza
65	Source code tampering	3 yr + 2 lakh
66C	Identity theft	3 yr + 1 lakh
66D	Personation se cheating	3 yr + 1 lakh
66E	Privacy violation (photos/videos)	3 yr + 2 lakh

67	Obscene content publish	3 yr + 5 lakh (1st), 5 yr + 10 lakh (2nd)
67B	Child porn	5 yr + 10 lakh
69	Unauthorized interception	7 yr + fine
70	Protected system access	10 yr + fine
72	Confidentiality breach	2 yr + 1 lakh
74	False digital cert / fraud publication	2 yr + 1 lakh

7.5 Important Points

IMPORTANT POINTS

1. **65 vs 66:** 65 = tampering (source code), 66 = hacking (system access).
2. **66C vs 66D:** 66C = pehchan churana (identity theft), 66D = pehchan se kaam karna (personation).
3. **67:** obscene — ek baar 3 saal, doosri baar 5 saal.
4. **70:** government protected system — sabse heavy (10 saal).
5. 66A (offensive messages) 2015 me **SC ne khatam** kar diya — free speech ke liye.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. 65 = tampering, 66C = identity theft, 66D = personation, 66E = privacy.
2. 67 = obscene (3 saal + 5 lakh), 67B = child porn (5 saal + 10 lakh).
3. 70 = protected system (10 saal), 72 = confidentiality breach (2 saal).
4. 66A Supreme Court ne 2015 me khatam kiya.

Viva Q1. 66C aur 66D me farak?

Ans: 66C = pehchan churana, 66D = us pehchan se cheating/kaam karna.

Viva Q2. Sabse zyada saza kis offence ki?

Ans: Section 70 (protected system) — 10 saal jail.

MEMORY TRICK

Trick: 6-6 ki line — 65 tampering, 66 hacking, 66C identity, 66D personation, 66E privacy, 67 obscene. Jaise **6 ke bhag** = andar ke kaandh.

PART 8 – MONETARY PENALTIES

Monetary Penalties — IT Act ke **penalty sections** (43, 43A) jisme jail nahi, **sirf paisa (fine/compensation)** lagta hai — zyada tar civil cases me.

8.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek hospital ne apne patients ki medical records ek app me rakhe the — par **proper security** nahi thi.

- Hacker ne hospital ki site hack kar ke saare **patient records** chura liye.
- Patients ne hospital par case kiya.

Court ne:

- **Hacker** par Section 66 (criminal — jail) ka case bana.
- **Hospital** par Section 43A + 43 lagaya — kyunki wo **data protect karne me fail** hua.

Penalty: hospital ko **crore rupaye ka compensation** dena pada (Section 43A — up to 5 crore).

Jail nahi, lekin bhari bharakam fine — Monetary Penalty.

8.2 Definition

DEFINITION

Monetary Penalty — IT Act ke under laga **paise ki saza** (fine / compensation) jo company ya person ko deni padti hai jab wo cyber law todta hai — bina jail ke, zyada tar civil offence me.

Simple: Jail ka khel nahi — paise ka khel.

8.3 Flowchart

FLOWCHART

```

graph TD
    A["Cyber law ka violation (civil)  
(data damage, data breach, access w/o permission)"] --> B["Section 43 / 43A laga hota hai"]
    B --> C["Court: penalty + compensation"]
    C --> D["Victim ko paisa milta hai"]
  
```

8.4 Key Penalty Sections (Table)

Section	Kya hai	Penalty
43	Computer/network/data damage ya unauthorized access (civil)	Up to 1 crore per incident
43A	Data privacy protect na karna (sensitive data breach)	Up to 5 crore
44	Certifying Authority rules na maanna	Up to 2 lakh per rule
45	Residuary penalty (baaki civil violations)	Up to 25,000

8.5 Important Points

IMPORTANT POINTS

1. **Section 43 vs 66:** 43 = civil penalty (paisa), 66 = criminal (jail). Ek hi kaam — dono lag sakte hain.
2. **43A** — company sensitive data (health, biometric, financial) leak hone par **up to 5 crore**.
3. Penalty victim ko **compensation** ke roop me jata hai.
4. Ab DPDP Act 2023 (naya data law) bhi companies par heavy fines deta hai.
5. Penalty = **civil**; Offence = **criminal** — ye dono alag concepts hain.

8.6 Prevention (Bachao Ke Tarike)

- Companies **data security** seriously lein — encryption, access control.
- **Audit** karte raho — data kahaan gayi, kaun access karta hai.
- Data breach ho to **report + fix** turant — deri se penalty badhti hai.
- User data **minimum** rakho — jitna chahiye utna.
- Privacy policy **clear** rakho aur users ko inform karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Monetary penalty = paisa ki saza (civil), jail nahi.
2. Section 43 = up to 1 crore (damage/unauthorized access).
3. Section 43A = sensitive data breach par up to 5 crore.
4. 43 = civil (paisa), 66 = criminal (jail) — dono ho sakte hain.

Viva Q1. 43 aur 66 me farak?

Ans: 43 = civil penalty (paisa), 66 = criminal offence (jail).

Viva Q2. 43A kab lagta hai?

Ans: Jab company sensitive personal data ko protect nahi karti aur breach ho jaye — up to 5 crore.

MEMORY TRICK

Monetary Penalty = Paisa ki saja. Trick: **43 = 1 crore, 43A = 5 crore** — A wala Ameer (5) zyada. Jail me bass 66, 67, 70.

PART 9 – JURISDICTION AND CYBER CRIMES

Jurisdiction — kaunsa court / kaunsa desh case sune ga. Cyber crime **ek desh me baith ke doosre desh** ho sakta hai — to case kis ke paas jayega? Ye cyber law ka sabse bada puzzle hai.

9.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek hacker **Mumbai** me baitha tha. Usne **USA ki bank website** hack kar li aur **London ke customer** ka paisa chura liya. Ab question: case kis desh me chalega?

- Crime ki website = USA.
- Victim (paise wala) = London.
- Hacker = India.

Solution (IT Act ke under):

- **Section 75** — agar crime me **India ki computer system/network** use hui, to Indian court case sunega.
- Agar hacker Indian hai to **IPC Section 4** bhi lagta hai — desh ke bahar bhi Indian par case.

Cyber crime me kaunsa desh/court case sune — Jurisdiction ka sawal.

9.2 Definition

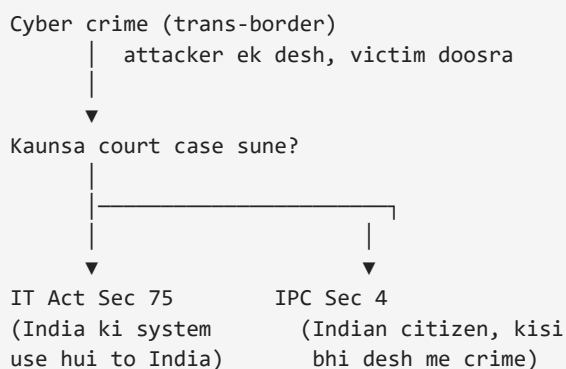
DEFINITION

Jurisdiction — law aur court ka **adhikar kshetr** — kaunsa court kis case ko sunne layak hai. Cyber crimes me ye **trans-border** hote hain, isliye jurisdiction decide karna mushkil hai.

Simple: Chor pakda gaya, par **kis daroghi ke thaane me** case hoga — yehi jurisdiction hai.

9.3 Flowchart

FLOWCHART



9.4 Jurisdiction Rules (Table)

Rule	Kya kehta hai
Territorial	Jis jagah crime hua, wahi court sunega
IT Act Sec 75	India ki computer system/network use hui → Indian court
IPC Sec 4	Indian citizen ne desh ke bahar crime kiya → India me case
Sec 1(2)	IT Act kisi bhi desh ke person par lagta hai agar India ki system involve hai
Cyber Convention	International cooperation se offender pakadna

9.5 Important Points

IMPORTANT POINTS

1. Cyber crime ki **asli jagah (location)** clear nahi hoti — isliye jurisdiction confusing.
2. **IT Act Sec 75** — India ki computer involve hui to Indian court case sun sakta hai (extra-territorial).
3. **IPC Sec 4** — Indian citizen bahar se crime kare to India me case.
4. Har desh apna law lagana chahta hai — **conflict of laws**.
5. Solution — **international cooperation, treaties** (Budapest Convention), mutual legal assistance.

9.6 Prevention (Bachao Ke Tarike)

- Case karne ke liye **proof + complaint** apne hi court me karo (Sec 75 ki madad se).
- Cross-border crime me **international cyber cell / CERT** ko batao.
- Websites ko **India me jurisdiction clause** rakhna chahiye.
- Companies apne users ke liye **clear jurisdiction** likhein.
- Cooperation ke liye **Budapest Convention** jaise agreements follow karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Jurisdiction = kaunsa court/country case sunega.
2. IT Act Sec 75 = India ki system involve → Indian court (extra-territorial).
3. IPC Sec 4 = Indian citizen bahar crime → India me case.
4. Cyber crime trans-border hai — isliye international cooperation zaroori.

Viva Q1. Jurisdiction problem kyu hoti hai?

Ans: Kyunki attacker aur victim alag desh me ho sakte hain — location clear nahi.

Viva Q2. IT Act Sec 75 kya kehta hai?

Ans: Agar India ki computer system/network involved hai to Indian court case sun sakta hai.

MEMORY TRICK

Jurisdiction = Thaane ka sawal. Trick: **75 = System India ki, case India me.** Jahan computer, wahan court.

PART 10 – NATURE OF CRIMINALITY

Nature of Criminality — cyber criminal ki pehchan, unke kaam karne ka tareeka aur kyu wo crime karte hain. Cyber crime aam crime se **bahut alag** hai — isko samjna zaroori hai.

10.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek 19 saal ka **Shaan** apne kamre me baitha tha. Uske paas sirf laptop + WiFi thi. Usne:

- Doosre desh ki company hack ki — **bina jaye, bina dastkhut ke.**
- Fake naam se account bana ke **koi pehchan nahi di.**
- Raat 2 baje attack kiya — **koi dekh nahi raha tha.**

Ab aam crime me: chor ko ghar me ghusna hota hai, pakda jata hai. Cyber crime me:

- **Koi physical presence nahi.**
- **Anonymity (chhupna aasaan).**
- **Risk kam, faida zyada.**
- **Ek kaam se crore logon ko nuksan.**

Isi wajah se cyber criminal ki nature badi alag hoti hai.

10.2 Definition

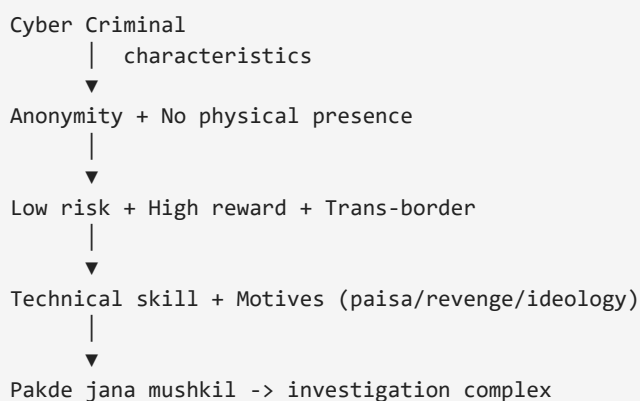
DEFINITION

Nature of Criminality — cyber criminals ki **characteristics aur crime karne ka tareeka**: anonymity, low risk-high reward, no physical presence, trans-border, technical skill, aur motive ke patterns.

Simple: Chor ke andar kya hai jo wo cyber world me itna aasan chalta hai.

10.3 Flowchart

FLOWCHART



10.4 Characteristics (Table)

Characteristic	Matlab
Anonymity	Chhupa rahta hai — fake identity se
No physical presence	Bina jaye bhi crime ho sakta hai
Low risk	Pakde jane ka dar kam
High reward	Ek kaam se bahut paisa

Trans-border	Ek desh se doosre desh ko nuksan
Technical skill	Tools + coding use karta hai
Young age (common)	Kai criminals 15-30 saal ke hote hain
Motive patterns	Paisa, revenge, ideology, thrill

10.5 Important Points

IMPORTANT POINTS

1. Cyber criminal **ghar me baith kar** bade-bade crime karta hai — isliye usse pakadna mushkil.
2. **Digital evidence** hi asli pakka saboot hai (IP, logs, devices).
3. Kaafi criminals **young aur skilled** hote hain — maang me skill, nahi to thodi badi.
4. Motives: **paisa (sabse zyada), revenge, show-off, ideology**.
5. Isliye police ko **digital forensics** aana zaroori hai.

10.6 Prevention (Bachao Ke Tarike)

- Criminals ke liye **maange kam** karne ke liye strong systems.
- **Digital forensics** teams ko train karo.
- **Anonymity** todne ke liye IP/logs tracking.
- Bachcho ko **ethical path** par le jao (cyber ethics + career).
- International cooperation se **trans-border** criminals pakdo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Nature of criminality = cyber criminal ki characteristics (anonymity, no presence, low risk-high reward).
2. Cyber crime trans-border hai, evidence digital hai.
3. Common motive = paisa, revenge, ideology, thrill.
4. Investigation ke liye digital forensics zaroori.

Viva Q1. Cyber crime aam crime se alag kya hai?

Ans: Anonymity + no physical presence + low risk high reward + trans-border.

Viva Q2. Sabse common motive?

Ans: Paisa (financial gain).

MEMORY TRICK

Nature of Criminality = Chor ka profile. Trick: **ALPHA** — Anonymity, Low risk, Presence nahi, High reward, Aaram (technical skill).

PART 11 – STRATEGIES TO TACKLE CYBER CRIME AND TRENDS

Strategies — cyber crime rokne ke tarike (law, tech, awareness). **Trends** — aaj kal cyber attacks kaise badal rahe hain (ransomware, AI, deepfakes). Dono exam me aane wale points hain.

11.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek city ke **hospital network** me ransomware aaya — saare computers lock, demand: bitcoin.

Hospital ne ghabra kar **paise de diye honge**, par ye hospital ne strategy use ki:

- **Backup** the — systems wapas banaye gaye.
- **CERT-In** ko turant report ki.
- Attack ka **source** trace hua (international).
- **Staff training** se pata chala — email link se virus aaya.

Ye **strategy** hai — rokna (prevent) + pakadna (detect) + theek karna (recover).

Aur aajkal **trends** — ransomware, AI attacks, deepfakes, IoT hacks — sabse zyada chal rahe hain.

11.2 Definition

DEFINITION

Strategies to tackle cyber crime — plan to **roko, pakdo, sulao** cyber crime: strong laws, technical security, awareness, enforcement, international cooperation.

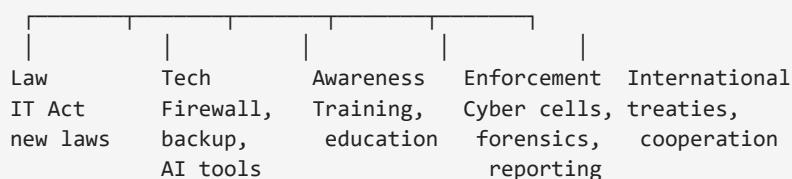
Cyber crime trends — naye-naye attack patterns jo ab chal rahe hain (ransomware, AI attacks, deepfakes, IoT, crypto frauds).

Simple: Strategy = cyber crime ka ilaaj; Trends = nayi bimariyan.

11.3 Flowchart

FLOWCHART

Strategies



11.4 Strategies (Detail)

Strategy	Kya hota hai
Legal	IT Act + naye laws (DPDP 2023), strong penalties
Technical	Firewall, encryption, backups, patching, AI threat detection
Awareness	Schools/companies me cyber safety training
Enforcement	Cyber police cells, digital forensics labs
International	Treaties, CERT-In cooperation, extradition

Reporting	cybercrime.gov.in, helpline 1930
-----------	----------------------------------

11.5 Trends (Aaj Kal Ke Attacks)

Trend	Kya hota hai
Ransomware	Data lock + paisa demand
Phishing 2.0	Spear phishing, AI se likhi fake emails
Deepfakes	AI se fake video/voice — fraud/defamation
AI-powered attacks	AI se fast + smart attacks
IoT hacks	Smart devices (camera, TV) hack hone
Crypto frauds	Fake crypto schemes, wallets
Cloud attacks	Cloud misconfig se data leak
Dark web	Stolen data market me

11.6 Prevention (Bachao Ke Tarike)

- Ransomware se bachav — **backups** + email links se savdhaan.
- Deepfakes se bachav — **verify karo** pehle, share baad me.
- IoT devices par **default password badlo**.
- Companies AI tools se **threat detection** karein.
- Har koi **report** kare — awareness hi asli bachav.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Strategies = Law + Tech + Awareness + Enforcement + International cooperation.
2. Reporting: cybercrime.gov.in, helpline 1930, CERT-In.
3. Trends = ransomware, AI attacks, deepfakes, IoT, crypto frauds, cloud hacks.
4. Ransomware se bachav = backup + training.

Viva Q1. Cyber crime se nipleine ke 3 tarike?

Ans: Law, technical security, awareness + enforcement.

Viva Q2. Ransomware me kya hota hai?

Ans: Data lock hota hai aur paisa (bitcoin) demand hota hai.

MEMORY TRICK

Strategy = Ilaj, Trend = Bimari. Trick: **L-T-A-E-I** — Law, Tech, Awareness, Enforcement, International. Trends ko yaad rakho: **R-A-D-I-C** — Ransomware, AI, Deepfakes, IoT, Crypto.

APPENDIX – QUICK REVISION SHEET (UNIT 3)

Sabse Pehle Yaad Rakho — One-Line Definitions

Topic	Ek Line Me
IT Act, 2000	India ka cyber law — cyber offences + penalties
Hacking	Bina permission system me ghus kar data churana/badalna/delete (Sec 66)
Teenage Web Vandals	Teens jo websites deface/hack karte hain mazak/show-off ke liye
Cyber Fraud	Internet par dhoka de kar paisa haasil karna (Sec 66D)
Defamation	Kisi ki izzat ko false + harmful baat se nuksan (IPC 499/500)
Harassment	Online pareshan karna — stalking, bullying, doxxing (IPC 354D, IT 67)
Monetary Penalty	Paisa ki saja — civil (Sec 43 up to 1 cr, 43A up to 5 cr)
Jurisdiction	Kaunsa court/country case sunega (IT Sec 75, IPC Sec 4)
Nature of Criminality	Criminal ki characteristics — anonymity, low risk-high reward
Trends	Naye attacks — ransomware, AI, deepfakes, IoT, crypto

Memory Tricks

- IT Act = **Digital gunahon ki book**.
- Section trick: **6-6 ki mithai** — 66 hacking, 66C identity, 66D personation, 67 obscene.
- 66A = **khatam** (SC 2015, Shreya Singhal case).
- Hacking = **Bina ijaazat ghusna**.
- Vandals = **Internet ke naughty bachche**.
- Fraud rule = **Pehle paisa maange to fraud**.
- Defamation = **Izzat ki LOL** (libel=written, slander=spoken).
- Penalty trick = **43 = 1 crore, 43A = 5 crore**.
- Jurisdiction = **Jahan computer, wahan court** (75).
- Nature trick = **ALPHA** — Anonymity, Low risk, Presence nahi, High reward, Aaram.
- Strategy trick = **L-T-A-E-I**; Trends = **R-A-D-I-C**.

Rapid Fire Viva (15 Questions)

Q	Answer
IT Act kab banaya?	2000
Section 66 ki saza?	3 saal jail + 5 lakh
66C ka matlab?	Identity theft (3 saal + 1 lakh)
66A ka kya hua?	SC ne 2015 me khatam kiya
Teenage vandals ka motive?	Show-off / mazak / attention
Cyber fraud me sabse common trick?	OTP nikalwana
Defamation ke 2 types?	Libel (written) + Slander (spoken)

Harassment ka sabse pehla step?	Block + reply nahi + screenshots
Cyberstalking ka section?	IPC 354D
Section 43 ki penalty?	Up to 1 crore (civil)
Section 43A ki penalty?	Up to 5 crore (sensitive data breach)
Jurisdiction ka IT Act section?	Sec 75 (India ki system → Indian court)
IPC ka jurisdiction section?	Sec 4 (Indian citizen bahar bhi)
Nature of criminality ka 1 point?	Anonymity / low risk high reward
Ransomware me kya hota hai?	Data lock + paisa demand

Note: Ye Unit 3 ka complete notes hai. Aage ke units same pattern par banenge.

CYBER SECURITY

UNIT 4 – COMPLETE DETAILED NOTES

Indian Evidence Act 1872 vs IT Act 2000 • Status of Electronic Records • Proof & Management of Electronic Records
Relevancy, Admissibility & Probative Value • Proving Digital Signatures • Proof of Electronic Agreements •
Proving Electronic Messages

Book-style notes with Practical Examples, Definitions, Flowcharts,
Diagrams, Important Points, Exam Notes & Memory Tricks

UNIT 4 – SYLLABUS (As per Examination)

S.No	Unit 4 Topics	Marks
1	The Indian Evidence Act of 1872 vs Information Technology Act, 2000	---
2	Status of Electronic Records as Evidence	---
3	Proof and Management of Electronic Records	---
4	Relevancy, Admissibility and Probative Value of E-Evidence	---
5	Proving Digital Signatures	---
6	Proof of Electronic Agreements	---
7	Proving Electronic Messages	---

How to read this book: Har topic me sabse pehle ek **long practical example** diya hai, phir **short definition**, phir **flowchart / diagram**, phir **important points** aur aakhir me **exam notes**. Isi order me padhna — definition yaad aasan ho jayegi.

CONTENTS

UNIT 4 – SYLLABUS (As per Examination)	2
PART 1 – INDIAN EVIDENCE ACT, 1872 vs IT ACT, 2000	5
1.1 Long Practical Example (Real Life)	5
1.2 Definitions	5
1.3 Flowchart	5
1.4 Comparison (Table)	5
1.5 How They Work Together	6
1.6 Important Points	6
PART 2 – STATUS OF ELECTRONIC RECORDS AS EVIDENCE	7
2.1 Long Practical Example (Real Life)	7
2.2 Definition	7
2.3 Flowchart	7
2.4 Kya-Kya Electronic Record Hota Hai (Table)	7
2.5 Important Points	8
2.6 Prevention / Guideline (Sahi Tarika)	8
PART 3 – PROOF AND MANAGEMENT OF ELECTRONIC RECORDS	9
3.1 Long Practical Example (Real Life)	9
3.2 Definition	9
3.3 Flowchart – Proof Process	9
3.4 Section 65B — 4 Conditions (Table)	9
3.5 Management Best Practices (Sambhalne Ke Tarika)	10
3.6 Important Points	10
PART 4 – RELEVANCY, ADMISSIBILITY AND PROBATIVE VALUE OF E-EVIDENCE	11
4.1 Long Practical Example (Real Life)	11
4.2 Definitions	11
4.3 Flowchart	11
4.4 Relevancy vs Admissibility vs Probative Value (Table)	12
4.5 Admissibility ke Rules (Electronic)	12
4.6 Probative Value Badhane Ke Tarika	12

PART 5 – PROVING DIGITAL SIGNATURES	13
5.1 Long Practical Example (Real Life)	13
5.2 Definition	13
5.3 Flowchart – Kaise Kaam Karta Hai	13
5.4 Proving Ke Sections (Table)	13
5.5 Important Points	14
5.6 Guideline (Sahi Use)	14
PART 6 – PROOF OF ELECTRONIC AGREEMENTS	15
6.1 Long Practical Example (Real Life)	15
6.2 Definition	15
6.3 Flowchart	15
6.4 E-Contract Ke Elements (Table)	15
6.5 Proving Ke Tarike (Section)	16
6.6 Important Points	16
PART 7 – PROVING ELECTRONIC MESSAGES	17
7.1 Long Practical Example (Real Life)	17
7.2 Definition	17
7.3 Flowchart	17
7.4 Proving Ke Points (Table)	17
7.5 Important Points	18
7.6 Guideline (Sahi Tarike)	18
APPENDIX – QUICK REVISION SHEET (UNIT 4)	20

PART 1 – INDIAN EVIDENCE ACT, 1872 vs IT ACT, 2000

Jab cyber crime ka case court me jata hai to **Evidence Act** aur **IT Act** dono kaam karte hain — Evidence Act batata hai **kya saboot chalta hai**, IT Act batata hai **computer/e-signature ko kaise maano**. Dono sath-sath chalte hain.

1.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek landlord ne tenant par case kiya — rent nahi diya. Tenant bola — “main diya tha.”

Landlord ke paas sirf **WhatsApp chats** thi jisme tenant ne likha: “paisa bhej diya hai.”

Court me sawal: **kya chat ka screenshot saboot (evidence) banega?**

Yahan dono laws mile:

- **Evidence Act** — ye kehta hai evidence kya hota hai aur electronic record kaise sabit hota hai (Section 65B).
- **IT Act** — ye kehta hai electronic record ka legal status kya hai (Sec 2(1)(t), 4).

Correct certificate (65B) ke saath chats **court me chali gayi** — tenant ko rent dena pada.

Isliye dono acts ek saath padhne padhte hain.

1.2 Definitions

DEFINITION

Indian Evidence Act, 1872 — wo law jo batata hai **kaunsa saboot court me admissibility hai**, kaise proof hota hai, aur court kya maan sakti hai.

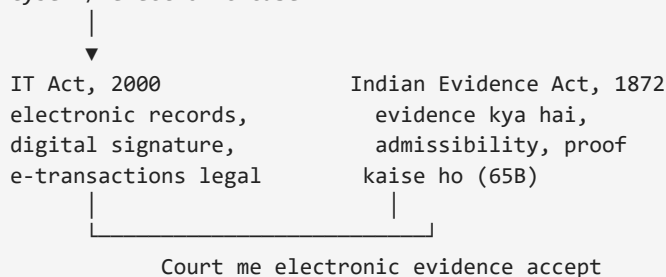
IT Act, 2000 — India ka cyber law jo **electronic records, digital signatures, e-transactions** ko legal maanyata deta hai aur cyber offences define karta hai.

Simple: Evidence Act = **Saboot ki book**; IT Act = **Computer ki book**. Cyber case me dono sath lagte hain.

1.3 Flowchart

FLOWCHART

Cyber / electronic case



1.4 Comparison (Table)

Basis	Evidence Act, 1872	IT Act, 2000
Kab bana	1872 (bahut purana)	2000 (computer era ke liye)
Kya batata hai	Evidence kya hai, kaise proof	Electronic records/signature legal
Main kaam	Admissibility + proof rules	Cyber offences + e-commerce legal
Electronic records	65A / 65B se prove hote hain	Sec 2(1)(t) me define
Digital signature	67A/73A/85B (opinion/presumption)	Sec 3, 3A (authentication)

Nature	Old + general law	New + special law (cyber)
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1.5 How They Work Together

- **IT Act 2000** ke through hi Evidence Act me **65A/65B, 67A, 73A, 85A/85B, 88A, 90A** jode gaye.
- IT Act ka **digital signature** hi Evidence Act me prove hota hai (67A/73A).
- Electronic record ka **legal status** IT Act se, **proof** Evidence Act se.
- Dono **supplement** karte hain — ek dusre ko replace nahi karte.

1.6 Important Points

IMPORTANT POINTS

1. **Evidence Act = general law**, IT Act = **special law** (cyber).
2. Electronic evidence ka **base IT Act** me hai, **admissibility Evidence Act** me.
3. Evidence Act 2000 me **amended** hua taaki electronic records admit ho sakein.
4. 65B certificate ke bina electronic evidence **admit nahi** hota (dubai ke maamle ke baad).

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Evidence Act 1872 = evidence kaise prove hoga (general law).
2. IT Act 2000 = electronic records/digital signatures ko legal maanyata (special law).
3. IT Act ke amendment se Evidence Act me 65A/65B, 67A, 73A, 85A/85B jode gaye.
4. Dono saath chalte hain — ek replace nahi karta.

Viva Q1. Electronic evidence kis act se prove hota hai?

Ans: Evidence Act Sec 65A/65B (IT Act ne in sections ko add kiya).

Viva Q2. Evidence Act aur IT Act ka sambandh?

Ans: IT Act electronic records ko legal banata hai; Evidence Act batata hai wo court me kaise prove honge.

MEMORY TRICK

Evidence Act = Saboot ki book, IT Act = Computer ki book. Trick: **IT Act ne Evidence Act ko upgrade** kiya — 1872 purani, 2000 nayi, dono sath-sath.

PART 2 – STATUS OF ELECTRONIC RECORDS AS EVIDENCE

Pehle court me **kaagaz aur gawah** hi chalta tha. Ab **email, chat, CCTV, bank records** bhi evidence hai — par unhe **specific tarike** se prove karna padta hai. Isi ko 'Status of Electronic Records as Evidence' kehte hain.

2.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek shop me **chor ne owner ko dhoka diya**. Police ne pakda to bande ne kaha — “main wahin tha hi nahi.”

Police ke paas the:

- Shop ki **CCTV footage** — usme wo banda dikh raha tha.
- Uske **bank statement** — usi time ka transaction.
- Uska **call record (CDR)** — usi area ka tower.

Court me jab ye electronic records dikhaye gaye — **65B certificate** ke saath — to court ne maan liya.

Banda pakda gaya.

Electronic records (CCTV, bank, CDR) ka evidence ban jana — Status of Electronic Records.

2.2 Definition

DEFINITION

Electronic Record — data, record, image, audio, video, signal ya message jo **electronic form** me hai — email, chat, SMS, CCTV, bank statement, call records, websites.

Evidence Act Sec 3 ke amendment ke baad '**evidence**' me **electronic records** bhi shamil hain — IT Act Sec 2(1)(t) me electronic record define hua hai.

Simple: Jaisa kaghaz ka record evidence hai — waise hi digital record bhi evidence hai (sahi tarike se prove hone par).

2.3 Flowchart

FLOWCHART

```

Electronic record banta hai
| (email, chat, CCTV, bank, CDR)
▼
Evidence Act Sec 3: 'evidence' me include
|
▼
IT Act Sec 2(1)(t): electronic record
|
▼
Sec 65A + 65B se prove karna padta hai
| (certificate + proper conditions)
▼
Court me admissible evidence ban jata hai
  
```

2.4 Kya-Kya Electronic Record Hota Hai (Table)

Record	Example
Emails	Business / personal emails
Chats / SMS	WhatsApp, Messenger, SMS
CCTV footage	Camera recordings
Bank records	Statement, transactions, UPI

Call records (CDR)	Call detail records, tower data
Photos / videos	Mobile / camera files
Website / social posts	Post, comments, profile
Logs	Server, login, IP logs

2.5 Important Points

IMPORTANT POINTS

1. Electronic record ka **status evidence jaise hi** hai — par **saboot conditionally** (65B ke through).
2. **Section 65A** — electronic record ke contents 65B ke under hi prove ho sakte hain.
3. **Section 65B** — computer output + certificate ke saath admissible.
4. Bina 65B certificate ke electronic evidence **reject** ho sakta hai.
5. Record **tamper-free + authentic** hona chahiye — tabhi weight milta hai.

2.6 Prevention / Guideline (Sahi Tarike)

- Electronic evidence **original device/se** aur **copy dono** sambhal ke rakho.
- Har record ke saath **65B certificate** banao.
- Record ki **hash value** note karo — tampering prove karne ke liye.
- **Metadata** (time, date, sender) preserve karo.
- Chains of custody — kaun-kis se record mila, sab likha rakho.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Electronic record = data in electronic form (Sec 2(1)(t) IT Act).
2. Evidence Act Sec 3 me 'evidence' me electronic records include hain (2000 amendment).
3. Proof sirf 65A + 65B se hota hai — certificate ke saath.
4. Bina 65B certificate electronic evidence reject ho sakta hai.

Viva Q1. Electronic record kya hai?

Ans: Data/record electronic form me — email, chat, CCTV, bank records.

Viva Q2. Electronic record evidence kaise banta hai?

Ans: Evidence Act 65A/65B ke through — 65B certificate ke saath.

MEMORY TRICK

Electronic Record = Digital kaghaz. Trick: **65B = 6+5 = 11 baat yaad rakho** — certificate + 4 conditions ke bina computer output court me nahi chalta.

PART 3 – PROOF AND MANAGEMENT OF ELECTRONIC RECORDS

Proof — court me electronic record ko sahi tarike se sabit karna. **Management** — record ko aise sambhalna ki wo court me kharab na ho (tamper-proof). Dono mil kar case jeete hain.

3.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek company ke against employee ne case kiya — “salary chhupi.” Company ke paas **email** thi jisme employee ne offer accept kiya tha.

Company ne court me **sirf email ka printout** dikhaya — **no 65B certificate, no chain of custody**.

Employee ke lawyer ne objection kiya:

- Email me **tamper ho sakti hai** — prove karo asli hai.
- Computer kiska tha, kaise nikala — kuch proof nahi.

Court ne email **admit nahi ki** — company case haar gayi.

Sahi management (certificate + custody) na hone par electronic record bekaar ho jata hai.

3.2 Definition

DEFINITION

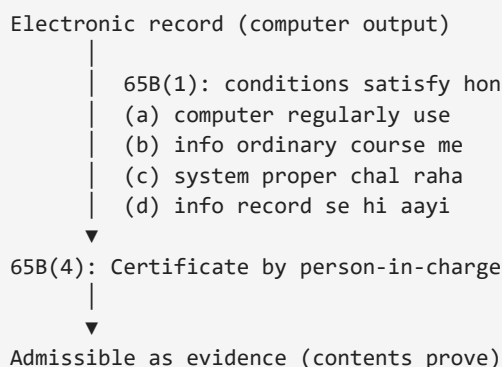
Proof of Electronic Records — electronic record ko **court me sabit karna** Evidence Act 65A/65B ke tarike se (computer output + 65B certificate).

Management of Electronic Records — record ko **safe, tamper-proof aur traceable** rakhna — backup, hash, metadata, access control, chain of custody.

Simple: Proof = court me **dikhaana**; Management = pehle se **sambhalna** ki kharab na ho.

3.3 Flowchart – Proof Process

FLOWCHART



3.4 Section 65B — 4 Conditions (Table)

Clause	Condition
65B(1)(a)	Computer regularly use hota tha us info ke liye
65B(1)(b)	Info normal course of activities me record hui
65B(1)(c)	Computer sahi (properly) chal raha tha

65B(1)(d)

Jo info dikha rahe, wo us record se copy/print hui

3.5 Management Best Practices (Sambhalne Ke Tarike)

- **Chain of custody** — record kahan-se kiska-paas gaya, har step likho.
- **Hash value** — original record ki hash lo; compare karke tampering catch karo.
- **Metadata preserve** — date, time, sender, device info.
- **Backup** — original + encrypted copies safe jagah.
- **Access control** — record ko kuch log hi touch karein.
- Record ko **seal/preserve** karo — jaisa mila, waise hi court tak.
- Forensic expert se **copy + analysis** karwao (write-protected).

3.6 Important Points

IMPORTANT POINTS

1. 65B(4) certificate **computer ka in-charge person** deta hai — jo us system ko control karta hai.
2. Certificate me likha hota hai ki conditions 65B(1) satisfy hui hain.
3. Proof ke waqt **original vs printout** — dono ko sath rakho.
4. **Management ki galti** = case ki haar — isliye pehle se sambhalna zaroori.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Proof = 65A/65B se computer output + certificate ke saath.
2. 65B ki 4 conditions: regular use, ordinary course, system proper, info se copy.
3. 65B(4) certificate in-charge person deta hai.
4. Management = chain of custody + hash + metadata + backup.

Viva Q1. 65B certificate kaun deta hai?

Ans: Computer system ka in-charge person.

Viva Q2. Hash value se kya pata chalta hai?

Ans: Record me tampering hui ya nahi.

MEMORY TRICK

Proof = Dikhaana, Management = Sambhalna. Trick: 65B ke **4 conditions = ABCD** — Agar regular use, B-normal kaam, C-system chal raha, D-info wahi se copy.

PART 4 – RELEVANCY, ADMISSIBILITY AND PROBATIVE VALUE OF E-EVIDENCE

Evidence court me tab tak bekaar hai jab tak wo **relevant**, **admissible** aur **weighty** na ho. Ye teen alag-alag cheezein hain — exam me sabse important part.

4.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Aman par case tha — usne online fraud kiya. Police ke paas the:

- **Bank transaction record** — jisme paisa Aman ke account me gaya.
- Uske doston ki **game chat** — jisme football ki baat thi.

Ab court me 3 sawal:

- **Relevancy** — bank record case se juda hai (relevant); game chat ka case se koi lena-dena nahi (irrelevant).
- **Admissibility** — bank record 65B certificate ke saath hai to admit hoga; bina certificate ke nahi.
- **Probative value** — certificate + metadata + matching amount ke saath bank record ka **weight** bada hai — court uspar bharosa karegi.

Yehi teen concepts — **Relevancy, Admissibility, Probative Value**.

4.2 Definitions

DEFINITION

Relevancy — kya ye fact case ke issues se **juda hai** (Evidence Act Sec 5-55).

Admissibility — kya evidence court me **legally accept** ho sakta hai (rules: hearsay, best evidence, 65B).

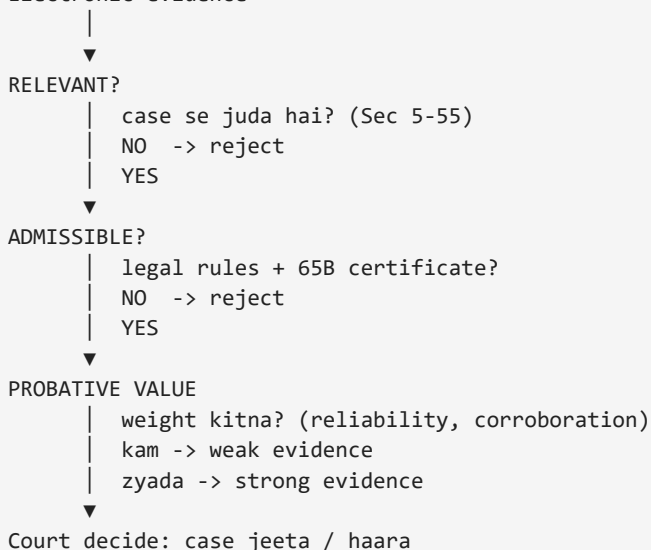
Probative Value — evidence ka **weight/bharosa** — kitna strong hai wo fact sabit karne ke liye.

Simple: Relevant = **juda hai**; Admissible = **andar aa sakta**; Probative = **kitna bolta hai**.

4.3 Flowchart

FLOWCHART

Electronic evidence



4.4 Relevancy vs Admissibility vs Probative Value (Table)

Basis	Relevancy	Admissibility	Probative Value
Sawal	Case se juda?	Law ke under accept?	Kitna bharosa?
Section	Sec 5-55 (IEA)	Evidence rules, 65B	Judge ki analysis
Nature	Logical link	Legal gate	Weight/reliability
Nahi ho to	Bahar (irrelevant)	Reject (inadmissible)	Weak evidence

4.5 Admissibility ke Rules (Electronic)

- **Best Evidence Rule** — original prove karo; copy par rely nahi (65B exception).
- **Hearsay** — computer output hearsay-like hai — isliye 65B certificate zaroori.
- **Authenticity** — prove karo ki record asli + tamper-free hai.
- **Produced properly** — sahi person + sahi tarike se court me pesh.

4.6 Probative Value Badhane Ke Tarike

- Record ko **corroborate** karo — 2 independent sources (bank + CCTV).
- **Hash + metadata** se authenticity prove karo.
- **Expert** (forensic / Examiner of E-Evidence Sec 79A) se analysis karwao.
- Witness se **human confirmation** — kisi ne dekha, kisi ne likha.
- Certificate **complete + correct** rakho — chhoti galti bhi reject karwa sakti hai.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Relevancy = case se juda hona (Sec 5-55).
2. Admissibility = legal rules + 65B ke through accept hona.
3. Probative value = evidence ka weight/bharosa (reliability).
4. Sabse strong = relevant + admissible + high probative value.

Viva Q1. Relevancy aur admissibility me farak?

Ans: Relevancy = logical link (juda hai); Admissibility = legal gate (accept ho sakta).

Viva Q2. Probative value kya hai?

Ans: Evidence ka weight — kitna strong wo fact sabit karne me.

MEMORY TRICK

Relevancy-Admissibility-Probative = Darwaza-Entry-Wazan. Trick: **R-A-P = Roko-Andar-Pesa** — Roko (relevancy check), Andar (admission), Pesa (weight).

PART 5 – PROVING DIGITAL SIGNATURES

Handwritten sign ka digital version = **Digital Signature**. Court me kisi electronic document par aaye sign ko **prove** karna — isliye Evidence Act me special sections (67A, 73A, 85B) hain.

5.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Rohit ne apni **income tax return** digital signature (DSC) se file ki. Baad me IT department ne kaha — “ye return tune nahi bhara tha.”

Ab court me sawal — **ye digital signature Rohit ka hai kaise prove hoga?**

Prove ke tarike:

- **Sec 67A (Evidence Act)** — Certifying Authority (CA) ki opinion — signature kis device/ka hai.
- **Sec 73A** — court digital signature verify karwa sakti hai.
- **Sec 85B** — presumption — secure digital signature ka signature, uske owner ka maana jata hai (jab tak ulta na sabit ho).

DSC record + CA record se Rohit ki baat pakki ho gayi — IT department ko maanna pada.

Digital signature ka owner aur validity court me sabit karna — Proving Digital Signatures.

5.2 Definition

DEFINITION

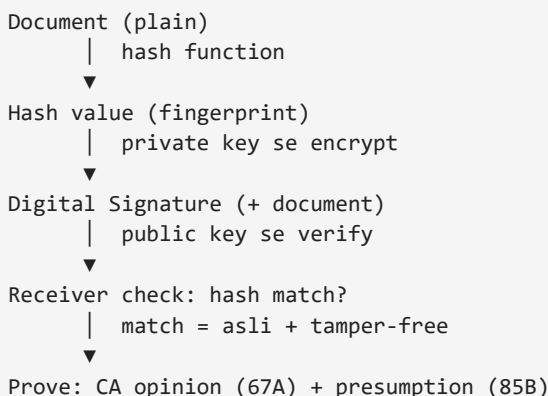
Digital Signature — electronic record ko authenticate karne ka tarika — **hash function + asymmetric crypto** (private + public key) se. (IT Act Sec 2(1)(p), 3)

Proving Digital Signature — court me sabit karna ki ye signature iska hai, valid hai aur record me koi tampering nahi (IEA 67A, 73A, 85B).

Simple: Hath ke sign ka digital twin — aur isko court me pehchan karwana.

5.3 Flowchart – Kaise Kaam Karta Hai

FLOWCHART



5.4 Proving Ke Sections (Table)

Section	Kya kehta hai
IEA 67A	Digital signature prove karne ke liye CA (Certifying Authority) ki opinion lena
IEA 73A	Court khud digital signature verify karwa sakti hai (comparison)

IEA 85B	Presumption — secure digital signature apne owner ka maana jata hai
IT Act 3	Digital signature ka process (hash + asymmetric crypto)
IT Act 2(1)(p)	Digital signature ki definition

5.5 Important Points

IMPORTANT POINTS

1. **Digital Signature = private key se signed, public key se verify** — jisne sign kiya wo deny nahi kar sakta (non-repudiation).
2. **Sec 85B** presumption ka matlab — court maan leti hai ki signature owner ka hai, jab tak opposite prove na ho.
3. **Sec 67A** — Certifying Authority ki opinion se proof.
4. Hash mismatch = document badal gaya — signature invalid.

5.6 Guideline (Sahi Use)

- DSC sirf **certified CA** se lo.
- Private key ko **safe** rakho — kisi se share na karo.
- Document sign karne se pehle **hash compare** karo.
- CA ka **record / certificate** rakho — court me prove ke liye.
- Expired / revoked certificate se sign **invalid** hota hai.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Digital signature = hash + asymmetric crypto (private/public key).
2. IEA 67A = CA opinion; 73A = court verification; 85B = presumption of owner.
3. Non-repudiation — sign karne wala deny nahi kar sakta.
4. Hash mismatch = tampering.

Viva Q1. Digital signature prove kaise hota hai?

Ans: IEA 67A (CA opinion) + 73A (court verify) + 85B (presumption).

Viva Q2. Sec 85B kya presumption deta hai?

Ans: Secure digital signature apne owner ka maana jata hai jab tak opposite prove na ho.

MEMORY TRICK

Digital Signature = Digital hath. Trick: **85B = bharosa** — B = Bharosa ki signature owner ka hai. Private key se sign, public key se verify.

PART 6 – PROOF OF ELECTRONIC AGREEMENTS

Online contract (agreement) kya **legal** hai? Haan — **IT Act Sec 10A** ke under. Par court me use **prove** karna bhi zaroori hai — Evidence Act Sec 85A se.

6.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Meena ne ek website se **laptop** order kiya. Website ka **terms & conditions** online accept kiya — “I agree” box par click kiya.

Paise gaye, laptop nahi aaya. Meena ne case kiya.

Court me:

- Website ka **order record** — uska agreement tha.
- Sec 85A (IEA) — presumption ki electronic agreement ka agreement asli hai (jab tak ulta na prove ho).
- IT Act 10A — electronic me bana contract valid hai.

Order record + payment proof se Meena jeet gayi — website ko refund dena pada.

Online contract ko court me prove karna — Proof of Electronic Agreements.

6.2 Definition

DEFINITION

Electronic Agreement / E-Contract — contract jo **electronic medium** (website, email, app) par bana — offer + acceptance + consideration, sab electronic.

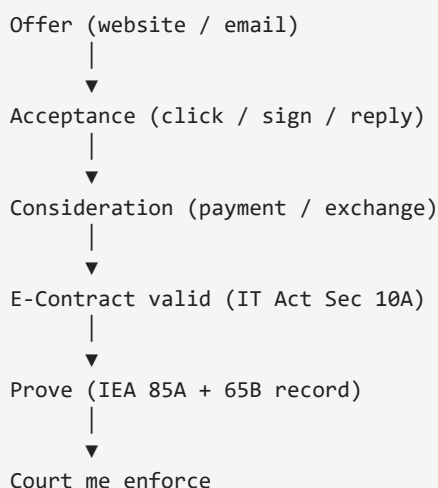
IT Act **Sec 10A** — e-contract valid hai, sirf isliye ki electronic me hua.

IEA **Sec 85A** — presumption ki electronic agreement ka agreement hai (jab tak opposite prove na ho).

Simple: Hath se sign ki jagah **click karna** — par utna hi legal.

6.3 Flowchart

FLOWCHART



6.4 E-Contract Ke Elements (Table)

Element	Electronic me kya hota hai
Offer	Website listing / email proposal
Acceptance	'I agree' click, order, reply

Consideration	Online payment
Consent	Terms & conditions accept karna
Authentication	Login / OTP / digital signature
Record	Server log, email, invoice

6.5 Proving Ke Tarike (Section)

Section	Kaam
IT Act 10A	E-contract valid — electronic hone se invalid nahi
IT Act 2(1)(h)	'data' + electronic record ki definition
IEA 85A	Presumption — electronic agreement asli + properly executed
IEA 65B	Agreement ke computer records prove karne ke liye
IEA 67A/73A	Agar digital signature use hua to usko prove karna

6.6 Important Points

IMPORTANT POINTS

1. **E-contract ka bhi valid** — bas aakhri (binding) hona chahiye: offer, acceptance, consideration.
2. **Sec 85A** — court presume karti hai ki electronic agreement ka agreement hi hai.
3. **Proof** ke liye records (server log, email, payment) 65B se.
4. Website par **terms clearly visible** honi chahiye — chhupa contract invalid ho sakta.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. E-contract valid = IT Act Sec 10A.
2. IEA 85A = presumption ki electronic agreement asli hai.
3. Elements = offer + acceptance + consideration + record.
4. Proof = 65B records + (digital signature ho to 67A/85B).

Viva Q1. E-contract valid kyu hai?

Ans: IT Act Sec 10A — electronic me bana contract valid, sirf electronic hone se invalid nahi.

Viva Q2. Sec 85A kya presume karta hai?

Ans: Electronic agreement ka agreement hai (asli + properly executed).

MEMORY TRICK

E-Contract = Online nibandh. Trick: **10A = asli** (IT Act) aur **85A = bharosa** (Evidence Act) — dono sath milke online contract court me kadak banate hain.

PART 7 – PROVING ELECTRONIC MESSAGES

Electronic Messages — email, SMS, WhatsApp, chat. Court me inhe **prove** karna — kaunsa message, kisne bheja, kya likha tha — iske liye 65B + 88A jaise sections hain.

7.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek manager ko employee se **dhamki wali email** mili — “job se nikala to anjaam buri.”

Manager ne case kiya. Court me prove karna tha:

- **Ye email kisne bheji?** — email header + IP + account se pata chala.
- **Message asli hai?** — 65B certificate + hash se.
- **Mil bhi gaya?** — IEA 88A presumption — message recipient ko pahuncha maana jata hai.

Email header me IP address aur account ki info se employee pakda gaya — saza mili.

Email/chat ko court me sabit karna — Proving Electronic Messages.

7.2 Definition

DEFINITION

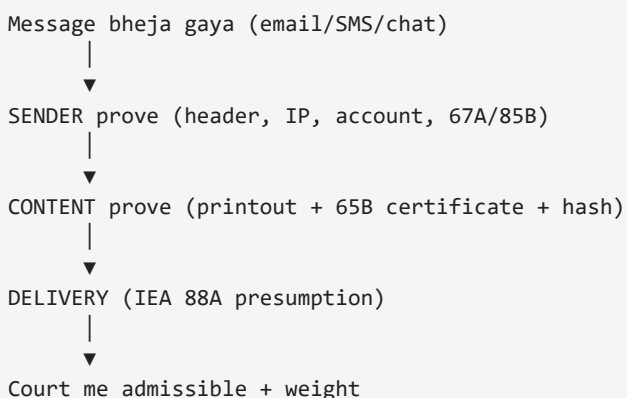
Electronic Message — koi bhi message jo electronic device se bheja gaya — email, SMS, WhatsApp/chat, voice note, video message.

Proving Electronic Messages — message ka **content, sender aur delivery** court me sabit karna — 65B (record), 88A (delivery presumption), headers/IP/logs (sender).

Simple: ‘Bolne wala kaun, baat kya, pahunchi kya’ — teeno prove karna.

7.3 Flowchart

FLOWCHART



7.4 Proving Ke Points (Table)

Kya prove karna	Tarika
Content kya tha	Printout/screenshot + 65B certificate + hash
Kisne bheja (sender)	Email header, IP address, account logs, digital signature
Message pahuncha	IEA 88A — presumption of delivery at destination
Tamper nahi hua	Hash value + forensic analysis (Sec 79A examiner)
Kis device/computer se	65B certificate (computer ka in-charge)

7.5 Important Points

IMPORTANT POINTS

1. **88A presumption** — electronic message apne address (destination) par pahuncha maana jata hai, jab tak ulta na prove ho.
2. **Sender** prove karna sabse mushkil — email header + IP + account se.
3. Screenshot **akela kaafi nahi** — 65B certificate chahiye.
4. Chats me **context + timestamps** bhi important hain.
5. Message ki **copy vs original** — dono sath rakho.

7.6 Guideline (Sahi Tarike)

- Message ka **screenshot + original device** dono sambhalo.
- **Header (email) / metadata** preserve karo.
- Har message ke saath **65B certificate** banwao.
- **Hash** note karo — tampering prove nahi ho sake.
- Chah to **forensic** se nikalo — unhe kaante jayenge.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Electronic message = email/SMS/chat (electronic form me).
2. Prove = sender (header/IP) + content (65B) + delivery (88A).
3. IEA 88A = presumption ki message destination par pahuncha.
4. Screenshot akela kaafi nahi — certificate + hash chahiye.

Viva Q1. Email ka sender kaise prove hota hai?

Ans: Email header, IP address, account logs se.

Viva Q2. Sec 88A kya presume karta hai?

Ans: Electronic message apne destination par pahuncha maana jata hai.

MEMORY TRICK

Electronic Message = Digital kaagaz. Trick: **88A = aayi** — message Aa gaya (pahuncha). Content = 65B, Sender = Header+IP. Teenon sath = case pakka.

APPENDIX – QUICK REVISION SHEET (UNIT 4)

Sabse Pehle Yaad Rakho — One-Line Definitions

Topic	Ek Line Me
Evidence Act, 1872	Saboot ki book — kaunsa evidence chalta hai, kaise prove hota hai
IT Act, 2000	Computer ki book — electronic records/digital signatures legal
Electronic Record	Data in electronic form — email, chat, CCTV, bank, CDR (Sec 2(1)(t))
Status as Evidence	Evidence Act Sec 3 me 'evidence' me include (2000 amendment)
Sec 65A	Electronic record ke contents sirf 65B se prove ho sakte
Sec 65B	Computer output + certificate ke saath admissible (4 conditions)
Relevancy	Case se juda hona (Sec 5-55)
Admissibility	Legal gate — 65B + rules ke through accept hona
Probative Value	Evidence ka weight/bharosa
Digital Signature	Hash + asymmetric crypto (private/public key)
IEA 67A	CA (Certifying Authority) ki opinion se sign prove
IEA 85B	Presumption — secure digital signature owner ka
E-Contract	IT Act 10A — valid; IEA 85A — presumption asli hai
Electronic Message	Email/SMS/chat — prove: sender(header/IP) + content(65B) + delivery(88A)

Memory Tricks

- Evidence Act = **Saboot ki book**, IT Act = **Computer ki book**.
- 65B ke 4 conditions = **ABCD** — Agar regular use, B-normal kaam, C-system chal raha, D-info wahi se copy.
- 65B certificate = **in-charge person** deta hai.
- R-A-P = **Roko-Andar-Pesa** — Relevancy, Admissibility, Probative value.
- Digital signature = **Digital hath** (private key se sign, public key se verify).
- 85B = **Bharosa** (sign owner ka maana jata hai).
- E-Contract = **Online nibandh** — 10A = asli, 85A = bharosa.
- 88A = **Aayi** — message pahuncha maana jata hai.

Rapid Fire Viva (14 Questions)

Q	Answer
Evidence Act kab bana?	1872
IT Act kab bana?	2000
Electronic record ka section (IT Act)?	Sec 2(1)(t)
Electronic evidence kaise prove?	IEA 65A + 65B
65B certificate kaun deta?	Computer ka in-charge person

65B ki kitni conditions?	4 (65B(1) a,b,c,d)
Relevancy ka section?	Sec 5-55 (IEA)
Admissibility ke 2 rules?	Best evidence + hearsay (65B se cure)
Digital signature process?	Hash + private key encrypt + public key verify
Sec 67A kya maangta?	Certifying Authority ki opinion
Sec 85B presumption?	Secure digital signature owner ka
E-contract valid kaunsa section?	IT Act Sec 10A
Sec 85A presumption?	Electronic agreement asli hai
Sec 88A presumption?	Message destination par pahuncha

Note: Ye Unit 4 ka complete notes hai — Evidence Act + IT Act ka electronic evidence wala hissa.

CYBER SECURITY

UNIT 5 – COMPLETE DETAILED NOTES

Proxy Servers & Anonymizers • Password Cracking • Keyloggers & Spyware • Virus & Worms • Trojan Horses • Backdoors

DoS & DDoS Attacks • Buffer Overflow • Attack on Wireless Networks • Phishing (Method & Techniques)

Book-style notes with Practical Examples, Definitions, Flowcharts, Diagrams, Important Points, Exam Notes & Memory Tricks

UNIT 5 – SYLLABUS (As per Examination)

S.No	Unit 5 Topics	Marks
1	Proxy Servers and Anonymizers	---
2	Password Cracking	---
3	Keyloggers and Spyware	---
4	Virus and Worms	---
5	Trojan Horses	---
6	Backdoors	---
7	DoS and DDoS Attacks	---
8	Buffer and Overflow (Buffer Overflow)	---
9	Attack on Wireless Networks	---
10	Phishing: Method of Phishing, Phishing Techniques	---

How to read this book: Har topic me sabse pehle ek **long practical example** diya hai, phir **short definition**, phir **flowchart / diagram**, phir **important points** aur aakhir me **exam notes**. Isi order me padhna — definition yaad aasan ho jayegi.

CONTENTS

UNIT 5 – SYLLABUS (As per Examination)	2
PART 1 – PROXY SERVERS AND ANONYMIZERS	6
1.1 Long Practical Example (Real Life)	6
1.2 Definition	6
1.3 Flowchart	6
1.4 Proxy vs Anonymizer vs VPN (Table)	6
1.5 Kyu Use Kiya Jata Hai	7
1.6 Prevention (Bachao Ke Tarika)	7
PART 2 – PASSWORD CRACKING	8
2.1 Long Practical Example (Real Life)	8
2.2 Definition	8
2.3 Flowchart	8
2.4 Techniques (Table)	8
2.5 Important Points	9
2.6 Prevention (Bachao Ke Tarika)	9
PART 3 – KEYLOGGERS AND SPYWARE	10
3.1 Long Practical Example (Real Life)	10
3.2 Definition	10
3.3 Flowchart	10
3.4 Types (Table)	10
3.5 Keylogger vs Spyware (Table)	11
3.6 Prevention (Bachao Ke Tarika)	11
PART 4 – VIRUS AND WORMS	12
4.1 Long Practical Example (Real Life)	12
4.2 Definition	12
4.3 Flowchart	12
4.4 Virus vs Worm (Table)	12
4.5 Types (Ek Nazar)	13
4.6 Prevention (Bachao Ke Tarika)	13

PART 5 – TROJAN HORSES	14
5.1 Long Practical Example (Real Life)	14
5.2 Definition	14
5.3 Flowchart	14
5.4 Types (Table)	14
5.5 Trojan vs Virus vs Worm (Table)	15
5.6 Prevention (Bachao Ke Tarike)	15
PART 6 – BACKDOORS	16
6.1 Long Practical Example (Real Life)	16
6.2 Definition	16
6.3 Flowchart	16
6.4 Types (Table)	16
6.5 Important Points	16
6.6 Prevention (Bachao Ke Tarike)	17
PART 7 – DoS AND DDoS ATTACKS	18
7.1 Long Practical Example (Real Life)	18
7.2 Definition	18
7.3 Flowchart	18
7.4 DoS vs DDoS (Table)	18
7.5 Types / Techniques (Table)	19
7.6 Prevention (Bachao Ke Tarike)	19
PART 8 – BUFFER OVERFLOW	20
8.1 Long Practical Example (Real Life)	20
8.2 Definition	20
8.3 Flowchart	20
8.4 Types (Table)	20
8.5 Important Points	20
8.6 Prevention (Bachao Ke Tarike)	21
PART 9 – ATTACK ON WIRELESS NETWORKS	22
9.1 Long Practical Example (Real Life)	22
9.2 Definition	22

9.3 Flowchart	22
9.4 Types of Wireless Attacks (Table)	22
9.5 Important Points	22
9.6 Prevention (Bachao Ke Tarike)	23
PART 10 – PHISHING (METHOD AND TECHNIQUES)	24
10.1 Long Practical Example (Real Life)	24
10.2 Definition	24
10.3 Flowchart – Method	24
10.4 Phishing Techniques (Table)	24
10.5 Important Points	25
10.6 Prevention (Bachao Ke Tarike)	25
APPENDIX – QUICK REVISION SHEET (UNIT 5)	27

PART 1 – PROXY SERVERS AND ANONYMIZERS

Cyber criminals apni **asli pehchan (IP)** chhupane ke liye **proxy server** aur **anonymizer** use karte hain — jaise chor mask pehen ke chalta hai.

1.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek hacker **Mumbai** me baitha tha. Usne ek bank website par attack kiya. Police ne **IP address** se pakadne ki koshish ki.

Par hacker ne pehle hi proxy server use kar liya tha:

- Uska request **proxy server** se hokar gaya — bank ko **proxy ka IP** dikha, hacker ka nahi.
- Proxy bhi kisi **doosre desh** me thi — trail toot gayi.
- Police ko pakadne me **bahut time laga**.

Isi wajah se proxy/anonymizers cyber crime me itne common hain.

1.2 Definition

DEFINITION

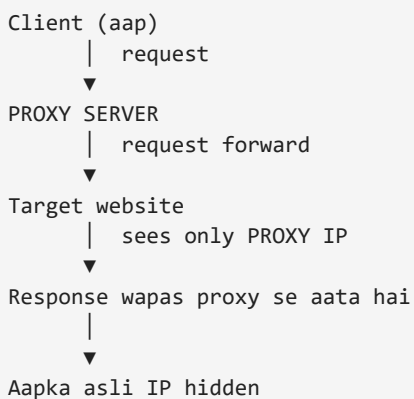
Proxy Server — beech ka server jo client (aap) aur internet ke darmiyaan request forward karta hai — aapka **asli IP server se chhupa** jaata hai.

Anonymizer — tool/service jo aapki **identity aur IP ko fully chhupati** hai — surfing par koi pata na chale ki aap kaun hain.

Simple: Proxy = **pehredar ka dikhna** (wo request aage badhata hai), Anonymizer = **inka baap** — identity bilkul chhupa.

1.3 Flowchart

FLOWCHART



1.4 Proxy vs Anonymizer vs VPN (Table)

Basis	Proxy Server	Anonymizer	VPN
IP chhupata hai	Haan (web traffic)	Haan (identity)	Haan (sab traffic)
Encryption	Aksar nahi	Hamesha nahi	Haan (encrypted)
Kaam	Request forward	Identity hide	Secure tunnel
Use (criminals)	IP chhupana	Identity chhupana	Fully anonymous

1.5 Kyu Use Kiya Jata Hai

- Cyber criminal **IP hide** karne ke liye — pakda na jaye.
- **Bypass restrictions** — blocked sites tak pahuchna.
- **Multi-layer proxy** (chain) — trail aur bhi mushkil.
- **Anonymizer** se email/chat bhi bina pehchan ke.
- Legal use bhi hai — privacy, corporate filtering.

1.6 Prevention (Bachao Ke Tarike)

- Investigators ko **proxy logs + ISP** se peechna padta hai — jaaldi karo.
- Multi-hop trails ke liye **timestamps + packet analysis**.
- Websites par **strong logging + anomaly detection**.
- Real-time **threat intel** se proxy-IP pools block karo.
- Companies me **reverse proxy** use karke attack block karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Proxy server = beech ka server — asli IP chhupata hai.
2. Anonymizer = identity/IP fully chhupane wala tool.
3. Criminal IP hide karne ke liye inka use karta hai — trail toot jati hai.
4. VPN = encrypted tunnel (sabse secure).

Viva Q1. Proxy server kya karta hai?

Ans: Client aur internet ke beech request forward karta hai — target ko asli IP nahi dikhta.

Viva Q2. Anonymizer ka kya kaam?

Ans: User ki identity/IP ko chhupana taaki pata na chale.

MEMORY TRICK

Proxy = Pehredar, Anonymizer = Inka baap. Trick: **Proxy = Paisa ka bichauliya** — jaise bichauliya aapke naam nahi batata, waise proxy aapka IP nahi batata.

PART 2 – PASSWORD CRACKING

Password Cracking — kisi ke account ka password nikalne ka tarika. Criminals iske liye weak passwords aur hash cracking tools use karte hain.

2.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek aadmi ka **Instagram account** hack ho gaya. Password tha — **123456**.

Hacker ne:

- Pehle **common passwords** try kiye (dictionary attack).
- Phir har letter-digit-combo **brute force** se try kiya.
- 123456 ek hi try me mil gaya.

Hacker ne account se paisa maangne wale messages bheje — aur account waapas karne ke liye paisa maanga.

Weak password hi sabse bada darwaza hai.

2.2 Definition

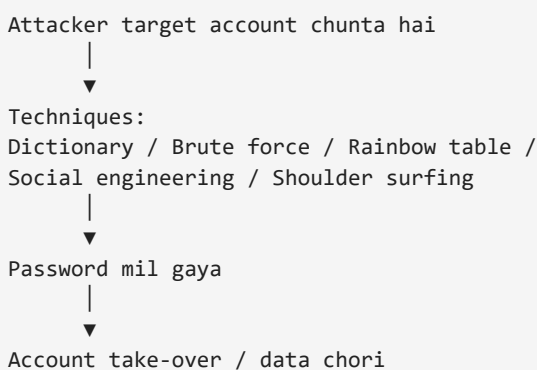
DEFINITION

Password Cracking — kisi user ke account ya system ka **password recover/guess** karne ka process — weak passwords, hash attacks, aur social engineering ke through.

Simple: Tala hai — par chabi itni kamzor ki koi bhi tod le.

2.3 Flowchart

FLOWCHART



2.4 Techniques (Table)

Technique	Kya hota hai
Dictionary attack	Common words/passwords ki list try karna (password, 123456)
Brute force	Har possible combo try karna — slow par full
Rainbow table	Pehle se bani hash table se password dhoondna
Credential stuffing	Doosri site ka leaked password yahan try karna
Social engineering	Logon se password poochna (call/message)
Shoulder surfing	Kandhe se dekh kar password yaad karna
Keylogging	Keylogger se password capture

2.5 Important Points

IMPORTANT POINTS

1. **Weak password** = seconds me toot jata hai.
2. **Hash attacks** — jab password hash me save hota hai — rainbow table se toota jata hai.
3. **Salt** (hash me random value) daalne se rainbow table bekaar ho jata hai.
4. **2FA** (OTP) hone par password ke baad bhi doosra guard rehta hai.
5. Criminals sabse pehle **common + leaked** passwords try karte hain.

2.6 Prevention (Bachao Ke Tarike)

- Password **12+ characters** — letters + numbers + symbols.
- **Alag-alag site** ke liye alag password (password manager).
- **2FA / OTP** har jagah ON karo.
- **Common passwords** (123456, password, qwerty) kabhi mat use karo.
- Leaked password check karo (haveibeenpwned jaisi sites).
- Cracking risk kam karne ke liye **account logout** + rate limiting.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Password cracking = password recover/guess karna.
2. Techniques = dictionary, brute force, rainbow table, social engineering, shoulder surfing.
3. Salt daalne se rainbow table kaam nahi karti.
4. Bachao = strong password + 2FA + alag-alag password.

Viva Q1. Brute force aur dictionary attack me farak?

Ans: Brute force har combo try karta hai; dictionary sirf common words/list try karta hai.

Viva Q2. Rainbow table kya hai?

Ans: Pehle se bani hash-value table jisse password turant dhoondha jata hai.

MEMORY TRICK

Password Cracking = Chabi ka todna. Trick: **Password = PassWORD** — jaise ghar ka darwaza strong lock, waise hi account strong password. 123456 = **khula darwaza**.

PART 3 – KEYLOGGERS AND SPYWARE

Keylogger har key (keystroke) record karta hai. **Spyware** aapki activities par nazar rakhta hai. Dono **chup-chaap** kaam karte hain aur passwords/data nikalte hain.

3.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek cyber cafe me hacker ne **keylogger software** install kar diya.

- Jaise hi koi logon ne **email/bank** me login kiya — uski har **key press record** ho gayi.
- Hacker ko saare **usernames + passwords** mil gaye.
- Usne account se paisa nikal liya — jab tak victim ko pata chalta, nuksan ho chuka tha.

Cyber cafe ke computer par **spyware** bhi tha jo screen capture karta tha.

Keylogger (keys) + Spyware (activities) — dono chhup ke data churate hain.

3.2 Definition

DEFINITION

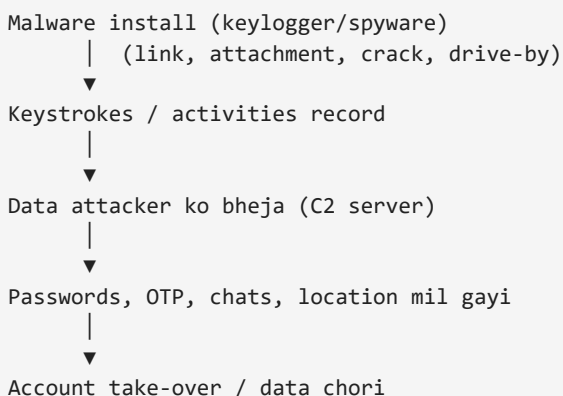
Keylogger — software/hardware jo user ki **har keystroke** record karta hai — password, OTP, messages sab capture ho jate hain.

Spyware — software jo **user ki activities par spy** karta hai — browsing, screenshots, camera, location, data — aur attacker ko bhejta hai.

Simple: Keylogger = **tap kiya hua phone** (typing sunai de); Spyware = **pichchhu** (sab dekh kar bata raha).

3.3 Flowchart

FLOWCHART



3.4 Types (Table)

Type	Kya karta hai
Software keylogger	App/rootkit se keys record karta hai
Hardware keylogger	USB/device jo keyboard se juda hota hai
Screen capture	Screenshots/screen record karta hai
Browser spyware	History, cookies, logins churata hai
Mobile spyware	SMS, calls, location, camera
Banking trojan	Bank pages par fake overlay banata hai

3.5 Keylogger vs Spyware (Table)

Basis	Keylogger	Spyware
Record karta hai	Keystrokes (typing)	Activities (browsing, screen, camera)
Target	Passwords, messages	Sab kuch (data, habits)
Detection	Mushkil (chhupa hota)	Antivirus se kabhi pakda jata
Goal	Login/password chori	Jasusi (espionage)

3.6 Prevention (Bachao Ke Tarike)

- **Antivirus + anti-spyware** install rakho (updated).
- Cyber cafe / public computer par **sensitive login na karo**.
- **Unknown attachments/links** na kholo.
- Apps **official store** se hi install karo.
- **2FA** — keylogger se password chori par bhi OTP guard.
- Task Manager/process list se **suspicious process** check karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Keylogger = har keystroke record karta hai (password chori).
2. Spyware = activities par spy (browsing, screen, camera).
3. Install = link/attachment/crack se; detection mushkil.
4. Bachao = antivirus + 2FA + public computer par login na karo.

Viva Q1. Keylogger ka target kya hota hai?

Ans: Passwords, OTP, messages — jo typing se aate hain.

Viva Q2. Spyware se kya kya pata chalta hai?

Ans: Browsing, screenshots, camera, location, data — sab.

MEMORY TRICK

Keylogger = Typing ka chhalla, Spyware = Screen ka bhediya. Trick: Keylogger me **KEY** hota hai — keyboard ki key. Jaise aap type karo, usne note kar liya.

PART 4 – VIRUS AND WORMS

Virus aur **worm** dono malware hain — par virus ko **host file chahiye**, worm **khud hi failta** hai network me. Ye computer ko slow, corrupt aur hackable banate hain.

4.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Office me ek employee ko **email ke sath attachment** mila — 'salary_details.xls'. Usne khola — file me **virus** tha.

- Virus **attach hua** files se — document, pictures corrupt ho gaye.
- Har file khulte hi doosri file me virus copy ho gaya.

Kuch hafta pehle hi usi office me ek **worm** tha — jisne **network me khud fail kar** saare computers par emails bheji.

Result:

- Virus = files/folders kharab + backup me bhi chhupa.
- Worm = network saturated, internet slow, sab systems infected.

Virus = host se chipakta hai; Worm = khud bhagta hai network par.

4.2 Definition

DEFINITION

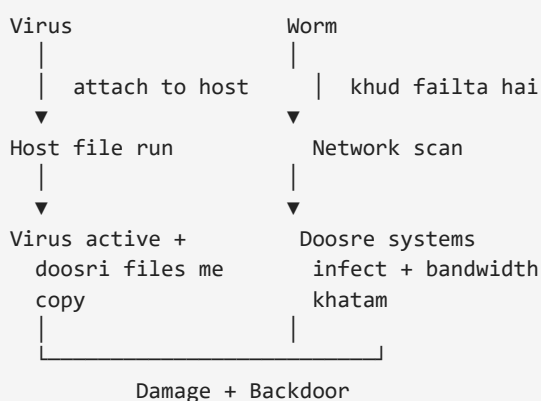
Virus — malware jo **host file/program me attach** hota hai aur jab file run hoti hai to failta hai — files corrupt, slow, damage.

Worm — **self-replicating** malware jo bina host ke **network me khud failta** hai — system slow, bandwidth khatam, backdoor bhi kholta hai.

Simple: Virus = **chipku** (file ke saath chipakta); Worm = **bhagdu** (khud daudta hai).

4.3 Flowchart

FLOWCHART



4.4 Virus vs Worm (Table)

Basis	Virus	Worm
Host chahiye?	Haan (file/program)	Nahi (standalone)
Failta kaise	File run hone par	Khud network me
Speed	Dheema (user action)	Bahut tez (automatic)

Bandwidth	Kam effect	Network saturated
Remove	File se delete	Network clean karna

4.5 Types (Ek Nazar)

Type	Kya karta hai
Boot sector virus	Startup me load hota hai
Macro virus	Office documents ke macro se
File infector	EXE/application files ko infect
Polymorphic virus	Har baar apna code badalta hai (detection mushkil)
Resident virus	Memory me reh kar failta hai

4.6 Prevention (Bachao Ke Tarike)

- **Antivirus updated** rakho + regular scan.
- **Email attachments** bina check kholo mat.
- Software **official sources** se lo.
- **USB/devices** scan karke use karo.
- Network par **firewall + endpoint security**.
- **Backup** — virus delete karke bhi data safe rahe.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Virus = host file me attach + file run par failta.
2. Worm = self-replicating, network me khud failta.
3. Virus slow + files corrupt; worm bandwidth + systems khatam.
4. Bachao = antivirus + attachments se savdhaan + backup.

Viva Q1. Virus aur worm me main farak?

Ans: Virus ko host file chahiye, worm khud network me failta hai.

Viva Q2. Polymorphic virus kya hai?

Ans: Har baar apna code badalta hai — antivirus ko confuse karta hai.

MEMORY TRICK

Virus = Chipku, Worm = Bhagdu. Trick: Virus me **host** = ghar — bina ghar ke nahi rahta; Worm me **W = Web** — web par daudta hai.

PART 5 – TROJAN HORSES

Trojan Horse — malware jo **accha/legal software ban ke** aata hai — andar danger. Naam usi purani kahaani se hai — ladai wala ghoda.

5.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek app thi — **‘Free Games 2026’** — bilkul sahi dikhne wali. Rohit ne download ki.

Game chali bhi — par sath-sath:

- Uske **bank app** ka data nikal raha tha.
- **Remote access** ka gate khul gaya — hacker Rohit ka phone control kar sakta tha.
- Uske **contacts** me fake messages gaye.

Rohit ko lagta raha — ‘bas ek game hai’.

Yahi trojan ka khel hai — bahar se sweet, andar se nuksan.

5.2 Definition

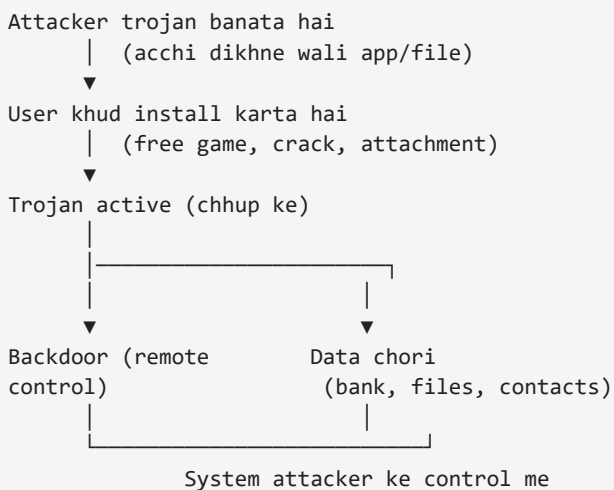
DEFINITION

Trojan Horse (Trojan) — malware jo **legitimate/useful program ke roop me chhupa** hota hai — user khud install karta hai — aur andar se **backdoor, data chori, remote access** karta hai.

Simple: Ghoda dakhil karo, **sipahi andar se nikle** — bilkul Trojan war wali story.

5.3 Flowchart

FLOWCHART



5.4 Types (Table)

Type	Kya karta hai
Remote Access Trojan (RAT)	Attacker ko system ka full control deta
Banking trojan	Bank pages fake bana kar credentials le
Keylogger trojan	Keystrokes record karta
Backdoor trojan	Hidden entry khulta
DDoS trojan	Botnet ka hissa banata

FakeAV trojan	Fake antivirus dikha kar paisa/access leta
---------------	--

5.5 Trojan vs Virus vs Worm (Table)

Basis	Trojan	Virus	Worm
Failta kaise	Khud failta nahi — user install karta	Host file se	Network me khud
Replicate?	Nahi	Haan	Haan (fast)
Main khatra	Backdoor/data chori	Files damage	Bandwidth + spread

5.6 Prevention (Bachao Ke Tarike)

- **Free apps/cracks** se bacho — trojan ka favourite rasta.
- Apps **official store** se hi install karo.
- **Permissions** dekho — game ko camera/contacts kyu chahiye?
- **Antivirus** + app verification ON rakho.
- Attachment/link jo **unsolicited** ho — na kholo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Trojan = acchi dikhne wali app/file jisme malware chhupa hai.
2. Ye khud failta nahi — user khud install karta hai.
3. Types = RAT, banking, keylogger, backdoor, DDoS trojan.
4. Bachao = official store + permissions + antivirus.

Viva Q1. Trojan ka naam kyu hai?

Ans: Trojan war wale ghode se — bahar accha, andar danger.

Viva Q2. RAT kya hai?

Ans: Remote Access Trojan — attacker ko system ka remote control deta hai.

MEMORY TRICK

Trojan = Ghar me aaya mehmaan, andar se loot. Trick: **Trojan = Tarike se ghusna** — T = Trusted dikhna, phir loot.

PART 6 – BACKDOORS

Backdoor — system ka **chhupa hua gate** — jisse attacker jab chaahe andar aa jata hai, bina normal login ke.

6.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Hacker ne pehli baar ek website ki **weak password** se login kiya aur kuch data nikala. Phir usne site me ek **backdoor** install kar diya — ek chhupa hua URL/function.

Website wale ne password badal liya, firewall laga diya — par:

- Hacker ab bhi **backdoor se andar** aa jaata hai.
- Har baar kuch na kuch le jaata hai.
- Backdoor ko dhoondhna **bahut mushkil** hota hai.

Ye chhupa hua gate hi Backdoor hai — aur cyber security ki sabse badi gaddari.

6.2 Definition

DEFINITION

Backdoor — system me chhupa hua **secret entry point** — jo normal authentication ko **bypass** karke attacker ko access deta hai — ya to developer ne chhoda, ya attacker ne install kiya.

Simple: Ghar ka **chhupa hua darwaza** — jo sirf chor jaanta hai.

6.3 Flowchart

FLOWCHART

Backdoor kaise aata hai?



6.4 Types (Table)

Type	Kya hota hai
Software backdoor	Malware/trojan se install
Hardware backdoor	Device/chip me built-in
Default credentials	admin/admin, root chhoda hua
Hidden accounts	Chhupa hua user account
Web backdoor	Website me chhupa hua script (webshell)

6.5 Important Points

IMPORTANT POINTS

1. Backdoor **authentication bypass** karta hai — password ki zaroorat nahi.
2. Detecting backdoor **mushkil** — kyunki wo normal traffic jaisa lagta hai.
3. **Rootkits** backdoor ko aur bhi chhupa dete hain.
4. Backdoor milne par poori **forensic investigation** + system rebuild chahiye.
5. Good practice: **default passwords + hidden services** check karna.

6.6 Prevention (Bachao Ke Tarike)

- **Default passwords** turant badlo.
- **Patch management** — exploits se backdoor na ban paye.
- Network **monitor** karo — suspicious connections dekho.
- **File integrity** check (hash) — chhupa script pakdo.
- Antivirus + **EDR** (endpoint detection) rakho.
- Backdoor milne par **system clean/reinstall** karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Backdoor = chhupa hua secret entry point — authentication bypass.
2. Aata kaise = attacker installs ya vendor chhode (default creds).
3. Detection mushkil; rootkit aur chhupata hai.
4. Bachao = default password badlo + patching + monitoring + hash check.

Viva Q1. Backdoor se kya hota hai?

Ans: Attacker bina normal login ke baar-baar system me ghus sakta hai.

Viva Q2. Backdoor kaise install hota?

Ans: Trojan/exploit se attacker install karta ya vendor ki default/galt setting se.

MEMORY TRICK

Backdoor = Chhupa hua darwaza. Trick: **Backdoor = Back ka door** — aage ka darwaza (login) lock, par pichwale se andar.

PART 7 – DoS AND DDoS ATTACKS

DoS — ek source se itna traffic ki website/service band ho jaye. **DDoS** — hazaaron computers (botnet) se saath-me attack — rokna mushkil.

7.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Diwali sale ke din ek **online shopping site** par lagaathaar orders aa rahe the.

Ek hacker group ne:

- Hazaaron **hacked computers** (botnet) ko control me liya.
- Sab computers ne **ek saath** site par fake requests bheji.
- Server ka traffic **itna badh gaya** ki:
- Real customers ka site **khulna band** ho gaya.
- Server crash ho gaya — site band.

Site ko 2 din lag gaye wapas chalane me — **crore ka nuksan**.

Bahut saare computers ka ek-saath attack — DDoS. Ek hi computer ka — DoS.

7.2 Definition

DEFINITION

DoS (Denial of Service) — ek system/source se itne requests bhejna ki target server/service **overload ho kar legit users ko service na de** sake.

DDoS (Distributed DoS) — **bahut saare computers (botnet)** se distributed attack — ek source se nahi, isliye block karna aasaan nahi.

Simple: DoS = ek aadmi darwaza tod raha; DDoS = poori bheed darwaza tod rahi.

7.3 Flowchart

FLOWCHART

DoS: Attacker ----1 system----> Target
single source, traffic flood

DDoS: Botnet (hazaaron hacked PCs)



Target server (overload)



Server down / legit users out

7.4 DoS vs DDoS (Table)

Basis	DoS	DDoS
Source	Ek system/connection	Hazaaron systems (botnet)
Block karna	Ek IP block karo	Mushkil — thousands IPs
Speed	Dheema possible	Tez aur powerful
Detect	Aasaan (single source)	Mushkil
Example	SYN flood ek IP se	Botnet ka attack

7.5 Types / Techniques (Table)

Type	Kya hota hai
SYN flood	Half-open connections bana kar server ko bekar karna
Ping of Death	Bade/badle ping packets se system crash
Smurf attack	Spoofed ping broadcast se network flood
UDP flood	UDP packets se bandwidth khatam
HTTP flood	Site par fake requests (application layer)
Botnet DDoS	Hacked devices se distributed attack

7.6 Prevention (Bachao Ke Tarike)

- **Rate limiting** — ek IP se limited requests.
- **Firewall / IPS** — abnormal traffic block.
- **CAPTCHA** — fake bots rokna.
- **CDN / DDoS protection** service (Cloudflare jaisi).
- **Server load balancing** + extra capacity.
- **Monitoring + alert** — attack shuru hote hi action.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. DoS = ek source se overload — service band.
2. DDoS = botnet se distributed attack — block mushkil.
3. Types = SYN flood, ping of death, smurf, UDP flood, HTTP flood.
4. Bachao = rate limit, firewall/IPS, CDN protection, monitoring.

Viva Q1. DoS aur DDoS me farak?

Ans: DoS ek source se; DDoS hazaaron sources (botnet) se.

Viva Q2. SYN flood me kya hota?

Ans: Half-open connections bana kar server ki capacity khatam karna.

MEMORY TRICK

DoS = Ek aadmi, DDoS = Bheed. Trick: DDoS me **D = Distributed = Duniya bhar ke** computers. Jaise bheed-bhad me darwaza koi nahi rok paata.

PART 8 – BUFFER OVERFLOW

Buffer Overflow — ek **chhote dabbe (buffer)** me zyada data daal diya — aur data **bahar leak** hoke program ki memory ko corrupt kar deta hai — attacker apna code chala leta hai.

8.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek website ke **login page** me username box sirf **20 letters** rakhta tha.

Hacker ne box me **200 letters** daal diye — usme se kuch letters **malicious code** the.

- Data buffer se **overflow** hokar memory ke doosre hisse me gaya.
- Program ne **attacker ka code** execute kar diya.
- Hacker ko **remote control** mil gaya.

Ye sab us program ki galti se hua — usne input ki **lambaai check nahi ki**.

Zyada input se memory corrupt + attacker code — Buffer Overflow.

8.2 Definition

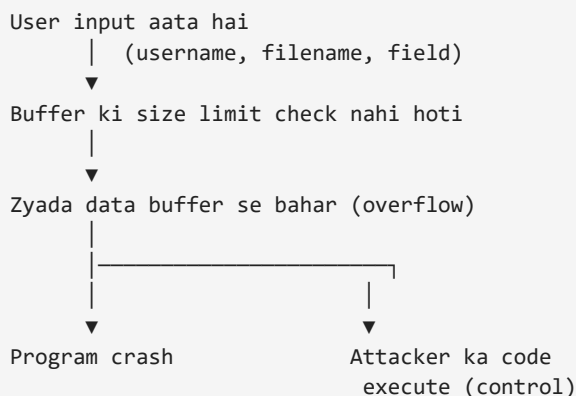
DEFINITION

Buffer Overflow — program ne ek fixed-size **buffer (memory area)** me capacity se **zyada data** likh diya — data adjacent memory me phail jata hai — program crash karta ya **attacker apna code execute** karwa leta hai.

Simple: Glass me paani se **zyada paani** daala — sab taraf phail gaya.

8.3 Flowchart

FLOWCHART



8.4 Types (Table)

Type	Kya hota hai
Stack overflow	Stack me buffer overflow — return address badalna
Heap overflow	Heap memory me overflow — data structures corrupt
Integer overflow	Bade number se variable bahar
Format string	Galat format se memory read/write

8.5 Important Points

IMPORTANT POINTS

1. Overflow tab hota hai jab program **input ki length validate nahi** karta.
2. Attacker overflow se **code injection** karta hai — system control.
3. **Stack overflow** me return address hijack hota hai — sabse common.
4. Ye **C/C++** me zyada hota hai (bina bound check); modern languages safe.
5. Patch lage na ho to **known buffer overflow** sabse khatarnak.

8.6 Prevention (Bachao Ke Tarike)

- **Input validation** — length aur type check karo.
- **Safe functions** (strcpy_s, gets nahi) use karo.
- **ASLR + DEP** — memory protection ON.
- **Compiler protections** (stack canary).
- Software **patch/update** karte raho.
- Code **review + testing** (fuzzing) karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Buffer overflow = zyada input, buffer se bahar, memory corrupt.
2. Attacker code execute karwa leta hai (code injection).
3. Common = stack overflow (return address hijack).
4. Bachao = input validation + ASLR/DEP + safe functions + patching.

Viva Q1. Buffer overflow me kya galat hota hai?

Ans: Program input ki length check nahi karta — data buffer se bahar jata hai.

Viva Q2. Stack overflow me kya hijack hota hai?

Ans: Return address — attacker apna code wahan daal deta hai.

MEMORY TRICK

Buffer Overflow = Paani ka glass zyada bharna. Trick: Buffer = **Bucket** — bucket chhoti, paani zyada — bahar phaila. Isliye hamesha **check input**.

PART 9 – ATTACK ON WIRELESS NETWORKS

WiFi par attack — kyunki WiFi ke signals **hawa me hote hain** — koi bhi paas khada ho kar pakad sakta hai. Weak security to bhi asaan ho jati hai.

9.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek cafe ki WiFi password thi — **WEP** (purani security). Hacker ne:

- **Packets sniff** kiye — WiFi ke signals me data pakda.
- WEP ki **weakness** use kar ke password tod diya.
- WiFi se jud kar **network me ghusa** — aur doosre devices ka data dekha.
- Ek **fake WiFi (evil twin)** bhi bana diya — 'Cafe_FreeWiFi' naam ka — jisse logon ke login churata raha.

Hawa me signal, asaan me attack — Wireless Network Attack.

9.2 Definition

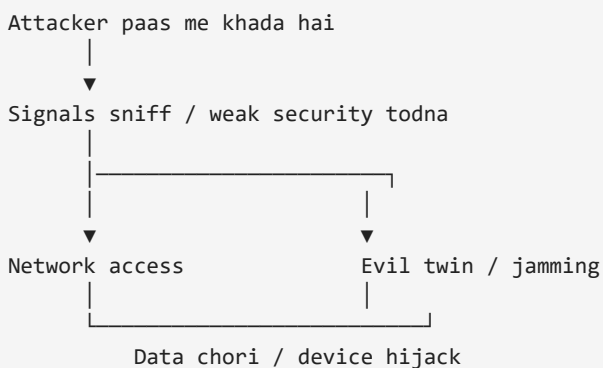
DEFINITION

Wireless Network Attack — WiFi ya wireless network ke **signals/security ko break** karna — password cracking, sniffing, fake access point, jamming — taaki data pakda jaye ya network me ghusa jaye.

Simple: Wired me taaar kaatna padta; wireless me **hawa hi rasta** hai.

9.3 Flowchart

FLOWCHART



9.4 Types of Wireless Attacks (Table)

Attack	Kya hota hai
Packet sniffing	Signals se data capture
WEP/WPS cracking	Weak security todna (WPS pin brute force)
Evil twin	Fake WiFi same naam — login churana
Rogue access point	Bina permission ka WiFi point
Deauthentication	Users ko WiFi se gira kar data pakadna
Jamming	Signals me interference — network band
KRACK	WPA2 ke handshake me vulnerability

9.5 Important Points

IMPORTANT POINTS

1. **WEP purani aur weak** hai — minutes me toot jati hai.
2. **WPA2/WPA3** strong hai — par WPS off rakho.
3. Public WiFi par **data plain** ja sakta hai — VPN use karo.
4. **Evil twin** same naam ka fake network — login churata hai.
5. Wireless attack ke liye attacker ko **network ke paas** hona padta hai.

9.6 Prevention (Bachao Ke Tarike)

- **WPA3/WPA2** use karo — WEP kabhi nahi.
- **WPS band** rakho.
- Router ka **default password** badlo.
- Public WiFi par **VPN** use karo.
- **SSID broadcast** + MAC filtering se thoda bachav.
- Router **firmware update** karte raho.

EXAM NOTES (2 MARKS / VIVA)**Exam Notes (2 Marks):**

1. Wireless attack = signals/security break karna.
2. Types = sniffing, WEP cracking, evil twin, rogue AP, deauth, jamming.
3. WEP weak; WPA2/WPA3 strong; WPS off rakho.
4. Public WiFi par VPN + sensitive kaam na karo.

Viva Q1. Evil twin attack kya hai?

Ans: Same naam ka fake WiFi banakar login/data churana.

Viva Q2. WEP kyu kharab hai?

Ans: Purani aur weak security — kuch hi minutes me toot jati hai.

MEMORY TRICK

Wireless = Hawa ka network. Trick: **WiFi = Wide-open** — signal hawa me hai, isliye hamesha **WPA2/3 + VPN**. Evil twin = **nakli bhai**.

PART 10 – PHISHING (METHOD AND TECHNIQUES)

Phishing — asli bank/company ban ke **fake link** bhejna — victim login kare to uske **credentials** churana. Ye sabse zyada successful attack hai.

10.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek aadmi ko email aayi:

- Sender: **support@hdfc-bank.com** (dhyan se dekho — dash hai, asli nahi).
- Message: “Aapka account block ho jayega. **Turant verify karo**” + link.

Usne link kholi — page bilkul **bank jaisa** dikhta tha. Usne **user + password + OTP** daal diye.

- Hacker ke paas saare **credentials** chale gaye.
- Account se paise nikal liye.

Fake page/link se credentials churana — Phishing.

10.2 Definition

DEFINITION

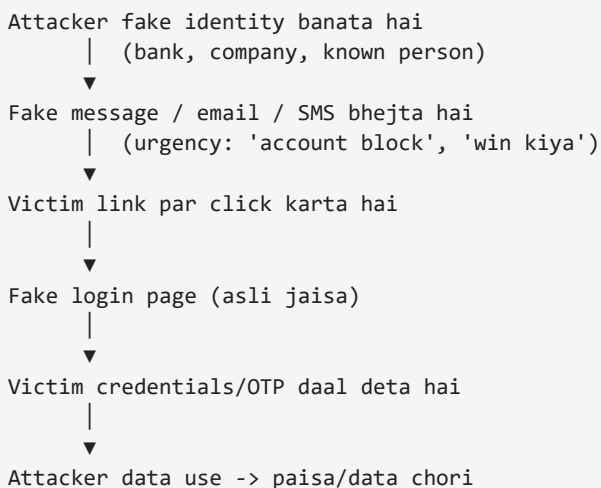
Phishing — ek **social engineering** attack jisme attacker **trusted entity (bank, company, person)** ban ke fake message/page bhejta hai — victim apne **login, OTP, card details** khud de deta hai.

Method: Fake message → Fake link → Fake page → Credentials → Nuksan.

Simple: Machhli pakadne ke liye **chara** (fake link) dalna — phishing = fishing.

10.3 Flowchart – Method

FLOWCHART



10.4 Phishing Techniques (Table)

Technique	Kya hota hai
Email phishing	Fake email + link
Spear phishing	Ek specific person ko targeted attack
Whaling	Bade official (CEO) ko targeted
Smishing	SMS ke through phishing

Vishing	Phone call ke through (voice phishing)
Pharming	DNS me chhed — asli site par bhi fake page bhi
Clone phishing	Asli email ki copy + badli hui link
Link manipulation	Galat URL — hdfc-bank.com jaisa nakli
Tab nabbing	Tab replace karke login churana

10.5 Important Points

IMPORTANT POINTS

1. Phishing **technique par nahi** — **bharosay par** chalti hai — isliye awareness sabse bada bachav.
2. **Urgency** wale messages par turant mat click karo.
3. **URL hamesha check** karo — asli vs nakli (dash, extra letter).
4. Pharming me DNS badal jata hai — URL sahi dikhne par bhi fake page.
5. Bank **kabhi OTP/ password email-call se nahi maangta**.

10.6 Prevention (Bachao Ke Tarike)

- Links par click karne se pehle **URL verify** karo.
- **2FA** — credentials churi hone par bhi guard.
- Bank/company ko **direct official channel** se call karo.
- **Email security** — sender address + spelling dekho.
- Unknown **attachments** na kholo.
- **Antivirus/anti-phishing** browser protection ON.
- Kisi ne credentials de diye to **turant password + bank card** change/block karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Phishing = fake trusted entity ban ke credentials churana.
2. Method = fake message + link + page + credentials.
3. Techniques = email, spear, whaling, smishing, vishing, pharming, clone.
4. Bachao = URL check + 2FA + urgency se savdhaan.

Viva Q1. Spear phishing kya hai?

Ans: Ek specific person (targeted) ke liye personal phishing attack.

Viva Q2. Pharming me kya badalta hai?

Ans: DNS — URL sahi dikhta hai par page fake hota hai.

MEMORY TRICK

Phishing = Fishing — fake link = **chara**, victim = machhli. Trick: **Urgency = Dhamki** — jo message 'turant karo' bole, socho phir click karo.

APPENDIX – QUICK REVISION SHEET (UNIT 5)

Sabse Pehle Yaad Rakho — One-Line Definitions

Topic	Ek Line Me
Proxy Server	Beech ka server — asli IP chhupata hai
Anonymizer	Identity/IP fully chhupane wala tool
Password Cracking	Password recover/guess karna (dictionary, brute force)
Keylogger	Har keystroke record karta hai
Spyware	Activities par spy (browsing, screen, camera)
Virus	Host file me attach + file run par failta
Worm	Self-replicating — network me khud failta
Trojan Horse	Acchi app/file ban ke andar malware
Backdoor	Chhupa hua secret entry point (auth bypass)
DoS	Ek source se overload — service band
DDoS	Botnet (hazaaron systems) se distributed attack
Buffer Overflow	Zyada input buffer se bahar — memory corrupt + code run
Wireless Attack	WiFi signals/security break (evil twin, sniffing)
Phishing	Fake trusted entity ban ke credentials churana

Memory Tricks

- Proxy = **Paisa ka bichauliya** (IP nahi batata).
- Password = **PassWORD** — 123456 = khula darwaza.
- Keylogger = **KEY** in keyboard.
- Virus = **Chipku**, Worm = **Bhagdu** (W = Web).
- Trojan = **Ghar me mehmaan, andar loot**.
- Backdoor = **Back ka door** — pichhla darwaza.
- DDoS = **Distributed = Duniya bhar ke** computers.
- Buffer Overflow = **Bucket chhoti, paani zyada**.
- Wireless = **Hawa ka network** — WPA2/3 + VPN.
- Phishing = **Fishing** — fake link = chara.

Rapid Fire Viva (15 Questions)

Q	Answer
Proxy server se kya chhupata hai?	Asli IP
Brute force kya hai?	Har possible password combo try karna
Rainbow table kya hai?	Pehle se bani hash-value table
Keylogger kya record karta?	Keystrokes (passwords, OTP)

Virus ko kya chahiye?	Host file (attach hone ke liye)
Worm kaise failta?	Network me khud (bina host ke)
Trojan kaise install hota?	User khud install karta hai (accha ban ke)
Backdoor kya bypass karta?	Normal authentication
DoS vs DDoS?	DoS ek source, DDoS botnet se
SYN flood me kya hota?	Half-open connections se server bekar
Buffer overflow me kya check nahi hota?	Input ki length
WEP kyu weak hai?	Purani security, minutes me toot jati
Evil twin kya hai?	Same naam ka fake WiFi — login churana
Pharming me kya badalta hai?	DNS
Phishing ka target?	Credentials (login/OTP/card)

Note: Ye Unit 5 ka complete notes hai — cyber crime ke tools aur methods.